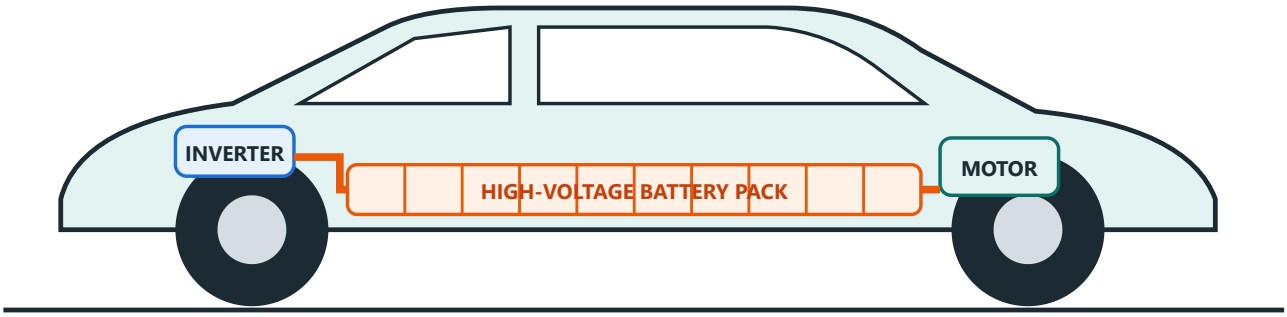
PRE-TRAINING STUDY GUIDE · PAKISTAN

EV & Hybrid Vehicle Technician Foundation Course

How electric and hybrid cars work, what is inside them, and how to work on them safely. Written for new technicians preparing for hands-on training.



Battery under the floor · inverter and motor at the axle · orange = high voltage

LEVEL
Beginner (GCSE / Matric)

STUDY TIME
About 6 weeks, 1 hour a day

CONTENTS
21 modules, 21 quizzes

EDITION
1.1 · November 2026

How to use this course

This course is the first step before hands-on training. It gives you the knowledge you need so that, when you stand in front of a real electric car in a training workshop, you already understand what you are looking at and why the safety rules exist.

You do not need to be an engineer. You need basic maths (multiply, divide, percentages), basic science, and the patience to read carefully.

How each module works

- **You will learn** — a short list at the start. Read it first.
- **The lesson** — short sections with diagrams. Read slowly. Draw the diagrams yourself in a notebook; drawing helps you remember.
- **Coloured boxes** — see the key below.
- **Picture it** — purple boxes and drawings that link a new idea to something you already know. These are the easiest way to remember. Close the book and try to see the picture in your mind.
- **Summary** — the main points to remember.
- **Quiz** — test yourself. Answers are at the back. On the web version you can tap an answer to check it.

DANGER
A safety rule. Breaking it can kill or injure you.

KEY IDEA
The most important point in the section.

WORKED EXAMPLE
A calculation shown step by step.

WORKSHOP TIP
Practical advice from real repair work.

IN PAKISTAN
How the topic applies to our roads, weather and market.

SUMMARY
What to remember from the module.

Study plan (6 weeks)

Week	Modules	Goal
1	1 – 3	Understand the opportunity, basic electricity, and high-voltage safety
2	4 – 7	Know the vehicle types, the main parts, and how batteries are built
3	8 – 11	Battery management, motors, power electronics and performance
4	12 – 14	Braking, hybrid systems and charging
5	15 – 18	Cooling, low-voltage systems, tools and fault finding
6	19 – 21, revision	Maintenance, bikes and rickshaws, career. Repeat every quiz.

THE PASS MARK
Aim for **80% or more** in every quiz before moving on. For Module 3 (High-voltage safety) the target is **100%**. In safety, one wrong answer is one too many.

READ THIS BEFORE ANYTHING ELSE

This book teaches **knowledge**. It does not make you qualified to work on high-voltage systems. Electric and hybrid cars carry **200 to 800 volts DC**. That is enough to kill instantly.

Do not open, disconnect or test any orange cable, battery pack, inverter or motor until you have completed supervised practical training with the correct safety equipment. Until then you may look, read, and learn. Nothing more.

About the vehicle data

Models, prices and specifications in this course were collected in November 2026 from Pakistani motoring press and manufacturer information. Companies change specifications often, and sources sometimes disagree. Figures are given to teach you how to *read and compare* specifications. Before working on any car, always check the official service information for that exact model and year.

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PART A · FOUNDATIONS

1 The opportunity: EVs and hybrids in Pakistan

Why these cars are arriving now, which ones you will see, and what work they will bring.

YOU WILL LEARN

- Why electric and hybrid cars are growing quickly in Pakistan
- Which brands and models are on our roads
- How an electric car is different to work on than a petrol car
- Why independent workshops will be needed

1.1 What is changing

For many years the Pakistani car market was a small number of Japanese petrol models. That has changed. Since 2024, Chinese manufacturers have entered with battery electric cars and plug-in hybrids, and most of them now assemble cars locally or plan to.

Three things are pushing this change:

- **Fuel cost.** Petrol is imported and expensive. Electricity per kilometre is much cheaper, especially for owners with solar panels at home.
- **Government policy.** The **New Energy Vehicle (NEV) Policy 2025–30** sets a target that 30% of new vehicles sold in 2030 should be electric, across bikes, rickshaws, cars, buses and trucks. It lowers duties on EV parts and supports local assembly.
- **Chinese technology and price.** China makes most of the world's EV batteries. Chinese brands can offer a well-equipped EV or plug-in hybrid at a price close to a Japanese petrol SUV.

IN PAKISTAN — THE NUMBERS (2026)

- Target: **30%** of new vehicle sales electric by 2030.
- Target: **3,000** public charging stations by 2030. In early 2026 there were only around 30–50.
- Local production of electric bikes and rickshaws rose from about **60,000** (2024–25) to about **210,000** (2025–26).
- The PAVE scheme gives buyers a cash subsidy on electric bikes and rickshaws that use lithium batteries.
- Regulator NEPRA set a special low electricity tariff for charging stations (base rate Rs 23.57 per unit, plus the station's margin).

1.2 The vehicles you will meet

You will learn what the letters BEV, PHEV, HEV and REEV mean in Module 4. For now: a **BEV** has only a battery and motor. A **hybrid** has an engine and a motor.

Table 1.1 — Main electrified brands in Pakistan (November 2026)

Brand (local partner)	Battery electric (BEV)	Plug-in / range-extender	Hybrid (HEV)
BYD (Mega Motor / HUBCO)	Atto 2, Atto 3, Seal, Sealion 7	Sealion 6, Shark 6 pickup	—
Changan Deepal (Master Changan)	S07, L07, E07	S05 (range extender)	—
MG (MG JW)	ZS EV, MG4, MG5 EV, Binguo	HS PHEV	—
GWM Haval (Sazgar)	Ora 03	H6 PHEV	H6 HEV, Jolion HEV
Chery / Jaecoo	Chery Q	Tiggo 7, 8, 9 PHEV; Jaecoo J7 PHEV	—

Brand (local partner)	Battery electric (BEV)	Plug-in / range-extender	Hybrid (HEV)
Toyota (Indus Motor)	—	—	Corolla Cross HEV; used imports: Aqua, Prius, Yaris
Honda (Honda Atlas)	—	—	HR-V e:HEV; used imports: Vezel, Fit
Hyundai / Kia	Ioniq 5, EV5 (small numbers)	—	Tucson, Santa Fe, Sportage HEV

There are also many small Chinese city EVs (for example Honri VE and other mini cars), and a very large number of electric motorcycles, scooters and rickshaws. Module 20 covers the two- and three-wheelers.

> **WORKSHOP TIP**
>
> Do not forget the **used Japanese hybrids**. Tens of thousands of imported Toyota Aqua, Prius and Honda Vezel hybrids are already 8–15 years old. Their hybrid batteries are wearing out now. This is work that exists today, not in five years.

1.3 Why independent workshops matter

A new EV usually has a vehicle warranty of about 4–6 years and a battery warranty of around 8 years or 160,000 km. During warranty, owners go to the dealer.

After warranty, things change. Dealer labour rates and parts prices are high. A middle-class family with limited spare income will look for a trusted private workshop — the same way they do today for a Corolla or a Mehran. The problem is that very few private workshops understand high-voltage cars. Many mechanics are afraid of them, and they are right to be afraid if they are not trained.

> **KEY IDEA**
>
> The cars are arriving faster than the technicians. A person who can **safely** diagnose and repair EVs and hybrids will have a skill that is in short supply for many years.

1.4 How is an EV different to work on?

Table 1.2 — Petrol car and electric car compared

Area	Petrol car	Battery electric car
Moving parts in the drive unit	Hundreds (pistons, valves, gearbox)	Around 20 (rotor, a few gears, bearings)
Regular service items	Engine oil, oil filter, air filter, spark plugs, belts	Cabin filter, brake fluid, coolant, tyres
Main danger	Fuel fire, hot parts, moving belts	**High voltage** (200–800 V), battery fire
Main tool	Spanners and experience	Scan tool, multimeter, insulation tester
Typical faults	Mechanical wear	Electrical, sensors, software, cooling, 12 V battery
Brake wear	Normal	Low (the motor does most of the braking)
Tyre wear	Normal	Higher (heavy car, strong torque)

An EV still has suspension, steering, brakes, tyres, air-conditioning, lights and body work. Your normal mechanical skills remain useful. What is new is the **high-voltage system** and the amount of **electronics and software**.

1.5 What an EV technician actually does

- Makes the car safe before any work (isolating the high-voltage system)
- Reads fault codes and live data with a scan tool, and understands what they mean
- Tests batteries, cables, motors and chargers with meters

- Replaces parts: 12 V batteries, coolant pumps, chargers, contactors, sensors, drive units
- Services cooling systems, brakes, air-conditioning and suspension
- Explains problems honestly to the customer

At a higher level, after more training: opening battery packs, replacing modules, and repairing power electronics.

SUMMARY

- Fuel cost, government policy and Chinese technology are driving EVs and hybrids into Pakistan.
- You will see Chinese BEVs and PHEVs, Japanese and Korean hybrids, and many electric bikes and rickshaws.
- After warranty, owners need affordable private workshops, and trained technicians are scarce.
- EVs have fewer mechanical parts but need electrical knowledge, diagnostic tools and strict safety.

Quiz 1

Choose one answer for each question.

1. **What is the NEV Policy target for electric vehicles as a share of new sales by 2030?**

A 10% | **B** 30% | **C** 60% | **D** 100%

2. **Why will many owners move to private workshops after a few years?**

A Dealers do not repair EVs | **B** EVs never break down | **C** The warranty ends and dealer repairs are expensive | **D** Private workshops are required by law

3. **Which of these is NOT a regular service item on a battery electric car?**

A Brake fluid | **B** Cabin filter | **C** Tyres | **D** Spark plugs

4. **What is the main new danger when working on EVs compared with petrol cars?**

A High voltage | **B** Hot exhaust | **C** Engine oil | **D** Heavy gearbox

5. **Which group of vehicles already needs hybrid battery work in Pakistan today?**

A New BYD cars | **B** Used Japanese hybrids such as Aqua, Prius and Vezel | **C** Diesel trucks | **D** Electric buses only

PART A · FOUNDATIONS

2 Electricity basics

Volts, amps, ohms, watts and kilowatt-hours — the language of every EV specification.

YOU WILL LEARN

- What voltage, current and resistance are, and how Ohm's law links them
- The difference between power (kW) and energy (kWh)
- The difference between AC and DC
- What happens when cells are connected in series and in parallel

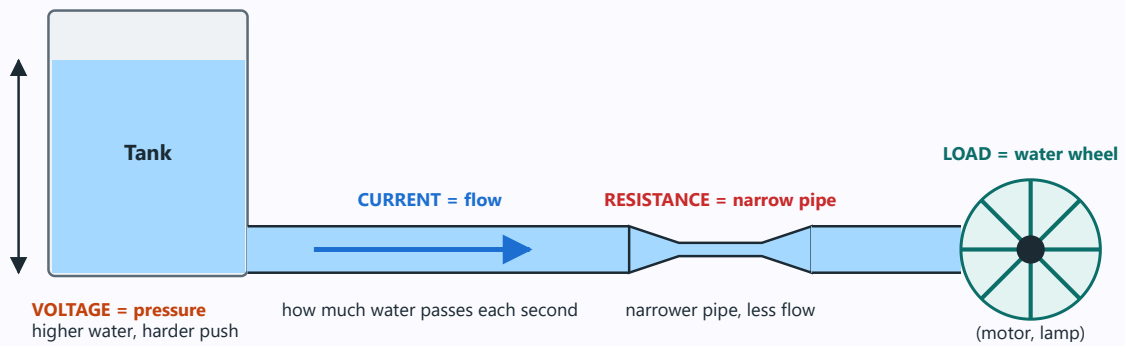
2.1 Voltage, current and resistance

Think of electricity as water in pipes.

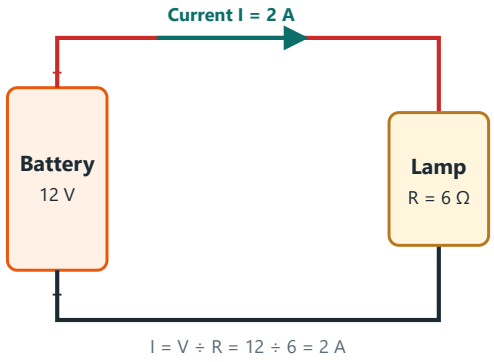
- **Voltage (V)** is the *pressure* pushing the electricity. Unit: **volt (V)**.
- **Current (I)** is the *flow* — how much electricity passes each second. Unit: **ampere or amp (A)**.
- **Resistance (R)** is anything that *restricts* the flow, like a narrow pipe. Unit: **ohm (Ω)**.

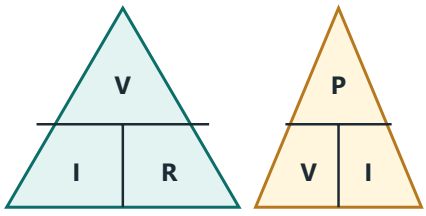
More pressure gives more flow. More restriction gives less flow. This is **Ohm's law**:

`V = I × R` `I = V ÷ R` `R = V ÷ I`



Picture it Electricity behaves like water. Raise the tank (more volts) and more flows. Narrow the pipe (more ohms) and less flows.





Cover the one you want.
Side by side = multiply. One over the other = divide.

Figure 2.1 A simple circuit, and the two triangles that help you remember the formulas.

2.2 Power: how fast energy is used

Power (P) is how quickly electrical energy is being used or delivered. Unit: **watt (W)**. One **kilowatt (kW)** is 1,000 watts.

`P = V × I`

WORKED EXAMPLE

An EV battery at **400 V** supplies **250 A** to the motor. What power is that?

P = V × I = 400 × 250 = 100,000 W = **100 kW**

For comparison, one horsepower (hp) is about 0.75 kW. So 100 kW ≈ 134 hp.

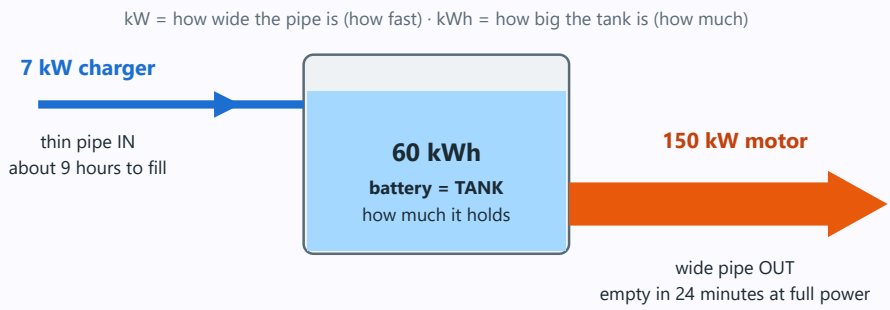
2.3 Energy: how much work can be done

Energy is power used over time. In EVs we measure it in **kilowatt-hours (kWh)**. It is the same "unit" that appears on your electricity bill.

`Energy (kWh) = Power (kW) × Time (hours)`

KEY IDEA

kW is like the size of the tap. kWh is like the size of the water tank.
Motor power (kW) tells you how strongly the car can accelerate. Battery energy (kWh) tells you how far it can go. Many people confuse them. A technician must not.



Picture it The same tank can be filled through a thin pipe and emptied through a wide one. Tank size and pipe size are different things: kWh and kW.

WORKED EXAMPLE

A car uses an average of 15 kW while cruising. Its battery holds 60 kWh. How long can it cruise?

Time = Energy ÷ Power = 60 ÷ 15 = **4 hours**

Amp-hours (Ah)

Battery cells are often rated in **amp-hours (Ah)**. A 100 Ah cell can give 100 A for 1 hour, or 10 A for 10 hours. To convert to energy, multiply by the voltage:

`Energy (Wh) = Voltage (V) × Capacity (Ah)`

Example: a 12 V, 50 Ah car battery stores 12 × 50 = 600 Wh = 0.6 kWh. An EV battery stores about 100 times more.

2.4 Why EVs use high voltage

Look again at `P = V × I`. To deliver 100 kW you can use high voltage and low current, or low voltage and high current.

Voltage	Current needed for 100 kW	Cable needed
12 V	8,333 A	Impossible — thick as your arm
400 V	250 A	About as thick as your thumb
800 V	125 A	Thinner and lighter again

Current causes heating in cables. The heat loss is $P_{loss} = I^2 \times R$. If you halve the current, the heat loss falls to a quarter. That is why EVs use 400 V or 800 V: thinner cables, less wasted heat, faster charging. The price is **danger** — see Module 3.

2.5 DC and AC

- **Direct current (DC)** flows in one direction only. Batteries store and give DC.
- **Alternating current (AC)** changes direction many times each second. Mains electricity in Pakistan is AC: **230 V, 50 Hz** (50 cycles per second). Factories and large chargers use **three-phase** AC at 400 V.

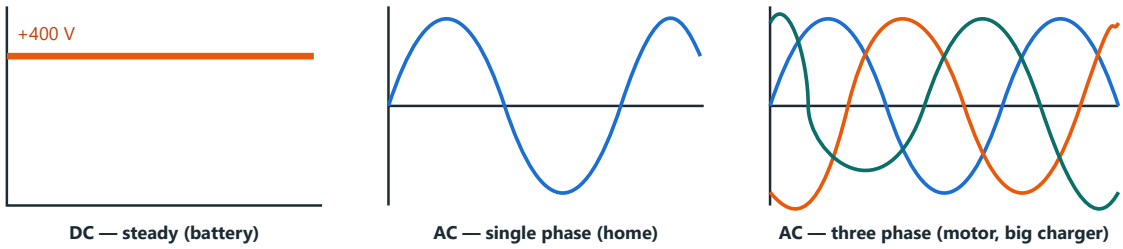


Figure 2.2 DC is steady. AC rises and falls. Three-phase AC is three waves, each one-third of a cycle apart.

KEY IDEA

The battery is **DC**. The mains is **AC**. The drive motor runs on **three-phase AC**. So every EV needs boxes that convert between them: a **charger** (AC → DC) and an **inverter** (DC → AC). You will meet them in Module 10.

2.6 Series and parallel

One battery cell gives only about 3 to 4 volts. To build a 400 V pack, cells are joined together. There are two ways.

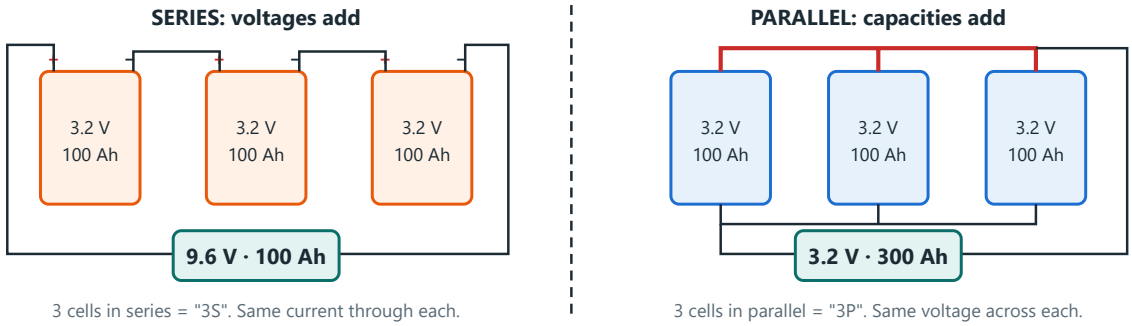


Figure 2.3 Series raises voltage. Parallel raises capacity. Both give the same total energy (here 960 Wh).

WORKED EXAMPLE

A pack has 120 cells in series. Each cell is 3.2 V and 150 Ah.

Pack voltage = 120 × 3.2 = **384 V**
Pack energy = 384 V × 150 Ah = 57,600 Wh = **57.6 kWh**

2.7 Useful units

Quantity	Symbol	Unit	EV example
Voltage	V	volt (V)	Pack voltage 400 V
Current	I	amp (A)	Fast charging at 200 A
Resistance	R	ohm (Ω), kΩ, MΩ	Insulation resistance 5 MΩ

Quantity	Symbol	Unit	EV example
Power	P	watt (W), kW	Motor power 150 kW
Energy	E	Wh, kWh	Battery 60 kWh
Capacity	C	amp-hour (Ah)	Cell 150 Ah
Frequency	f	hertz (Hz)	Mains 50 Hz

Prefixes: **m** (milli) = ÷1,000 · **k** (kilo) = ×1,000 · **M** (mega) = ×1,000,000.

2.8 Colour codes: what the colours mean

Colour is the fastest safety information you have. These conventions are used on real vehicles and tools, and the diagrams in this book follow them wherever a convention exists.

Table 2.1 — Colour conventions you must know

Colour	Where	Meaning
Orange	Cables, connectors and covers on every EV and hybrid	**High voltage.** The one colour rule that is the same on every make in the world.
Red	12 V battery positive (+) terminal and cable; meter positive lead	Positive
Black	12 V battery negative (−), earth straps; meter COM lead	Negative / earth (the car body)
Yellow	Airbag (SRS) wiring and connectors	Do not probe or unplug with the 12 V battery connected
Red	A/C gauge and hose	High-pressure side
Blue	A/C gauge and hose	Low-pressure side
Brown	Mains cable (modern international code)	Live. Older local wiring often uses red, yellow and blue for the three phases.
Blue	Mains cable (modern international code)	Neutral. Older local wiring often uses black.
Green / yellow	Mains cable; charger earth pin (PE)	Protective earth
Yellow triangle	Labels on packs, inverters, compressors	Warning: electrical hazard (black lightning flash)
Green · **Amber** · **Red**	Dashboard lights	Information · caution · stop

PICTURE IT — A TRAFFIC LIGHT FOR YOUR HANDS

Orange = stop and think. Treat every orange cable like a red traffic light: you do not pass until you have isolated, waited and proved dead. **Red = positive, black = negative** on every battery and every meter: "red to plus".

WORKSHOP TIP

Inside a 12 V wiring loom, wire colours are **different on every make**. Never guess a wire's job from its colour; use the wiring diagram, which lists the colour code and pin number of every wire. The three motor phase cables (U, V, W) are all orange, so they are identified by labels, not colours.

In this book's diagrams: orange lines are high voltage, red is 12 V positive or a meter's positive lead, black is negative, thin blue lines are low-voltage signal and control wires, and grey is mechanical drive.

SUMMARY

- Voltage pushes, current flows, resistance restricts. `V = I × R`.
- Power `P = V × I` in kW. Energy = power × time in kWh.
- kW = how strong. kWh = how far.
- High voltage lets EVs carry big power with small current.
- Batteries are DC. Mains and motors are AC.
- Series adds voltage. Parallel adds capacity.
- Orange = high voltage. Red = positive. Black = negative.

Quiz 2

Choose one answer for each question.

1. A heater has a resistance of 23 Ω and is connected to 230 V. What current flows?

- **A** 0.1 A
- **B** 10 A
- **C** 253 A
- **D** 5,290 A

2. A battery supplies 300 A at 350 V. What is the power?

- **A** 650 W
- **B** 10.5 kW
- **C** 105 kW
- **D** 1,050 kW

3. Which unit tells you how much energy a battery stores?

- **A** kW
- **B** kWh
- **C** Ω
- **D** Hz

4. Four 3.7 V, 50 Ah cells are connected in series. What do you get?

- **A** 3.7 V, 200 Ah
- **B** 14.8 V, 50 Ah
- **C** 14.8 V, 200 Ah
- **D** 3.7 V, 50 Ah

5. Why do EVs use 400 V or more instead of 12 V?

- **A** High voltage is safer
- **B** Batteries cannot be made at 12 V
- **C** The same power needs much less current, so cables are thinner and losses lower
- **D** Motors only work above 300 V

6. What type of electricity does a battery store?

- **A** DC
- **B** Single-phase AC
- **C** Three-phase AC
- **D** All three

PART A · FOUNDATIONS · MOST IMPORTANT MODULE

3 High-voltage safety

The rules that keep you alive. Learn this module until you can explain it to someone else.

YOU WILL LEARN

- Why EV voltage is deadly and how electricity injures the body
- How the car protects you, and where that protection ends
- The personal protective equipment (PPE) you must use
- The safe isolation procedure, step by step
- What to do in an emergency

3.1 How much voltage is dangerous?

International vehicle standards define **high voltage** as anything above **60 V DC** or **30 V AC**. Below this, dry skin normally protects you. Above this, current can pass through your body.

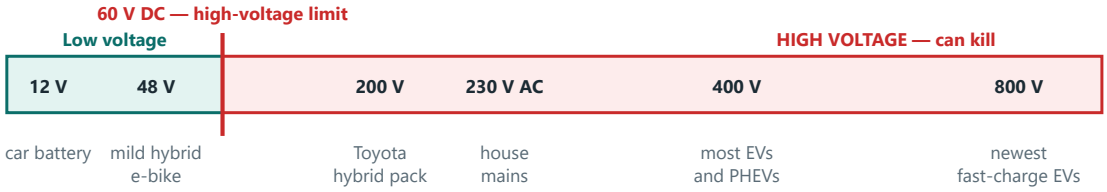


Figure 3.1 Where common voltages sit. Everything to the right of the red line needs high-voltage precautions.

3.2 What electricity does to the body

It is the **current** through your body that injures you. The voltage decides how much current flows. Your body's resistance is roughly 1,000 Ω when skin is damp or broken.

WORKED EXAMPLE

You touch 400 V across your body with sweaty hands (R ≈ 1,000 Ω).

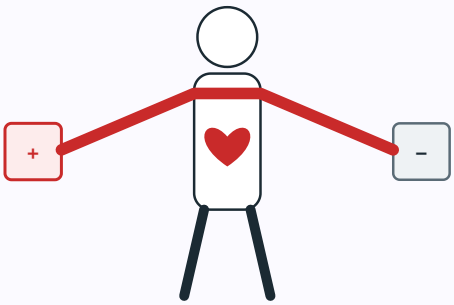
I = V ÷ R = 400 ÷ 1,000 = 0.4 A = **400 mA**. The table below shows this is many times a deadly current.

Table 3.1 — Effect of current through the body

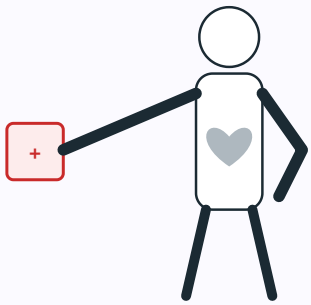
Current	Effect
1 mA	Slight tingle
10–20 mA	Muscles lock. You cannot let go.
50–100 mA	Heart rhythm is disturbed (fibrillation). Death is likely without fast help.
Over 200 mA	Severe burns, heart stops, damage to internal organs

There are three ways high voltage hurts you:

1. **Electric shock** — current through the body, especially hand to hand across the chest.
2. **Arc flash** — if a metal tool bridges two battery terminals, thousands of amps flow. The metal explodes into a flash hotter than a welding arc. It burns skin and eyes in a fraction of a second.
3. **Fire and chemical** — a damaged lithium cell can overheat, vent toxic gas and burn (thermal runaway, Module 8).



Hand to hand: the current crosses the heart



One hand only: no path across the chest

Picture it The deadliest path is hand to hand. Working with one hand where possible lowers the risk; it does not remove it. Gloves and "prove dead" still apply, every time.

IN PAKISTAN

Summer heat means sweat. Sweaty skin has low resistance, so the same voltage pushes more current through you. Monsoon rain and flooding add water to the mix. Our conditions make high-voltage discipline **more** important, not less.

3.3 How the car protects you

Manufacturers build several layers of protection into every EV and hybrid.

Protection	What it does
Orange cables	All high-voltage cables and connectors are coloured orange. Orange means: do not touch unless the system is proven dead.
Isolated system	The high-voltage circuit is not connected to the car body. Both + and − are insulated from the chassis ("floating"). The 12 V system, in contrast, uses the body as its negative.
Contactors	Large relays inside the battery pack. When the car is off, they open and disconnect the battery from everything outside the pack.
HVIL (high-voltage interlock loop)	A small low-voltage wire loop running through every HV connector and cover. If a connector is unplugged, the loop breaks and the contactors open.
Insulation monitoring	The car constantly measures the resistance between the HV system and the body. If it falls too low, it warns the driver or refuses to start.
Service disconnect	A plug or switch that a technician removes to break the HV circuit, often in the middle of the pack.
Crash cut-off	When airbags fire, the contactors are opened at once, sometimes by a small explosive fuse (pyro-fuse).
Discharge circuit	Capacitors in the inverter store charge. After switch-off, the car drains them, usually within seconds to a few minutes.

DANGER — WHERE THE PROTECTION ENDS

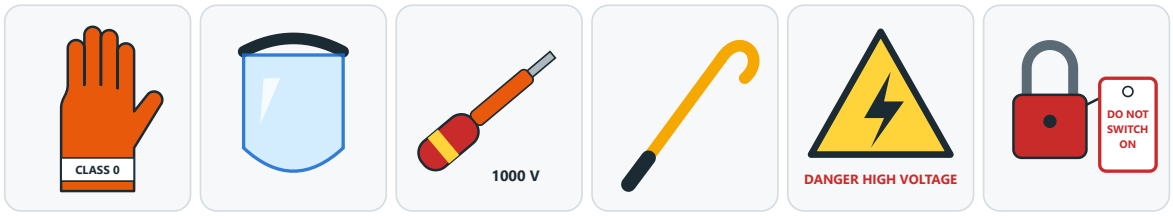
- **Inside the battery pack is always live.** Contactors and the service plug only disconnect the pack from the rest of the car. The cells themselves cannot be switched off.
- Protection systems can fail, especially after a **crash, flood or fire**, or after a bad repair.
- Capacitors can hold charge if the discharge circuit has failed.
- So we never trust the car. **We always measure.**

3.4 Personal protective equipment (PPE)

Table 3.2 — Minimum PPE for high-voltage work

Item	Requirement	Check before every use

Item	Requirement	Check before every use
Insulating gloves	Rubber, **Class 0** (rated 1,000 V AC / 1,500 V DC), marked with standard IEC 60903 or ASTM D120	Roll the cuff to trap air and squeeze: any leak means throw them away. Check date stamp. No cuts, cracks or oil.
Leather over-gloves	Worn over the rubber gloves to protect them from cuts	Clean, no metal parts
Eye and face protection	Safety glasses always; face shield when a live arc is possible	No cracks
Safety shoes	Insulating soles, dry	Soles not worn through
Clothing	Cotton, long sleeves. No synthetic clothes that melt onto skin.	Dry
Insulated tools	Marked 1,000 V (VDE)	Insulation not damaged
Test meter	Rated **CAT III 1,000 V** with matching leads	Leads undamaged; prove it on a known voltage



Insulating glove class and voltage on cuff | **Face shield** against arc flash | **Insulated tool** marked 1000 V | **Rescue hook** non-conducting pole | **Warning sign** around the work bay | **Lock and tag** only you hold the key

Know your tools High-voltage safety equipment. The leather over-glove (not shown) is worn on top of the rubber glove.

Remove all metal from your body: rings, watch, bracelet, chain, keys in pockets, metal belt buckle near the work.

> **WORKSHOP TIP**
>
> Ordinary rubber or mechanic's gloves are **not** insulating gloves. Real Class 0 gloves have the class and voltage printed on the cuff. They cost more, they are worth more, and they must be re-tested or replaced regularly (typically every 6 months).

3.5 The workshop area

- A clean, dry bay. No water on the floor.
- Barrier or cones around the car, and a sign: **"DANGER — HIGH VOLTAGE — DO NOT TOUCH"**.
- A **second person** nearby who knows what you are doing and how to help.
- An insulated **rescue hook** (a non-conducting pole) to pull a person away from a live part.
- Fire extinguisher (CO_2 or dry powder for small electrical fires) and access to plenty of water.
- Insulating rubber mat to stand on.

3.6 The safe isolation procedure

Every manufacturer has its own exact procedure. You **must** follow the service manual for the car in front of you. But all procedures follow the same logic: **switch off → disconnect → secure → wait → prove dead**.

1. **Prepare the car.** Park on a level, dry surface. Select P, apply parking brake, chock the wheels. Unplug any charging cable.
2. **Switch off and remove the key.** Take the smart key at least 5 metres away so the car cannot wake up. Check the "READY" light is off.
3. **Put on PPE and set up the area.** Gloves checked, glasses on, signs and barrier in place.

4. **Disconnect the 12 V battery negative terminal.** Without 12 V, the contactors cannot close.
5. **Remove the service disconnect** (wearing gloves). Keep it in your pocket or a locked box so nobody can refit it. This is "lock out, tag out".
6. **Wait.** The manual gives a time, usually 5 to 10 minutes, for capacitors to discharge.
7. **Prove dead — the three-step test.** (a) Test your meter on a known live source, such as the 12 V battery. (b) Measure at the HV point you will work on: + to −, + to body, − to body. All must read near 0 V. (c) Test your meter on the known source again to prove it still works.
8. **Only now begin work.** If you leave the car and return, prove dead again.

> **KEY IDEA — "PROVE, TEST, PROVE"**
>
> A meter with a broken lead or a flat battery reads 0 V on a live circuit. That is why you check the meter **before and after** the dead test. Many electricians have died trusting a faulty meter.

3.7 Special danger situations

- **Accident-damaged cars.** Insulation may be cut and the body may be live. Treat every orange cable and the pack case as live. Look for leaking fluid, smoke, a sweet chemical smell, or hissing sounds.
- **Flood-damaged cars.** Water inside a pack causes corrosion and short circuits that can start a fire days or weeks later. Store such cars outdoors, away from buildings and other cars.
- **Charging.** Never work on a car while it is plugged in.
- **Towing.** Many EVs must be moved on a flatbed truck. Turning the driven wheels spins the motor, which generates voltage.
- **Hybrids.** The engine can start by itself when the car is in READY mode. Always switch fully off.

3.8 Emergencies

If someone is receiving a shock

1. **Do not touch them.** You will become the second victim.
2. Switch off the source if you can do it instantly.
3. Pull them clear with the insulated rescue hook, or dry wood or thick dry cloth — never metal, never wet.
4. Call for help: **Rescue 1122**.
5. If they are not breathing, start CPR and continue until help arrives. Every EV workshop should have at least one person trained in first aid.
6. Anyone who has had an electric shock must see a doctor, even if they feel fine. Heart problems can appear hours later.

If a battery is smoking or burning

1. Get everyone out and away. The smoke is toxic.
2. Call the fire service (Rescue 1122 / 16) and say clearly: **"electric vehicle battery fire"**.
3. If safe, move other cars away. Do not try to be a hero.
4. A lithium battery fire needs a very large amount of water for a long time to cool it. It can re-ignite hours later.

SUMMARY

- Above 60 V DC is high voltage. EVs use 200–800 V DC. This kills.
- Orange = high voltage. Never touch unless proven dead.
- The car has many protections, but the inside of the battery is always live and protections can fail.
- Class 0 gloves, eye protection, insulated tools, CAT III 1,000 V meter, no metal jewellery.
- Switch off → disconnect 12 V → remove service plug → wait → prove dead (prove, test, prove).
- Never work alone. Never touch a shock victim with bare hands.

Quiz 3 — target 100%

Choose one answer for each question.

1. Above what DC voltage is a vehicle system classed as high voltage?

- **A** 12 V
- **B** 30 V
- **C** 60 V
- **D** 230 V

2. What colour are high-voltage cables?

- **A** Red
- **B** Yellow
- **C** Blue
- **D** Orange

3. After removing the service disconnect, which part is still live?

- **A** The inverter
- **B** The motor
- **C** The inside of the battery pack
- **D** Nothing is live

4. Which gloves are correct for high-voltage work?

- **A** Thick leather welding gloves
- **B** Nitrile mechanic's gloves
- **C** Class 0 rubber insulating gloves with leather over-gloves
- **D** Any dry gloves

5. What is the correct order?

- **A** Remove service plug, then switch off
- **B** Switch off, disconnect 12 V, remove service plug, wait, prove dead
- **C** Prove dead, then switch off
- **D** Disconnect orange cables, then wait

6. Why do you test the meter on a known live source before AND after measuring?

- **A** To charge the meter
- **B** To save time
- **C** To be sure a 0 V reading is real and not a faulty meter
- **D** It is not necessary

7. A colleague is stuck to a live part. What do you do first?

- **A** Grab and pull them away
- **B** Pour water on them
- **C** Do not touch them; cut the power or use an insulated hook
- **D** Run to find the owner

8. **What does the HVIL do?**

- **A** Charges the 12 V battery
- **B** Opens the contactors if a high-voltage connector or cover is opened
- **C** Cools the battery
- **D** Measures the battery's state of charge

PART B · THE VEHICLES

4 Types of electrified vehicle

MHEV, HEV, PHEV, REEV, BEV — what the letters mean and how each one is built.

YOU WILL LEARN

- The five main types of electrified vehicle
- The difference between series, parallel and series-parallel hybrids
- How to recognise each type from its specification

4.1 The electrification ladder

"Electrified" means the car uses an electric motor for at least part of its driving. The more electric it is, the bigger its battery and the higher its voltage.

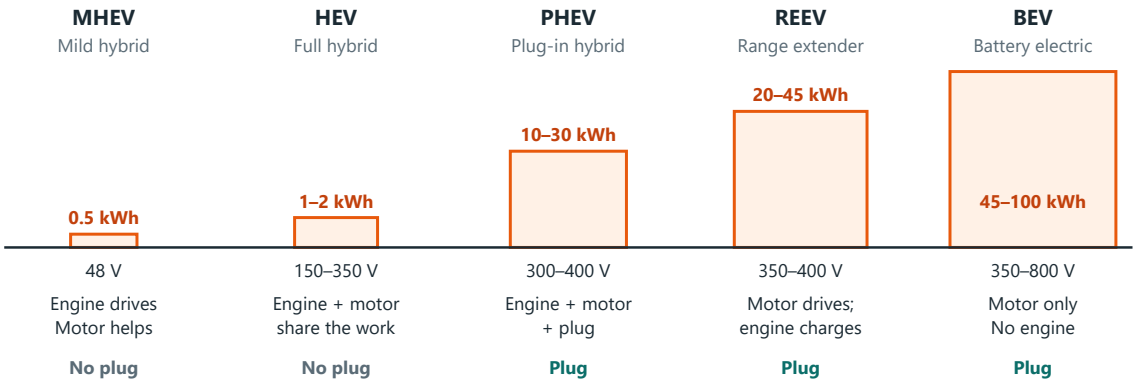


Figure 4.1 The electrification ladder. Orange bars show typical battery size (not to scale).

Table 4.1 — The five types

Type	How it works	Electric-only driving	Examples in Pakistan
MHEV Mild hybrid	A small 48 V motor-generator helps the engine and restarts it smoothly. Cannot drive the car alone.	None	Some European and Korean imports
HEV Full ("self-charging") hybrid	Engine plus one or two motors and a small HV battery. Charged only by the engine and by braking.	1–2 km at low speed	Toyota Corolla Cross, Aqua, Prius; Honda HR-V e:HEV, Vezel; Haval H6 HEV; Hyundai Tucson; Kia Sportage
PHEV Plug-in hybrid	Like an HEV but with a much bigger battery that can be charged from a socket. Runs as an EV for daily trips, as a hybrid for long ones.	50–120 km	BYD Sealion 6, Shark 6; Haval H6 PHEV; Jaecoo J7; Chery Tiggo 7/8/9; MG HS
REEV / EREV Range-extender EV	The wheels are always driven by the electric motor. A small engine only turns a generator to make electricity.	100–200 km	Deepal S05 REEV
BEV Battery electric	Battery and motor only. No engine, no fuel tank, no exhaust.	All (250–600 km)	BYD Atto 2/3, Seal; Deepal S07/L07; MG ZS EV, MG4

You may also hear **FCEV** (fuel-cell electric vehicle). It makes electricity on board from hydrogen gas. There are none in normal use in Pakistan, so this course does not cover them.

IN PAKISTAN

PHEVs suit our conditions well. Most daily city driving is under 50 km, which a PHEV can do on cheap electricity. For Lahore–Islamabad or Karachi–Hyderabad, where chargers are still rare, the petrol engine removes the worry. This is why Chinese PHEV SUVs have sold strongly. Expect to see a lot of them in your workshop.

4.2 Three ways to arrange a hybrid

Every hybrid is one of three basic layouts, or a mixture that can switch between them.

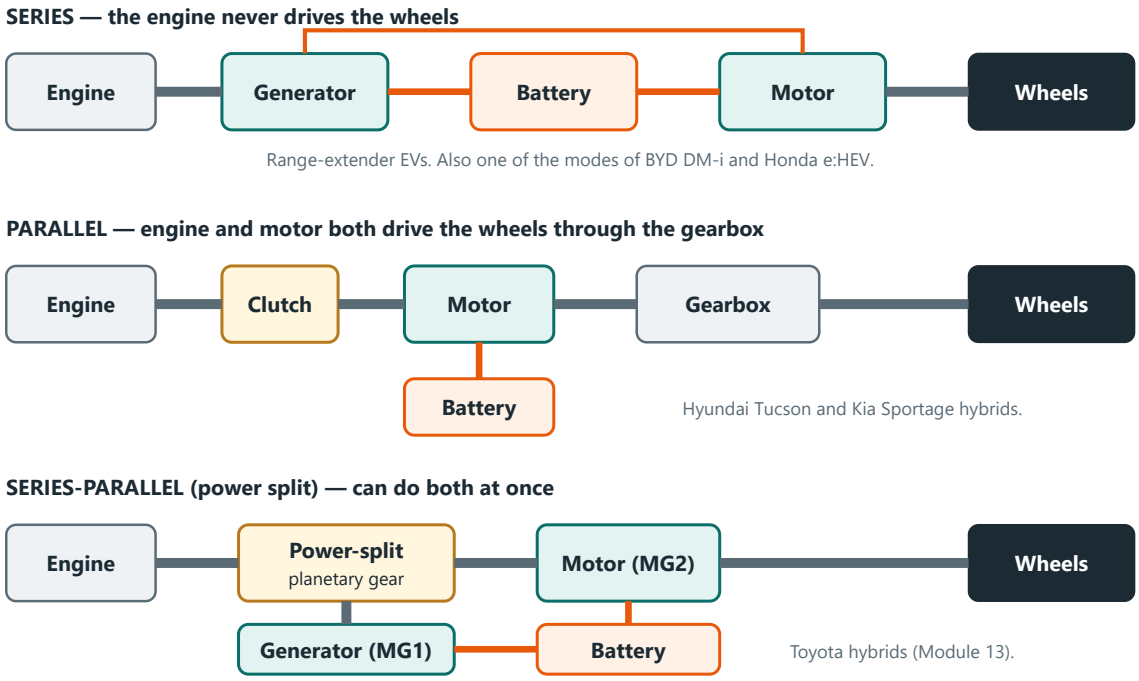


Figure 4.2 Hybrid layouts. Grey = mechanical drive. Orange = high-voltage electrical power.

4.3 Reading a specification sheet

When a customer or a parts supplier describes a car, these clues tell you its type:

- **Battery in kWh.** About 1–2 kWh → HEV. 10–30 kWh → PHEV. Over 40 kWh → BEV.
- **A charging port** → PHEV, REEV or BEV. HEVs and MHEVs have none.
- **An exhaust pipe** → it has an engine, so not a BEV.
- **"EV range" and "combined range"** quoted together → PHEV or REEV.

4.4 Buses, coaches and trucks

Electric commercial vehicles use the same principles as cars, on a larger scale. Chinese training courses often include a day on them, because China runs the world's largest fleets of electric buses.

Table 4.2 — Passenger car and electric bus compared

	Passenger EV	Electric bus or coach
System voltage	300–450 V (some 800 V)	Typically 500–750 V
Battery energy	40–100 kWh	200–450 kWh, in several packs (roof, floor or rear)
Chemistry	LFP or NMC	Almost always LFP
Drive	One or two motors, 100–400 kW	Large central motor or motors in the axle

	Passenger EV	Electric bus or coach
Extra high-voltage consumers	A/C compressor, heater	Also an electric **air compressor** (brakes, doors, suspension) and electric **power-steering pump**, fed from a "multi-in-one" controller
Low-voltage system	12 V	24 V
Charging	AC and DC	Mostly high-power DC, sometimes two guns at once

Everything in Module 3 applies with more force: higher voltage, far more stored energy, components mounted on the roof (working at height), and a much heavier vehicle.

SUMMARY

- MHEV → HEV → PHEV → REEV → BEV: battery and voltage grow at each step.
- HEVs do not plug in. PHEVs, REEVs and BEVs do.
- Series: engine makes electricity only. Parallel: engine and motor both drive. Series-parallel: both.
- Battery size in kWh is the quickest clue to the type of car.

Quiz 4

Choose one answer for each question.

1. **Which type of vehicle has no engine at all?**

A HEV
B PHEV
C REEV
D BEV

2. **A car has a 1.5-litre engine, an 18 kWh battery and a charging port. What is it?**

A MHEV
B HEV
C PHEV
D BEV

3. **In a series hybrid, what does the engine do?**

A Drives the wheels directly
B Turns a generator to make electricity
C Charges the 12 V battery only
D Cools the battery

4. **Which of these cannot be charged from a wall socket?**

A Toyota Corolla Cross HEV
B BYD Sealion 6
C Deepal S07
D Haval H6 PHEV

5. **What voltage does a mild hybrid (MHEV) system usually use?**

A 12 V
B 48 V
C 400 V
D 800 V

PART B · THE VEHICLES

5 Main components: a map of the car

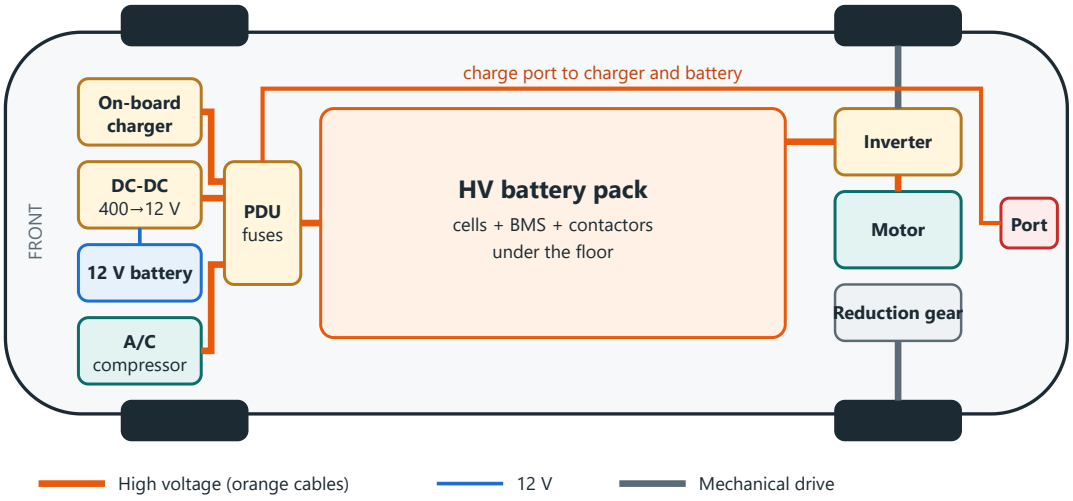
Every major part of an electric car, what it does, and how energy flows between them.

YOU WILL LEARN

- The names and jobs of the main high-voltage components
- Where they are usually found in the car
- How energy flows when charging, driving and braking

5.1 The big picture

Most modern EVs use a "**skateboard**" layout. The battery is a flat slab under the floor between the axles. The motor and its electronics sit at one or both axles. This keeps the weight low and leaves the cabin free.



This example is rear-wheel drive. Front-wheel-drive EVs have the motor at the front. AWD cars have one at each axle.

Figure 5.1 Top view of a typical battery electric car. Positions vary by model, but the same parts are always there.

5.2 What each component does

Table 5.1 — Main components of an electric car

Component	Job	Module
HV battery pack	Stores the energy (DC, 300–800 V). Contains the cells, the BMS, contactors, fuses and sensors.	6–8
BMS (battery management system)	The battery's brain. Measures every cell, controls contactors, protects the pack.	8
Inverter (motor controller, MCU)	Changes battery DC into three-phase AC to drive the motor, and back again during braking.	10
Traction motor	Turns electrical power into rotation. Also works as a generator when braking.	9
Reduction gear	A single-speed gearbox with differential. Reduces motor speed and multiplies torque.	12
On-board charger (OBC)	Changes AC from the mains into DC to charge the battery.	10, 14
DC-DC converter	Steps HV down to about 14 V to run the 12 V system and charge the 12 V battery. It replaces the alternator.	10, 16
PDU (power distribution unit)	The high-voltage junction box with fuses and relays for each HV consumer.	10
Charge port	Where the charging cable plugs in. AC and DC pins.	14

Component	Job	Module
Electric A/C compressor	Runs on high voltage, so the air-conditioning works with no engine. Also cools the battery on many cars.	15
PTC heater / heat pump	Heats the cabin and sometimes the battery.	15
12 V battery	Powers computers, lights, locks — and wakes the car up. Without it, the HV system cannot start.	16
VCU (vehicle control unit)	The master computer. Reads the pedals and decides how much torque to ask from the motor.	16

WORKSHOP TIP — "X-IN-1" UNITS

To save space and cost, makers combine parts into one housing. A "3-in-1" drive unit is motor + inverter + reduction gear. BYD's "8-in-1" adds the charger, DC-DC, PDU, VCU and BMS controller. For a technician this means fewer cables to check — but if one part fails, the whole expensive unit may need to come out.

KEY IDEA — THE "THREE-ELECTRIC" SYSTEM

Chinese trainers, manuals and parts suppliers group the main parts under one name: the **three-electric system** (三电, *sān diàn*).

- **The "big three":** battery (电池), motor (电机) and electronic control (电控: the inverter and vehicle controller).
- **The "small three":** on-board charger (OBC), DC-DC converter and power distribution unit (PDU).

You will hear this phrase on the first day of any course in China. It simply means the parts in Table 5.1.

5.3 How energy flows

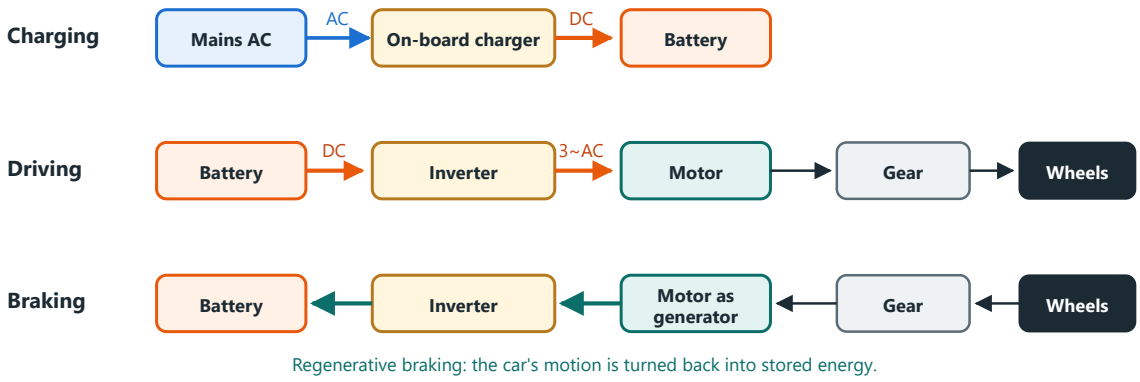


Figure 5.2 The three energy paths. Notice that the inverter and motor work in both directions.

5.4 How a hybrid differs

A hybrid has all of the above (with a smaller battery in an HEV), **plus** a complete petrol engine with its fuel, exhaust, cooling and lubrication systems, and a special transmission that combines engine and motor. A hybrid is therefore **more** complicated than either a petrol car or a pure EV. Module 13 explains hybrids in detail.

SUMMARY

- Battery under the floor, drive unit at the axle: the "skateboard".
- Battery (DC) → inverter → motor (3-phase AC) → reduction gear → wheels.
- Mains AC → on-board charger → battery.
- The DC-DC converter replaces the alternator and keeps the 12 V battery charged.
- Braking sends energy back to the battery through the same motor and inverter.

Quiz 5

Choose one answer for each question.

1. Which component changes battery DC into three-phase AC for the motor?

- **A** On-board charger
- **B** DC-DC converter
- **C** Inverter
- **D** PDU

2. What replaces the alternator in an electric car?

- **A** The inverter
- **B** The DC-DC converter
- **C** The on-board charger
- **D** The BMS

3. What is the job of the on-board charger?

- **A** Convert AC from the mains to DC for the battery
- **B** Convert DC to AC for the motor
- **C** Cool the battery
- **D** Charge the 12 V battery from the HV battery

4. Where is the battery pack in most modern EVs?

- **A** Under the bonnet
- **B** In the boot
- **C** Under the floor between the axles
- **D** On the roof

5. During regenerative braking, the motor works as a…

- **A** heater
- **B** charger for the mains
- **C** generator
- **D** brake pad

6. What is a "3-in-1" drive unit?

- **A** Three motors in one housing
- **B** Motor, inverter and reduction gear in one housing
- **C** Three batteries joined together
- **D** A charger with three plugs

PART C · THE BATTERY

6 Battery cells and chemistry

The battery is the most expensive part of the car. It starts with a single cell.

YOU WILL LEARN

- How a lithium-ion cell stores and releases energy
- The main chemistries: LFP, NMC, NiMH and lead-acid
- The shapes cells come in, including BYD's Blade
- Why chemistry matters for safety, life and range

6.1 Inside a lithium-ion cell

A cell has four main parts:

- **Negative electrode (anode)** — graphite coated on copper foil.
- **Positive electrode (cathode)** — a lithium metal compound coated on aluminium foil. The compound gives the chemistry its name (LFP, NMC…).
- **Separator** — a very thin plastic sheet with tiny holes. It keeps the electrodes apart but lets lithium ions pass.
- **Electrolyte** — a liquid that carries the ions. It is flammable.

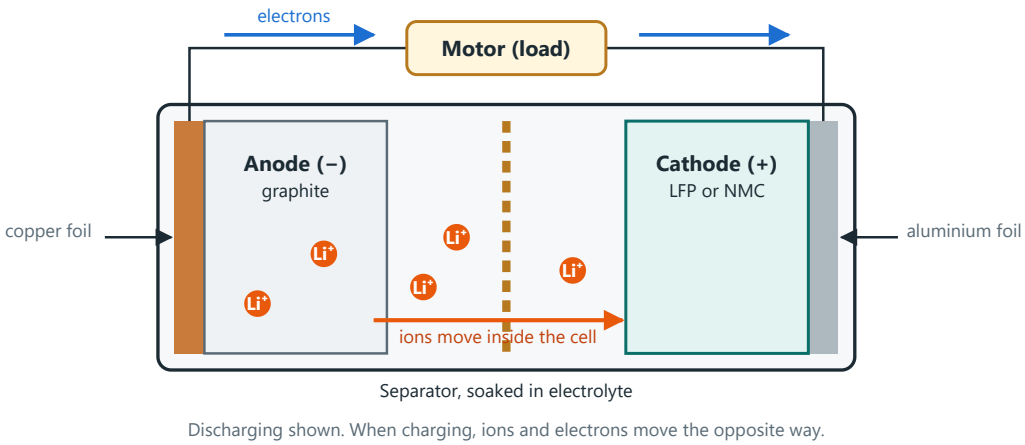


Figure 6.1 A lithium-ion cell while discharging. Lithium ions travel through the separator; electrons travel through the outside circuit and do the work.

When you **charge**, the charger pushes lithium ions from the cathode into the graphite anode. When you **discharge**, they flow back and the electrons flow round the outside circuit, driving the motor. Nothing is burned or used up — ions simply move back and forth. That is why the cell can be recharged thousands of times.

DANGER

If the separator is damaged — by a crash, a nail, a manufacturing fault or overheating — the two electrodes touch. This is an **internal short circuit**. The cell heats very quickly and can catch fire. Never drop, crush, puncture or open a lithium cell.

6.2 The main chemistries

Table 6.1 — Battery chemistries you will meet

	LFP	NMC ("ternary")	NiMH	Lead-acid
Full name	Lithium iron phosphate	Lithium nickel manganese cobalt oxide	Nickel-metal hydride	Lead-acid (incl. "graphene" type)

	LFP	NMC ("ternary")	NiMH	Lead-acid
Nominal cell voltage	3.2 V	3.6–3.7 V	1.2 V	2.0 V
Working range per cell	2.5–3.65 V	3.0–4.2 V	1.0–1.5 V	1.75–2.4 V
Energy per kg	Medium (150–190 Wh/kg)	High (220–280 Wh/kg)	Low (60–80)	Very low (30–40)
Cycle life	Very long (3,000+)	Good (1,000–2,000)	Long in hybrid use	Short (300–500)
Fire safety	Best of the lithium types	Needs more protection	Good	Good (acid and gas hazards)
Cost	Lower	Higher	Medium	Lowest
Where used	All BYD cars (Blade), most affordable Chinese EVs and PHEVs	Deepal S07/L07, long-range and performance EVs, Korean EVs	Toyota Aqua, Prius, older hybrids	12 V batteries, cheap e-bikes and rickshaws

What this means in practice

- **LFP** is tough, safe and long-lived, and it tolerates being charged to 100% often. It stores less energy per kilogram, so the pack is heavier for the same range. Its voltage hardly changes between 20% and 90% charge, which makes it hard for the car to know the exact charge level; makers ask owners to charge to 100% regularly so the BMS can re-calibrate.
- **NMC** stores more energy in less weight, so it gives longer range and stronger power. It is less tolerant of heat and of sitting at 100% charge. Makers usually recommend charging to 80% for daily use. Chinese documents call it **NCM** or "ternary lithium" (三元锂); it is the same chemistry.
- **NiMH** is old but reliable. It is heavy and holds little energy, which is fine for a hybrid that only needs 1–2 kWh.
- **Lead-acid** is cheap and heavy and wears out quickly when deeply discharged.

> **IN PAKISTAN**
>
> Our summers are hard on batteries. LFP copes with heat better than NMC, which is one reason it suits this market. But **no** lithium battery likes being parked in direct sun at 45 °C. Advising customers to park in shade is real, useful battery care.

Coming next: *sodium-ion* cells (cheaper, no lithium, lower energy) are starting to appear in small Chinese EVs and scooters. *Solid-state* cells, with no liquid electrolyte, are in development. The principles in this course apply to both.

6.3 Cell shapes

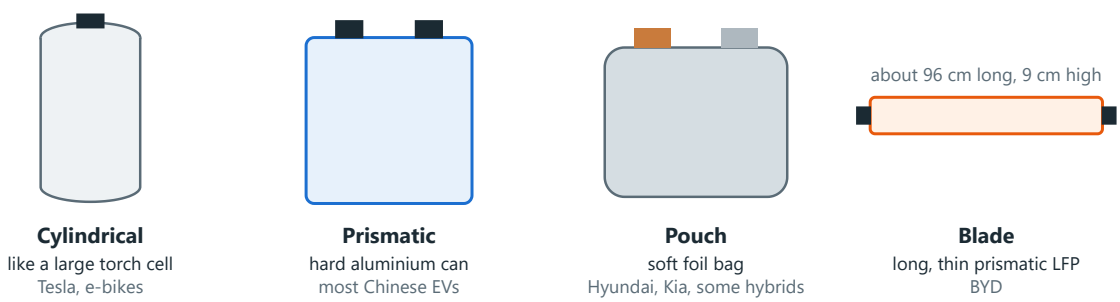


Figure 6.2 Cell formats. The chemistry inside can be the same; only the packaging differs.

Format	Strong points	Weak points
Cylindrical	Cheap to make, strong case, easy to cool	Thousands of cells and welds in one pack
Prismatic	Large cells, fewer connections, easy to stack	Can swell with age

Format	Strong points	Weak points
Pouch	Light, flexible shape	Easily damaged, swells, needs strong support
Blade	Cells act as structural beams; very good safety record in nail tests	Individual cells are hard to replace

SUMMARY

- A lithium-ion cell has an anode, cathode, separator and electrolyte. Ions move inside; electrons do work outside.
- LFP: 3.2 V, safer, longer life, heavier. NMC: 3.7 V, more energy, needs more care.
- NiMH is used in Toyota hybrids. Lead-acid in 12 V batteries and cheap e-bikes.
- Cells are cylindrical, prismatic, pouch or blade.
- A damaged separator means an internal short and possible fire.

Quiz 6

Choose one answer for each question.

1. What is the nominal voltage of one LFP cell?

- **A** 1.2 V
- **B** 2.0 V
- **C** 3.2 V
- **D** 12 V

2. What is the job of the separator?

- **A** Stores the lithium
- **B** Keeps the electrodes apart while letting ions pass
- **C** Carries electrons to the motor
- **D** Cools the cell

3. Which chemistry does BYD's Blade battery use?

- **A** NMC
- **B** NiMH
- **C** Lead-acid
- **D** LFP

4. Compared with LFP, an NMC cell generally has…

- **A** more energy per kg but needs more care with heat and full charge
- **B** lower voltage and longer life
- **C** no fire risk
- **D** a lower price

5. Which battery type is found in a Toyota Aqua hybrid?

- **A** LFP
- **B** Lead-acid
- **C** NiMH
- **D** Sodium-ion

6. Why do makers ask owners of LFP cars to charge to 100% regularly?

- **A** LFP is damaged below 100%
- **B** Its flat voltage curve makes charge level hard to measure, so the BMS needs to re-calibrate
- **C** To heat the battery
- **D** To make the car faster

PART C · THE BATTERY

7 Battery pack design and specifications

How hundreds of cells become one pack, and how to read the numbers that describe it.

YOU WILL LEARN

- How cells are grouped into modules and packs
- How to calculate pack voltage and energy
- The meaning of SOC, SOH, C-rate and usable capacity
- How packs are cooled, and the difference between 400 V and 800 V systems

7.1 Cell → module → pack

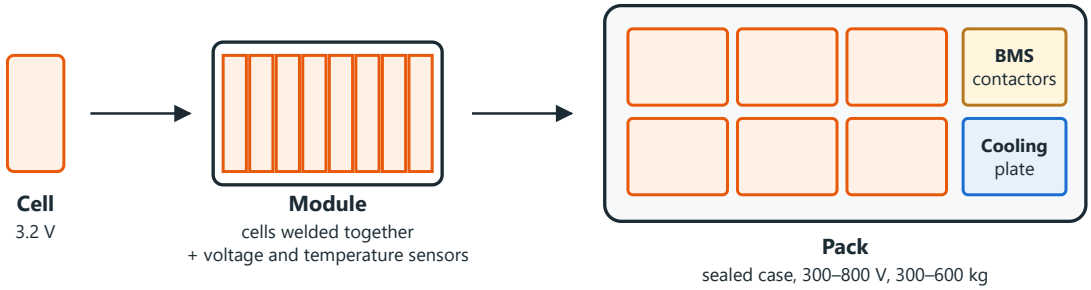


Figure 7.1 The traditional build: cell, module, pack.

Newer designs remove steps to save weight and space:

- **Cell-to-pack (CTP)** — no modules. Cells go straight into the pack. BYD Blade and many CATL packs do this.
- **Cell-to-body (CTB)** — the top of the pack is the floor of the car (BYD Seal).

WORKSHOP TIP

Older module-type packs can be repaired by swapping one bad module. CTP and CTB packs are much harder to repair at cell level; often the whole pack is exchanged. Knowing which type you have tells you what repair is realistic before you quote a price.

7.2 Pack voltage and energy

Pack layout is written as **xS yP**: x groups in series, each group having y cells in parallel.

```
Pack voltage = cells in series × cell voltage
Pack capacity (Ah) = cells in parallel × cell Ah
Pack energy (kWh) = pack voltage × pack Ah ÷ 1,000
```

WORKED EXAMPLE — BYD ATTO 3 (EXTENDED RANGE)

Published data: 126 Blade cells in series (126S1P), each 3.2 V and 150 Ah.

Voltage = 126 × 3.2 = **403.2 V**
Energy = 403.2 × 150 ÷ 1,000 = **60.48 kWh** — exactly the figure on the brochure.

WORKED EXAMPLE — A TYPICAL NMC PACK

96S2P using 3.65 V, 78 Ah cells.

Nominal voltage = 96 × 3.65 = 350 V. Capacity = 2 × 78 = 156 Ah. Energy = 350 × 156 ÷ 1,000 = **54.6 kWh**.

Fully charged: 96 × 4.2 = **403 V**. Empty: 96 × 3.0 = **288 V**.

KEY IDEA

Pack voltage is **not fixed**. It rises as the pack charges and falls as it empties. A "400 V" car may measure anything from about 290 V to 420 V. Always think in terms of *minimum, nominal and maximum* voltage.

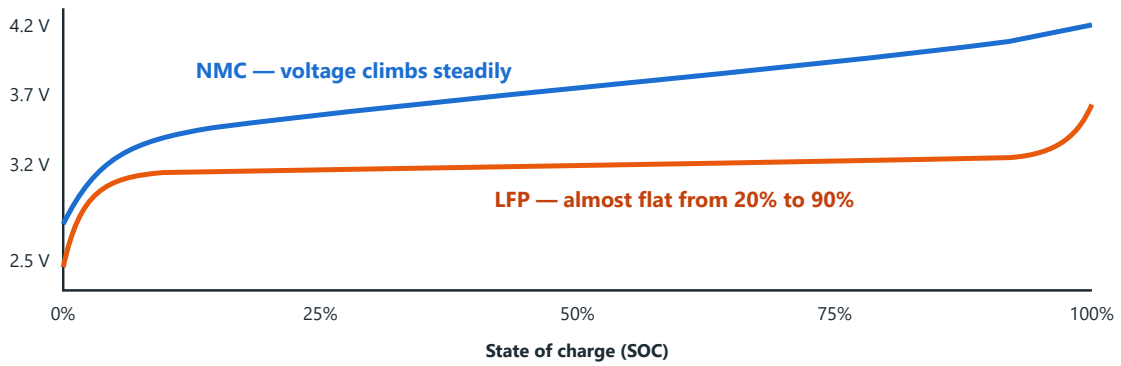


Figure 7.2 Cell voltage against state of charge. The flat LFP curve is why voltage alone cannot tell you the charge level of an LFP pack.

7.3 The words used to describe a battery

Term	Meaning	Example
SOC — state of charge	How full the battery is now, in %	Like the fuel gauge: 80%
SOH — state of health	Present full capacity compared with when new, in %	A 60 kWh pack that now holds 54 kWh has SOH 90%
DoD — depth of discharge	How much has been used: 100% − SOC	SOC 30% = DoD 70%
Gross capacity	Total energy the cells can hold	64 kWh
Usable (net) capacity	The part the car lets you use. The rest is a hidden safety buffer at top and bottom.	60 kWh
C-rate	Charge or discharge current compared with capacity. 1C empties or fills the battery in 1 hour.	60 kWh pack at 120 kW = 2C
Cycle	One full charge and discharge (two half-charges count as one cycle)	LFP: 3,000+ cycles
Internal resistance	Resistance inside the cells. Causes heat and voltage drop under load. Rises with age and cold.	Milliohms per cell

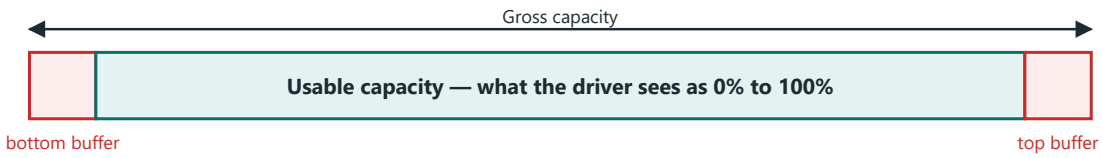


Figure 7.3 The car hides a small part of the battery at each end to protect the cells. "0%" on the dashboard is not truly empty.

WORKED EXAMPLE — C-RATE

A 50 kWh pack is charged at a 100 kW fast charger. C-rate = 100 ÷ 50 = **2C**. In theory 0–100% in half an hour (in practice the car slows down above 80%).

The same pack on a 7 kW home charger: 7 ÷ 50 = 0.14C — very gentle, about 7 hours.

7.4 Battery power: why a small pack limits performance

A battery can only supply a certain maximum current safely. Since $P = V \times I$, the battery has a **maximum power**, and the motor cannot use more than the battery can give.

- A **bigger pack** (more cells in parallel) can give more current, so more power.
- A **cold, hot, old or nearly empty** pack can give less. The BMS tells the car to reduce power — the driver feels this as weak acceleration or sees a "tortoise" warning light.
- Under heavy current, voltage sags: $V_{drop} = I \times R_{internal}$. An aged pack with high internal resistance sags more and heats more.

This is why a hybrid with a 1.5 kWh battery can only drive electrically at low power, while a 60 kWh BEV pack can deliver 150 kW easily. Module 11 returns to this.

7.5 Keeping the pack at the right temperature

Lithium cells are happiest between about **15 °C and 35 °C**. Too hot and they age quickly or become unsafe. Too cold and they give less power and cannot be charged quickly.

Table 7.1 — Battery cooling methods

Method	How it works	Found on	Service points
Air cooling	A fan blows cabin air through the pack	Toyota and Honda hybrids, older or cheaper EVs	Intake grille and filter under/behind the rear seat block with dust and pet hair → battery overheats. Very common fault.
Liquid cooling	Coolant flows through a plate under the cells, then to a radiator or a chiller on the A/C system	Most modern EVs and PHEVs	Coolant level and condition, pump, leaks, special low-conductivity coolant
Refrigerant (direct) cooling	A/C refrigerant flows directly through plates in the pack	BYD and some others	A/C system health directly affects the battery

> **IN PAKISTAN**
>
> With 45 °C summers, battery cooling works very hard. A weak air-conditioning system is not only a comfort problem — on many EVs it will slow charging, cut power and shorten battery life. Dust is our other enemy: blocked battery fan filters on used Aqua and Prius cars are one of the most common causes of early hybrid battery failure here.

7.6 400 V and 800 V systems

Most EVs today are called "400 V" (roughly 300–450 V). Newer high-end platforms use "800 V" (roughly 550–900 V).

- **Advantage of 800 V:** half the current for the same power, so thinner cables, less heat, and much faster DC charging (250 kW and above).
- **Cost:** more cells in series, more expensive power electronics (silicon carbide), and stricter insulation.
- **For you:** the safety procedure is the same, but your meter and gloves must be rated for the voltage. A CAT III 1,000 V meter and Class 0 gloves cover both.

7.7 The pack case

The pack is a sealed metal case, usually rated **IP67** (dust-tight and able to survive short immersion in water). On it you will find the HV connectors, low-voltage signal connector, coolant pipes, a pressure-relief vent, the service disconnect on many models, and a **data label**.

Table 7.2 — Reading a battery pack label (example)

On the label	What it tells you
Type: Li-ion (LFP)	Chemistry

On the label	What it tells you
Rated voltage: 403.2 V	Nominal pack voltage → 126 LFP cells in series
Rated capacity: 150 Ah	Pack amp-hours
Rated energy: 60.48 kWh	403.2 × 150 ÷ 1,000
Mass: 420 kg	You need a proper lifting table, not a trolley jack
Warning symbols	High voltage, do not open, do not burn, recycle

SUMMARY

- Cells → modules → pack, or cells straight to pack (CTP).
- Voltage = series count × cell voltage. Energy = voltage × Ah.
- Pack voltage moves between a minimum and maximum as SOC changes.
- SOC = how full. SOH = how healthy. C-rate = how hard it is charged or discharged.
- The battery limits the maximum power available to the motor.
- Cooling is by air, liquid or refrigerant. Heat and dust are the big enemies in Pakistan.

Quiz 7

Choose one answer for each question.

1. A pack has 100 LFP cells (3.2 V, 200 Ah) in series. What is its energy?

A 32 kWh
B 64 kWh
C 100 kWh
D 640 kWh

2. A battery that held 50 kWh when new now holds 40 kWh. What is its SOH?

A 40%
B 50%
C 80%
D 125%

3. A 40 kWh battery is charged at 80 kW. What is the C-rate?

A 0.5C
B 1C
C 2C
D 40C

4. What does "usable capacity" mean?

A The capacity when the battery is old
B The part of the total capacity the car allows the driver to use
C The capacity at 800 V
D The 12 V battery capacity

5. A used Toyota hybrid has an overheating battery. What simple thing should you check first?

A Engine oil
B Tyre pressure
C Blocked battery cooling fan intake or filter
D Wiper blades

6. What is the main benefit of an 800 V system over a 400 V system?

A It is safer to touch
B It needs no cooling
C Half the current for the same power, allowing faster charging and thinner cables
D It uses fewer cells

PART C · THE BATTERY

8 Battery management and protection

The BMS, contactors and pre-charge circuit: how the pack looks after itself.

YOU WILL LEARN

- What the battery management system (BMS) measures and controls
- How contactors and the pre-charge circuit connect the battery to the car
- What cell balancing is and why one weak cell limits the whole pack
- What thermal runaway is, and what shortens battery life

8.1 What the BMS does

A lithium cell is damaged by over-charging, over-discharging, too much current and extreme temperature. The **BMS** exists to stop these things happening.

Job	How
Measure	Voltage of every cell (or cell group), pack voltage, pack current, and temperatures at many points
Calculate	SOC, SOH, and how much power the pack can safely give or accept right now
Protect	Opens the contactors if voltage, current or temperature go outside safe limits
Balance	Keeps all cells at the same charge level
Control temperature	Requests cooling or heating
Monitor insulation	Checks resistance between the HV system and the car body
Communicate	Talks to the vehicle computer and the charger over CAN bus; stores fault codes

The BMS usually has a **master** controller and several **slave** boards, one on each module or group of cells, which read the cell voltages and temperatures.

8.2 Inside the pack: contactors and pre-charge

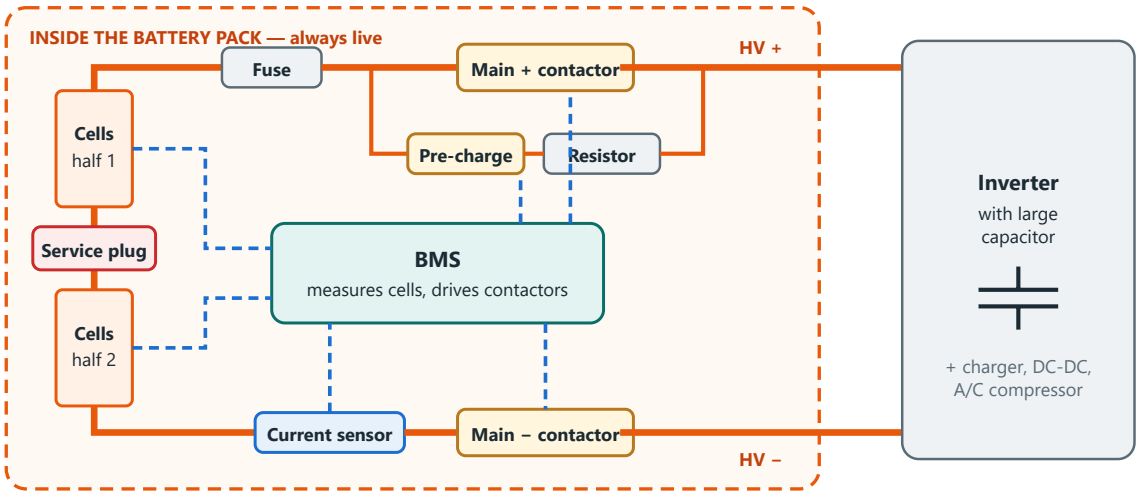


Figure 8.1 Simplified battery pack circuit. Dashed blue lines are low-voltage sensing and control wires.

Contactors are heavy-duty relays. They are opened and closed by 12 V signals from the BMS. When the car is off, all contactors are open and no voltage leaves the pack.

Why pre-charge is needed

The inverter contains a large capacitor. An empty capacitor behaves like a short circuit for an instant. If the main contactors closed straight onto it, a huge current would rush in and weld the contacts together.

> **PICTURE IT — THE CANAL GATE**
>
> Open a big gate suddenly onto a dry canal and the first rush of water tears at the banks. A careful operator opens a **small side gate** first, lets the canal fill gently, and only then lifts the main gate. Nothing rushes, because the levels are already equal. The pre-charge resistor is the small side gate; the inverter capacitor is the dry canal.

So the car connects in stages:

1. **Main negative contactor closes.** Click.
2. **Pre-charge contactor closes.** Current flows gently through the resistor and fills the capacitor. This takes a fraction of a second.
3. **BMS checks the voltage** on the inverter side. When it reaches about 95% of pack voltage, pre-charge is complete.
4. **Main positive contactor closes.** Click. The pre-charge contactor opens. The car shows **READY**.

> **WORKSHOP TIP**
>
> Listen when a car is switched to READY: you can hear the contactors click in sequence. If the voltage does not rise during pre-charge (for example, because of a short circuit or a failed resistor), the BMS stops and stores a **"pre-charge failure"** fault code. If a contactor has welded shut, the BMS will detect voltage where there should be none and refuse to start. These are common, logical faults you will learn to diagnose.

8.3 Cell balancing: the weakest cell rule

Cells in series are like links in a chain. All carry the same current. But no two cells are perfectly identical; over time some hold slightly more charge than others.

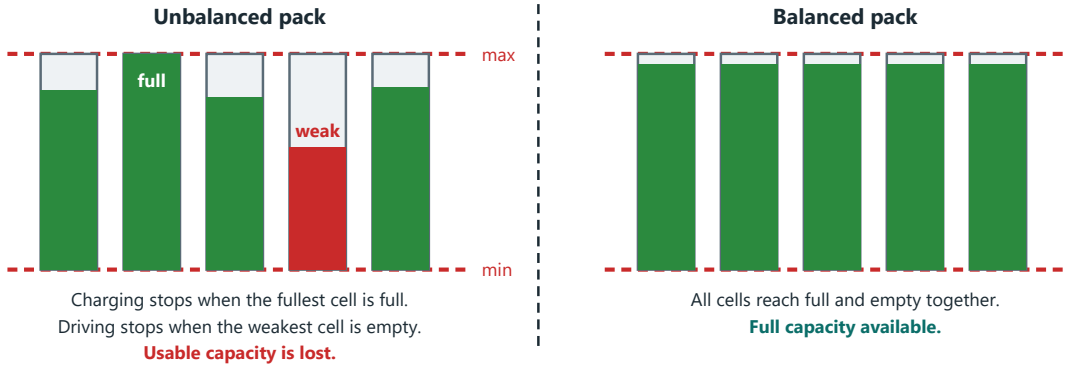


Figure 8.2 A pack is only as good as its weakest cell.

Balancing corrects this. In *passive* balancing (the common type), the BMS switches a small resistor across the highest cells to bleed off a little charge, usually near the end of charging. In *active* balancing, charge is moved from high cells to low cells.

> **KEY IDEA**
>
> On a scan tool, look at the **difference between the highest and lowest cell voltage**. A healthy pack at rest shows a spread of a few millivolts (for example under 0.02–0.03 V). A large spread, or one cell that drops quickly under load, points to a failing cell or module. This single reading is one of the most useful in battery diagnosis.

8.4 Insulation monitoring

Remember that the HV system is insulated from the car body. The BMS (or a separate unit) keeps measuring the resistance between HV+ and the body, and HV− and the body. Healthy values are in **megohms**. Regulations require at least 100 Ω per volt for DC parts and 500 Ω per volt where AC is present — for a 400 V car that is 40 kΩ to 200 kΩ as an absolute minimum.

If the resistance falls — because of water, damaged cable insulation, coolant leaking into the pack, or a failed A/C compressor — the car logs an **"insulation fault"** (also called isolation fault or leakage fault) and may refuse to drive or charge. Finding these is a core EV diagnostic skill (Module 18).

8.5 Thermal runaway

Thermal runaway is when a cell heats itself faster than it can cool. Heat triggers chemical reactions that make more heat. The cell vents hot, toxic, flammable gas and may burn, and the heat can spread to the next cell.

Causes

- Crash or impact damage from below
- Internal short from a manufacturing defect
- Over-charging (failed BMS or wrong charger)
- External short circuit
- Outside fire or extreme heat
- Water inside the pack

Warning signs

- Sweet, solvent-like or burning smell
- Hissing or popping from the pack
- White or grey "smoke" (vent gas)
- Pack case hot to the back of the hand
- Swollen case or modules
- Battery temperature warning on dash

> **DANGER**
>
> If you see any of these signs: stop work, move people away, move the car outside if it is safe to do so, and call the fire service. Do not try to open the pack. LFP is much harder to push into runaway than NMC, but **no chemistry is immune**.

8.6 What makes a battery age

All batteries slowly lose capacity. A well-treated EV battery typically keeps more than 80% after 8–10 years. These things speed up ageing:

Bad for the battery	Advice you can give customers
High temperature	Park in shade or a garage. Keep the cooling system serviced.
Sitting at 100% for long periods (especially NMC)	For daily use charge NMC to 80%. Charge to 100% just before a long trip. Follow the maker's advice for LFP.
Sitting near 0%	Do not leave the car parked empty. Store at 40–60% if leaving it for weeks.
Frequent DC fast charging	Use home AC charging for normal days; fast chargers for journeys.
Poor-quality or wrong chargers	Especially on e-bikes: use the correct charger for the battery.

8.7 Inside a battery repair: what gets measured

Opening a pack is Level 4 work (Module 21) and is taught only under supervision. But you should understand the measurements, because they explain every battery report you will read.

Table 8.1 — Battery repair measurements

Measurement	What it is	What it tells you
OCV — open-circuit voltage	Cell voltage at rest, with no current flowing	The cell's state of charge; cells that sit lower than the rest

Measurement	What it is	What it tells you
Internal resistance (IR)	Measured in milliohms with a special battery tester	Cell health. A worn or damaged cell has higher resistance, sags under load and runs hot.
Capacity test	Charge fully, discharge at a set current, and count the amp-hours delivered	True SOH: measured capacity ÷ rated capacity. Slow, but the most reliable.
Insulation resistance	Pack + and pack − to the metal case, with an insulation tester	Leakage inside the pack
Airtightness test	After the lid is refitted, the pack is filled with low-pressure air and the pressure is watched	Whether the seal will keep water out (IP67)

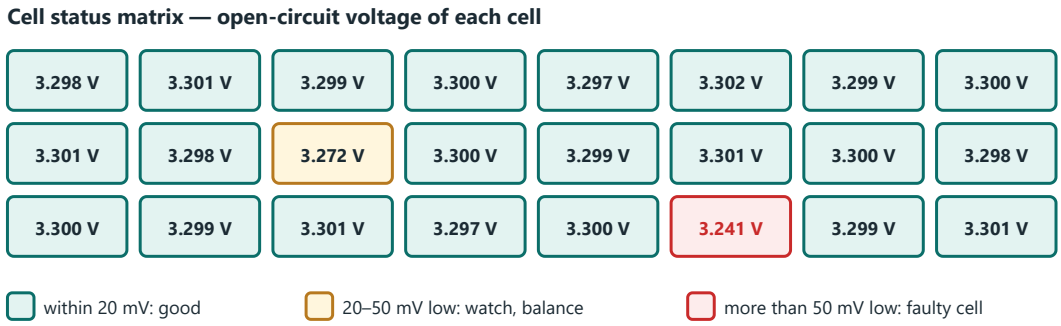


Figure 8.3 A cell status matrix. Writing every reading into a grid makes the bad cell obvious at a glance.

Table 8.2 — Typical limits used in Chinese battery repair training

Check	Typical limit
Cell voltage spread on live data	Investigate above 50 mV
Cell temperature spread	Investigate above 5 °C
Single cell on the bench	Abnormal if more than 30 mV from the others, or internal resistance more than 50% above the others
After balancing	Spread of 20 mV or less
Pack insulation resistance	At least 20 MΩ
Bolted bus-bar joints	Tightened with a torque wrench to the specified value

These figures come from training-provider material; the maker's values for a particular pack always take priority.

Why cells must be matched

A replacement module or cell must match the others in **capacity, internal resistance and state of charge**. A new, strong cell among old, tired ones does not rescue the pack: the pack is still limited by the weakest cell, and the mismatch makes balancing harder. This is "grading and matching", and it is why a proper repair measures every cell and does not simply swap the one that failed.

SUMMARY

- The BMS measures every cell, calculates SOC and SOH, protects, balances and communicates.
- Contactors connect the pack to the car. Pre-charge fills the inverter capacitor gently first.
- A pack is limited by its weakest cell. Check the cell voltage spread.
- Insulation monitoring watches for leakage between HV and the body.
- Thermal runaway is rare but serious. Know the warning signs.
- Heat, extreme SOC and constant fast charging shorten battery life.

Quiz 8

Choose one answer for each question.

1. What is the purpose of the pre-charge circuit?

- **A** To charge the 12 V battery
- **B** To heat the cells
- **C** To fill the inverter capacitor gently so the main contactors are not damaged
- **D** To balance the cells

2. In a pack of cells in series, when does discharging have to stop?

- **A** When the average cell is empty
- **B** When the weakest cell reaches its minimum voltage
- **C** When the strongest cell is empty
- **D** When half the cells are empty

3. Which of these is NOT a job of the BMS?

- **A** Measuring cell voltages
- **B** Controlling the contactors
- **C** Calculating SOC
- **D** Converting DC to AC for the motor

4. What does an "insulation fault" mean?

- **A** The battery is too cold
- **B** Resistance between the HV system and the car body has fallen too low
- **C** The 12 V battery is flat
- **D** The tyres are worn

5. Which is a warning sign of thermal runaway?

- **A** A click when the car is switched on
- **B** A cold battery case
- **C** A sweet chemical smell and hissing from the pack
- **D** Low tyre pressure

6. What operates the contactors?

- **A** High voltage from the motor
- **B** 12 V signals from the BMS
- **C** The driver's foot
- **D** The charging cable

7. Which habit is best for the life of an NMC battery in daily use?

- **A** Always charge to 100% and leave it parked in the sun
- **B** Always run it down to 0%
- **C** Charge to about 80% and park in shade
- **D** Use only DC fast chargers

PART D · THE ELECTRIC DRIVE

9 Electric motors

How a motor turns electricity into motion, and the types used in cars, bikes and rickshaws.

YOU WILL LEARN

- The basic principle of every electric motor
- The parts of a traction motor
- The main motor types and where each is used
- The shape of the motor's torque–speed curve
- Common motor faults

9.1 The principle

You know that magnets push and pull each other: like poles repel, unlike poles attract. You also know from school that a coil of wire carrying current becomes a magnet (an **electromagnet**), and that you can reverse its poles by reversing the current.

A motor uses these two facts. The outer part, the **stator**, has coils of copper wire. The inner part, the **rotor**, has magnets. By switching current through the stator coils in the right order, we create a magnetic field that **rotates** around the stator. The rotor magnets are dragged round with it. That rotation is what turns the wheels.

> **PICTURE IT — THE DONKEY AND THE CARROT**
>
> Hold a carrot on a stick in front of a donkey and walk in a circle: the donkey follows the carrot round and round. The stator's rotating magnetic field is the carrot. The rotor magnets are the donkey. Move the carrot faster (higher frequency) and the donkey turns faster. That is all a motor is.

Three-phase AC (Module 2) is perfect for this: three sets of coils, fed by three currents one-third of a cycle apart, naturally produce a smoothly rotating field.

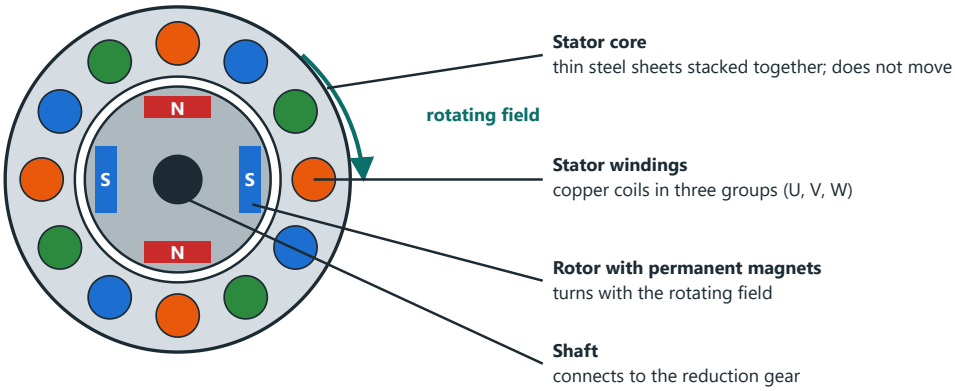


Figure 9.1 A permanent-magnet motor in cross-section. The gap between rotor and stator is less than a millimetre.

9.2 Parts of a traction motor

Part	Description
Stator	Stack of thin steel sheets with copper windings in slots. Modern motors use flat "hairpin" copper bars instead of round wire.
Rotor	Steel core carrying magnets (or aluminium/copper bars in an induction motor). Mounted on the shaft.
Bearings	Support the shaft at each end. They spin at up to 16,000–20,000 rpm.
Housing and cooling	Aluminium case with a water jacket around the stator, or oil sprayed directly on the windings.

Part	Description
Resolver (position sensor)	Tells the inverter the exact angle of the rotor so it can switch the coils at the right instant.
Temperature sensors	Buried in the windings. If the motor overheats, power is reduced.
Three-phase terminals	Three thick orange cables or bus bars (U, V, W) from the inverter.

9.3 Motor types

Table 9.1 — Traction motor types

Type	How the rotor works	Good	Less good	Used in
PMSM Permanent-magnet synchronous motor	Strong rare-earth magnets in the rotor. Rotor turns exactly in step with the field.	Most efficient (up to 97%), small, powerful	Magnets are expensive; generates voltage whenever it spins	Almost all EVs and hybrids: BYD, Deepal, MG, Toyota, Honda, Hyundai
Induction motor (asynchronous)	No magnets. Aluminium or copper bars in the rotor; the stator field *induces* current in them.	Simple, robust, cheap, no drag when not powered	Slightly less efficient, rotor runs hotter	Some Tesla and Audi motors; second axle of some AWD EVs; many industrial motors
BLDC hub motor Brushless DC	Permanent magnets on a rotor built into the wheel. Simple Hall sensors for position.	Very simple, no gearbox or chain	Heavy wheel, lower efficiency, limited power	E-bikes, scooters, some rickshaws
Wound-rotor synchronous (EESM)	Rotor is an electromagnet fed through slip rings	No rare-earth magnets	Brushes or extra parts wear	Some BMW and Renault EVs

DANGER — A SPINNING MOTOR IS A GENERATOR

A permanent-magnet motor produces voltage at its terminals **whenever the rotor turns**, even with the battery disconnected. If the driven wheels are turned — by towing, pushing fast, or spinning a wheel on a lift — the orange cables can become live. Never turn the wheels of an EV with HV parts exposed.

9.4 The torque–speed curve

Torque is turning force, measured in **newton-metres (Nm)**. This graph is the single most important picture for understanding how an EV drives.

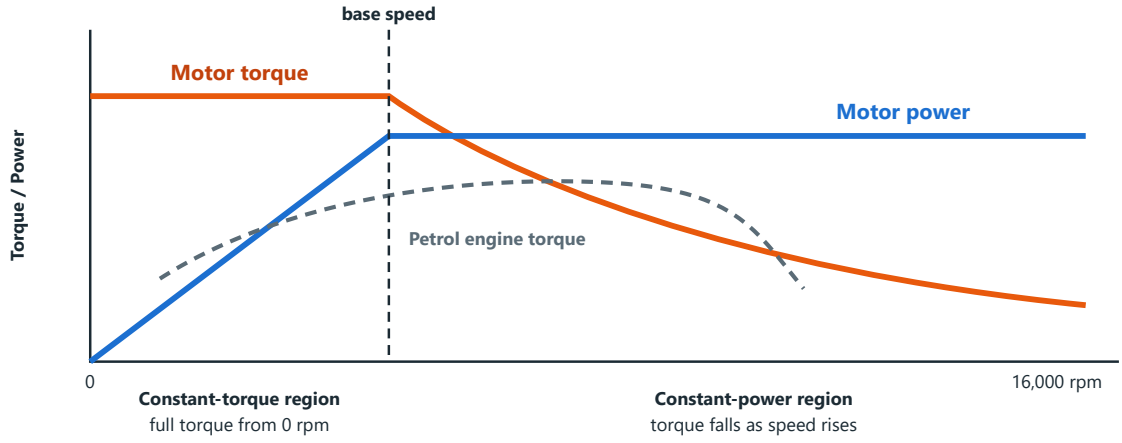


Figure 9.2 An electric motor gives full torque from standstill. A petrol engine must rev up to make torque and needs a multi-speed gearbox to stay in its useful range.

- From 0 rpm up to the **base speed**, the motor can give its **maximum torque**. The limit here is the maximum current the inverter and battery can supply. Power rises steadily with speed.
- Above base speed, the motor has reached its **maximum power**. The limit here is voltage. Power stays roughly level, so torque must fall as speed rises.

This is why an EV feels so quick away from traffic lights, and why it needs only one gear.

9.5 Motor ratings on a specification sheet

- **Peak power (kW)** — the maximum, available for a short time (typically 30 seconds). This is the advertised figure.
- **Continuous (rated) power (kW)** — what it can give all day without overheating. Often about half the peak.
- **Peak torque (Nm)** — maximum turning force.
- **Maximum speed (rpm)** — typically 14,000–20,000.

9.6 What goes wrong with motors

Fault	Symptom	Typical cause
Bearing wear	Whine or rumble that changes with road speed	High mileage; electrical currents through the bearing; water
Resolver fault	Jerky drive, no drive, warning light, fault code	Sensor, connector or wiring problem
Winding insulation breakdown	Insulation fault code; car will not go READY	Overheating, coolant or water inside the motor
Coolant leak into motor	Insulation fault, coolant loss with no visible leak	Internal seal failure
Oil seal leak	Gear oil at the drive unit joint	Seal wear
Overheating	Reduced power in hot weather or on hills	Low coolant, failed pump, blocked radiator

Motor windings are tested (with the HV system safely isolated) in two ways: the **resistance between phases** should be equal on all three pairs, and the **insulation resistance** from windings to the case, measured with an insulation tester, should be very high. Module 17 covers the tools.

SUMMARY

- A rotating magnetic field in the stator pulls the rotor round.
- PMSM is the standard car motor. Induction motors have no magnets. BLDC hub motors drive bikes.
- Full torque from zero speed up to base speed; constant power above it.
- A resolver tells the inverter the rotor position.
- A turning permanent-magnet motor generates voltage — never spin the wheels with HV parts exposed.

Quiz 9

Choose one answer for each question.

1. **Which part of the motor carries the copper windings and does not move?**

A Rotor | **B** Stator | **C** Resolver | **D** Shaft

2. **Which motor type is used in most electric cars today?**

A Brushed DC | **B** Induction | **C** Permanent-magnet synchronous (PMSM) | **D** Stepper

3. At what speed can an electric motor produce its maximum torque?

- **A** Only at maximum rpm
- **B** Only above 3,000 rpm
- **C** From 0 rpm up to base speed
- **D** Only at base speed exactly

4. What does the resolver do?

- **A** Cools the motor
- **B** Measures the rotor's position
- **C** Converts DC to AC
- **D** Stores energy

5. Why is it dangerous to spin the driven wheels of an EV with HV cables exposed?

- **A** The tyres may burst
- **B** The brakes will overheat
- **C** The permanent-magnet motor generates voltage when it turns
- **D** The battery will drain

6. Which motor type has no permanent magnets in its rotor?

- **A** PMSM
- **B** BLDC hub motor
- **C** Induction motor
- **D** All of them have magnets

PART D · THE ELECTRIC DRIVE

10 Power electronics

The boxes that convert and control electrical power: inverter, DC-DC converter, on-board charger and PDU.

YOU WILL LEARN

- How an inverter makes three-phase AC from battery DC
- How the inverter controls motor speed and torque
- What the DC-DC converter, on-board charger and PDU do
- Common faults and simple checks

10.1 The inverter

The inverter (also called motor control unit, MCU, or power control unit, PCU) sits between the battery and the motor. Inside are **six fast electronic switches** arranged in three pairs — one pair for each motor phase.

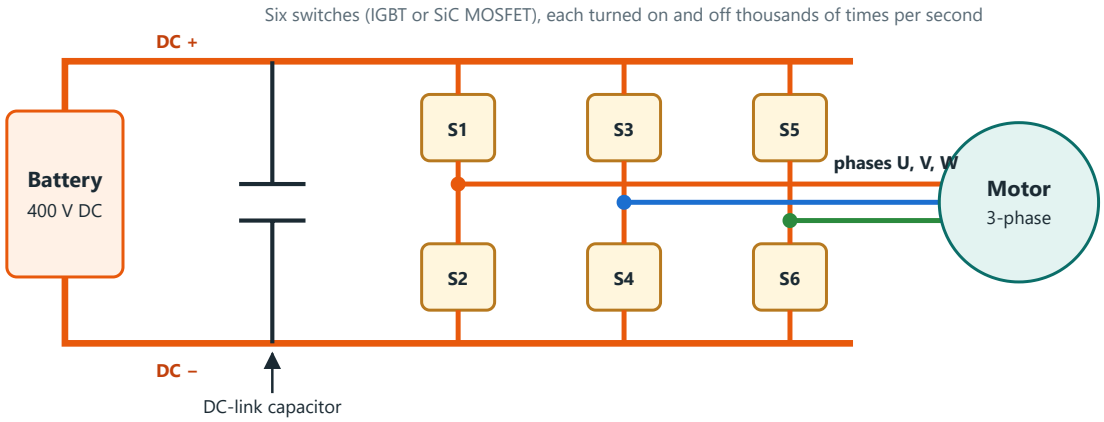
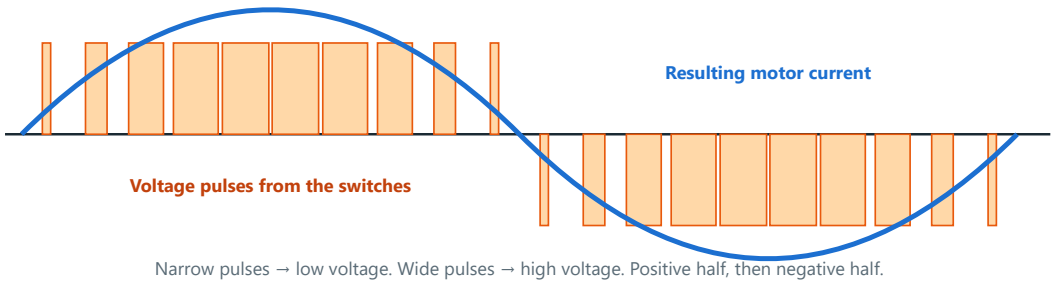


Figure 10.1 The inverter's power stage. By choosing which switches are on, each motor phase is connected to battery + or battery −.

How switching makes AC

The switches can only be fully on or fully off. To make a smooth AC wave, the inverter turns them on and off very quickly (around 5,000–20,000 times per second) and varies how long each pulse lasts. Wide pulses give a high average voltage; narrow pulses give a low one. This is **pulse-width modulation (PWM)**. The motor windings smooth the pulses into a near-perfect sine-wave current.



Narrow pulses → low voltage. Wide pulses → high voltage. Positive half, then negative half.

Figure 10.2 Pulse-width modulation (simplified).

PICTURE IT — THE TAP YOU CAN ONLY OPEN FULLY

Imagine a tap that can only be fully open or fully shut, filling a bucket with a small hole in it. Open it for long bursts and the water level rises; open it for short bursts and the level falls. By changing only the **length of the bursts**, you can hold the water at any level you like. The inverter's switches are that tap, and the level in the bucket is the voltage the motor feels.

KEY IDEA — HOW THE DRIVER'S FOOT CONTROLS THE MOTOR

The inverter controls two things:
• The **frequency** of the AC sets the motor **speed**.
• The **current** (amplitude) sets the motor **torque**.
Press the accelerator → the vehicle computer asks for more torque → the inverter sends more current. To reverse, it simply swaps the order of the phases.

Inside the inverter

- **Power module** — the six switches. Older and cheaper designs use **IGBTs** (silicon). Newer ones use **SiC MOSFETs** (silicon carbide), which waste less energy as heat and are essential for 800 V systems.
- **DC-link capacitor** — a large capacitor across the DC input that smooths the supply. **It stores a dangerous charge.**
- **Gate driver and control board** — the electronics that decide when to fire each switch.
- **Current sensors** — on the phase outputs.
- **Cooling plate** — coolant flows under the power module.

DANGER

The DC-link capacitor holds battery voltage after switch-off until the discharge circuit drains it. This is the reason for the waiting time in the isolation procedure, and the reason you **measure** before touching.

10.2 The DC-DC converter

A petrol car has an alternator driven by the engine to charge the 12 V battery. An EV has no alternator. Instead a **DC-DC converter** takes high voltage from the main battery and steps it down to about **13.5–14.8 V**. Typical output is 1.5–3 kW (over 100 A).

WORKSHOP TIP — A SIMPLE, SAFE TEST

Measure the 12 V battery voltage with the car **off** (expect about 12.4–12.8 V) and again with the car in **READY** (expect about 13.5–14.8 V). If the voltage does not rise in READY, the DC-DC converter is not charging. This is a low-voltage test you can do without touching any orange cable.

10.3 The on-board charger (OBC)

The battery needs DC. The mains supplies AC. The **on-board charger** converts one to the other when you plug into an AC socket or wallbox. Its power rating sets the **maximum AC charging speed**:

OBC rating	Supply needed	Time to add 50 kWh
3.3 kW	Single-phase, 16 A	about 15 hours
6.6–7 kW	Single-phase, 32 A	about 7.5 hours
11 kW	Three-phase, 16 A per phase	about 4.5 hours
22 kW	Three-phase, 32 A per phase	about 2.5 hours

During **DC fast charging** the big converter is inside the charging station, and current goes straight to the battery. The on-board charger is bypassed (Module 14).

Many Chinese EVs have a **bidirectional** charger that can also work backwards, giving 230 V AC from the car battery through an adapter. This is called **V2L** (vehicle-to-load).

IN PAKISTAN

V2L is a real selling point here: during load-shedding an EV can run fans, lights, a fridge or tools for many hours. Customers will ask you about it. They may also damage it by overloading it or using poor-quality adapters, which gives you repair work.

10.4 The power distribution unit (PDU)

The PDU is the HV junction box. From the battery, power is shared out — through fuses and sometimes relays — to the inverter, on-board charger, DC-DC converter, A/C compressor and heater. When one HV consumer stops working (for example the A/C compressor), a blown HV fuse in the PDU is a possible cause. HV fuses are only checked or changed with the system isolated and proven dead.

10.5 The boost converter in Toyota hybrids

Toyota hybrid batteries are only about 150–250 V. To make the motors more powerful and efficient, a **boost converter** inside the inverter assembly raises this to as much as 600–650 V for the motors. So the voltage inside a Toyota inverter is far higher than the battery voltage written on the label.

10.6 Cooling and common faults

Power electronics are about 95–98% efficient, but 2–5% of 150 kW is still several kilowatts of heat. They share a coolant loop with an electric pump and a radiator.

Fault	Symptom	First checks
Inverter overheating	Power reduced, warning light, worse in summer	Coolant level, electric coolant pump running, air in system, radiator blocked
Failed inverter coolant pump	Same as above; fault code for pump	12 V supply and fuse to pump; pump noise; coolant movement in the reservoir
DC-DC failure	12 V battery goes flat; many warning lights; car dies while driving	12 V voltage in READY (see tip above)
OBC failure	Will not charge on AC but charges on DC	Try another cable and socket first; read fault codes
Power module failure	No drive, HV fuse may blow, fault codes	Specialist diagnosis; usually unit replacement

WORKSHOP TIP

On older Toyota hybrids, a failed **inverter coolant pump** is a classic fault. It is cheap. If ignored, it destroys the inverter, which is not cheap. Checking that the coolant in the inverter reservoir is moving (with the car in READY) takes ten seconds.

SUMMARY

- The inverter uses six switches and PWM to turn DC into three-phase AC.
- Frequency sets speed; current sets torque.
- The DC-DC converter charges the 12 V system: expect 13.5–14.8 V in READY.
- The on-board charger converts mains AC to DC and sets the AC charging speed.
- The PDU distributes and fuses the high voltage.
- Power electronics need liquid cooling. Check coolant and pumps first.

Quiz 10

Choose one answer for each question.

1. How many main power switches does a three-phase inverter have?

- **A** 2
- **B** 3
- **C** 6
- **D** 12

2. To make the motor produce more torque, the inverter increases the…

- **A** frequency
- **B** current
- **C** battery capacity
- **D** gear ratio

3. With the car in READY, the 12 V battery measures 12.1 V and is falling. What is the likely fault?

- **A** On-board charger
- **B** Resolver
- **C** DC-DC converter not charging
- **D** Tyre pressure sensor

4. A car charges at a DC fast charger but not from an AC wallbox. Which part is suspect?

- **A** Inverter
- **B** On-board charger
- **C** Motor
- **D** Reduction gear

5. Why must you wait after switching off before working on the inverter?

- **A** To let the motor cool
- **B** The DC-link capacitor holds a dangerous charge until it is discharged
- **C** To let the coolant settle
- **D** To save the 12 V battery

6. What is V2L?

- **A** A type of battery cell
- **B** A fast-charging plug
- **C** Using the car's battery to power 230 V AC appliances
- **D** A two-speed gearbox

PART D · THE ELECTRIC DRIVE

11 Performance: how battery and motor shape the car

Acceleration, top speed, range and running cost — and the numbers behind them.

YOU WILL LEARN

- How torque, power and speed are related
- How motor power and battery power together limit acceleration
- How battery energy and consumption decide range
- Why real range differs from the brochure, and how to explain it to customers

11.1 Torque, speed and power

Power is torque multiplied by rotational speed. In workshop units:

```
Power (kW) = Torque (Nm) × Speed (rpm) ÷ 9,550
```

WORKED EXAMPLE

A motor makes 310 Nm at 4,600 rpm. Power = 310 × 4,600 ÷ 9,550 = **149 kW**.

The same motor at 12,000 rpm can still make about 150 kW, so its torque there is 150 × 9,550 ÷ 12,000 = **119 Nm**. Same power, much less torque — the constant-power region from Figure 9.2.

- **Torque** (multiplied by the gear ratio) decides how hard the car pushes off from rest and at low speed.
- **Power** decides how well it keeps accelerating at higher speed, how it climbs long hills, and its top speed.

11.2 From motor torque to acceleration

WORKED EXAMPLE — WHY EVS LAUNCH SO HARD

Motor torque 310 Nm. Reduction gear ratio 10:1. Wheel radius 0.33 m. Car mass 1,750 kg.

1. Torque at the wheels = 310 × 10 = 3,100 Nm (ignoring small losses)
2. Driving force = torque ÷ wheel radius = 3,100 ÷ 0.33 ≈ 9,400 N
3. Acceleration = force ÷ mass = 9,400 ÷ 1,750 ≈ **5.4 m/s²**

That is more than half of gravity, available instantly from 0 km/h. A typical family petrol car manages about half of this in first gear, and only once the engine has revved up.

11.3 The weakest link sets the limit

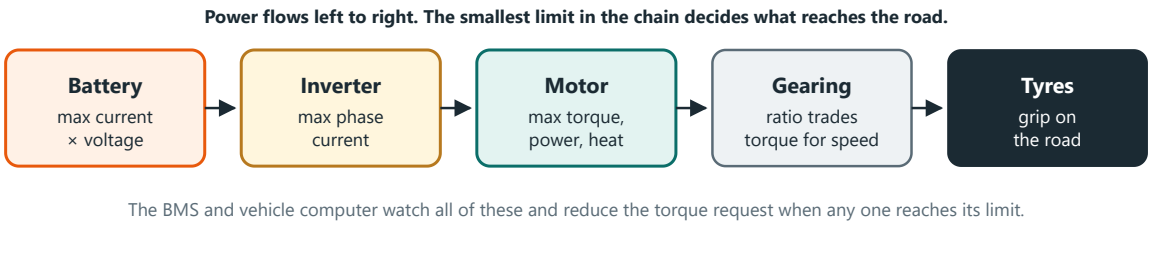


Figure 11.1 The performance chain.

How the battery limits power

The motor may be rated at 150 kW, but it can only deliver what the battery can supply.

WORKED EXAMPLE

A 60 kWh pack at 400 V can safely give 400 A. Maximum battery power = 400 × 400 = **160 kW**. The 150 kW motor can be fully used.

Now the same motor with a small 20 kWh pack of the same cells: one-third of the cells in parallel, so about 133 A. Maximum = 400 × 133 = **53 kW**. The motor is still a 150 kW motor, but the car performs like a 53 kW car.

This explains several things you will see:

- **Long-range versions are often faster.** A bigger battery can supply more current, so the maker fits or unlocks a more powerful motor.
- **Hybrids need "power cells".** An HEV battery of 1.5 kWh giving 40 kW is working at about 27C. These cells are designed for high power, not high energy.
- **PHEVs feel strongest with charge in the battery.** With the battery depleted, less electrical power is available to help the engine.
- **Power is reduced when the battery is cold, very hot, nearly empty or aged.** Internal resistance is higher, so the BMS lowers the current limit. The customer complains of "low power"; the scan tool shows the BMS limit as the reason.

KEY IDEA

Motor kW is what the car *could* do. **Battery kW** (which depends on battery size, chemistry, temperature, charge level and health) is what it is *allowed* to do right now. Performance is whichever is lower.

11.4 Power-to-weight

A heavy car needs more power for the same acceleration. Dividing power by mass gives a quick comparison.

Table 11.1 — Power-to-weight examples (approximate figures)

Car	Power	Mass	kW per tonne	0–100 km/h
Toyota Corolla Cross HEV	90 kW	1,400 kg	64	about 11 s
BYD Atto 3	150 kW	1,750 kg	86	7.3 s
Deepal S07	190 kW	2,000 kg	95	about 7 s
BYD Seal AWD (two motors)	390 kW	2,185 kg	178	3.8 s

All-wheel-drive EVs have a motor on each axle. Their powers add together, and four driven tyres can put more force on the road without slipping.

11.5 Range

```
Range (km) = Usable battery energy (Wh) ÷ Consumption (Wh per km)
```

Consumption is the EV's "fuel average". A small city car may use 110–130 Wh/km; a compact SUV 150–180 Wh/km; a large SUV or pickup 200–280 Wh/km. An electric motorcycle uses only 25–40 Wh/km.

WORKED EXAMPLE

Usable battery 60 kWh = 60,000 Wh. Consumption 160 Wh/km.

Range = 60,000 ÷ 160 = **375 km**

On the motorway at 120 km/h with A/C, consumption rises to 230 Wh/km: range = 60,000 ÷ 230 = **261 km**.

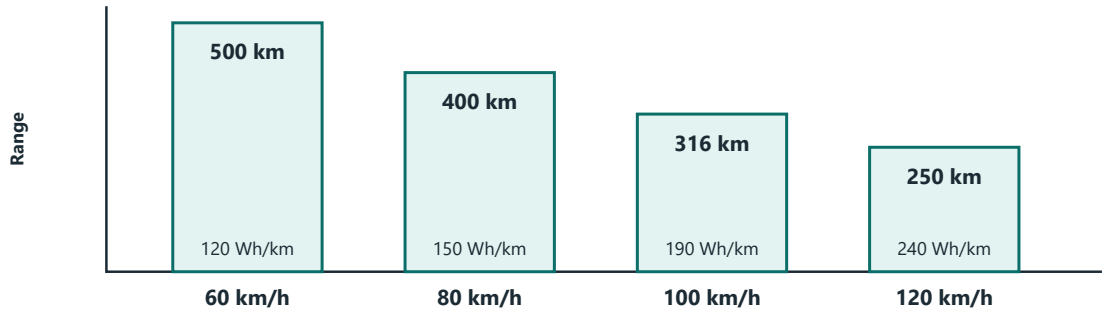


Illustration for a compact electric SUV with 60 kWh usable, steady speed, mild weather

Figure 11.2 Speed is the biggest factor in range. Air resistance rises with the *square* of speed.

What reduces range

Factor	Why	Typical effect
High speed	Air drag rises with speed squared	Large: see Figure 11.2
Air-conditioning in extreme heat	Compressor uses 1–3 kW; also cools the battery	5–20%
Hard acceleration and braking	Regeneration recovers only part of the energy	10–20%
Weight and roof loads	More rolling resistance and drag	5–15%
Under-inflated or wrong tyres	More rolling resistance	3–10%
Battery age (lower SOH)	Less energy stored	Equal to the SOH loss
Dragging brakes, poor alignment	Wasted energy	Varies — and it is your job to find it

Brochure range and real range

Range is measured on standard laboratory tests. Different tests give different answers for the same car:

- **NEDC** and **CLTC** — older and Chinese tests. Gentle and optimistic. Often quoted in Pakistani advertising.
- **WLTP** — the current European test. More realistic.
- **Real mixed driving** in Pakistan is usually around **70–80% of the NEDC or CLTC figure**, and less on the motorway in summer.

> **WORKSHOP TIP**
>
> "My range has dropped" is one of the most common EV complaints. Before blaming the battery, check: tyre pressures, brake drag, wheel alignment, driving speed, A/C use, and the season. Then read the **SOH** on the scan tool. Most of the time the battery is fine and the cause is one of the simple things.

11.6 Efficiency and running cost

From battery to wheels, an EV is about **85–90%** efficient. A petrol engine turns only about **25–35%** of the fuel's energy into motion; the rest is heat. This is the basic reason an EV costs less per kilometre.

> **WORKED EXAMPLE — COST PER 100 KM (PUT IN TODAY'S PRICES)**
>
> **EV:** 16 kWh per 100 km × Rs 50 per unit = **Rs 800**
> **Petrol car:** 12 km per litre → 8.3 litres per 100 km × Rs 270 per litre = **Rs 2,240**
>
> With home solar, the EV figure can be far lower. Prices change, so always redo this sum with current rates.

SUMMARY

- Power (kW) = torque (Nm) × rpm ÷ 9,550.
- Torque × gear ratio gives launch; power gives high-speed performance.
- Performance is limited by the weakest of: battery power, inverter current, motor, tyres.
- Battery power depends on size, chemistry, temperature, SOC and health.
- Range = usable energy ÷ consumption. Speed, A/C, tyres and driving style change consumption.
- Expect about 70–80% of NEDC/CLTC range in real use.

Quiz 11

Choose one answer for each question.

1. A motor produces 200 Nm at 9,550 rpm. What is its power?

A 20 kW
B 100 kW
C 200 kW
D 955 kW

2. A car has 50 kWh usable and uses 200 Wh/km. What is its range?

A 100 km
B 250 km
C 400 km
D 1,000 km

3. A 150 kW motor is fitted to a car whose battery can supply only 80 kW. What is the maximum drive power?

A 150 kW
B 230 kW
C 80 kW
D 70 kW

4. Why does a cold or aged battery reduce the car's power?

A The motor shrinks
B Higher internal resistance makes the BMS lower the current limit
C The tyres lose grip
D The gear ratio changes

5. Which factor usually has the biggest effect on motorway range?

A Radio volume
B Headlights
C Speed
D Colour of the car

6. A car is advertised with 500 km NEDC range. What real mixed-driving range is realistic?

A 600 km
B 500 km
C About 350–400 km
D 100 km

7. A customer says range has dropped. What should you check before suspecting the battery?

A Nothing — replace the battery
B Tyre pressures, brake drag, alignment and driving conditions, then SOH
C The paint
D The wiper motor

PART D · THE ELECTRIC DRIVE

12 Transmission, regenerative braking and chassis

The mechanical side of an EV: one gear, two kinds of braking, and a heavy car on four tyres.

YOU WILL LEARN

- Why EVs have a single-speed reduction gear
- How regenerative braking works and when it is limited
- How regenerative and friction brakes are blended
- Brake, tyre and lifting points specific to EVs

12.1 The reduction gear

A petrol engine works well only in a narrow speed band, so it needs 5 or more gears. An electric motor gives useful torque from 0 to over 16,000 rpm, so **one fixed gear ratio** is enough. A typical ratio is between **8:1 and 12:1**: the motor turns about ten times for each turn of the wheels.

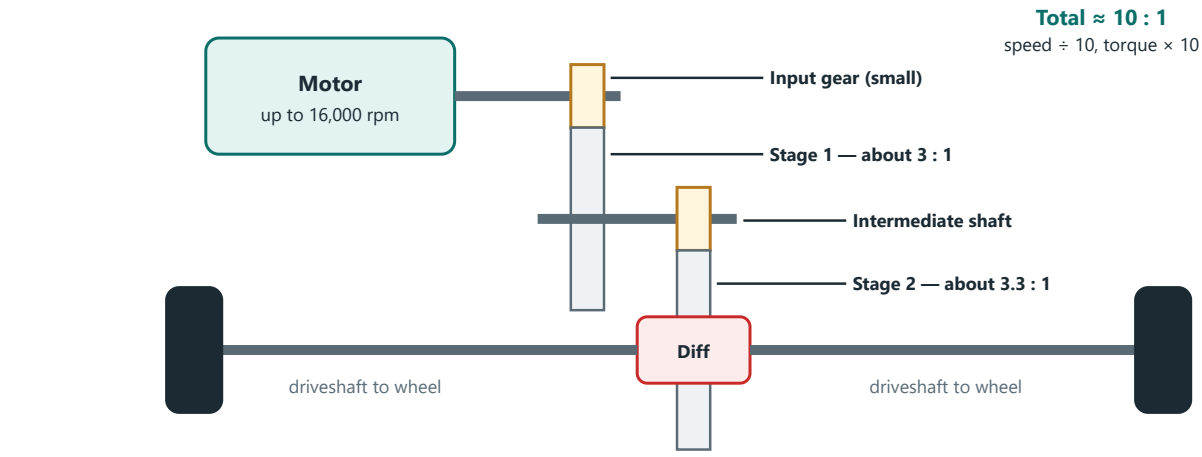


Figure 12.1 A typical two-stage reduction gear with differential, shown schematically.

- **Reverse** is done by spinning the motor backwards. There is no reverse gear.
- **Park** uses a locking pawl in the gear unit, usually moved by a small electric actuator, plus the electric parking brake.
- **Neutral** simply means the inverter sends no current.
- The unit has its own **gear oil** (the specific fluid named by the maker). Some are "filled for life"; many specify a change at set intervals. Follow the service schedule.

Hybrids are different: they have special transmissions that mix engine and motor power. See Module 13.

12.2 Regenerative braking

When the driver lifts off the accelerator or presses the brake, the inverter makes the motor work as a **generator**. The wheels drive the motor; the motor resists; the car slows; and electricity flows back into the battery. Energy that a normal car throws away as brake heat is recovered.

> **PICTURE IT — THE BICYCLE DYNAMO**
>
> Switch on a bicycle's dynamo lamp and pedalling becomes harder. The dynamo is taking energy from your legs and turning it into electricity. Regenerative braking is the same thing on purpose: the motor becomes the dynamo, the "harder to pedal" feeling is what slows the car, and the electricity goes back into the battery instead of a lamp.

- In city driving, regeneration can recover roughly 15–25% of the energy used.
- Most cars let the driver choose the strength (low / standard / high). Strong regeneration allows **one-pedal driving**.
- Brake lights come on automatically when regeneration is slowing the car firmly.

When regeneration is reduced or switched off

Situation	Reason
Battery at or near 100%	There is nowhere to put the energy
Battery very cold or very hot	Cells cannot accept high charge current safely
Very low speed (below about 5–10 km/h)	The motor generates too little
Emergency stop or ABS/ESC active	Friction brakes give faster, more controllable braking at each wheel
Fault in the HV system	Regeneration disabled for safety

> **WORKSHOP TIP**
>
> A customer who charges to 100% at the top of a hill station and then drives down will find the car "does not slow like before". That is normal: a full battery cannot take regeneration, so the friction brakes do all the work. Explain it; do not go looking for a fault.

12.3 Brake blending

Every EV and hybrid still has normal hydraulic **friction brakes** with discs and pads. The car's brake computer decides, moment by moment, how much braking comes from the motor and how much from the pads. The driver should feel one smooth pedal.

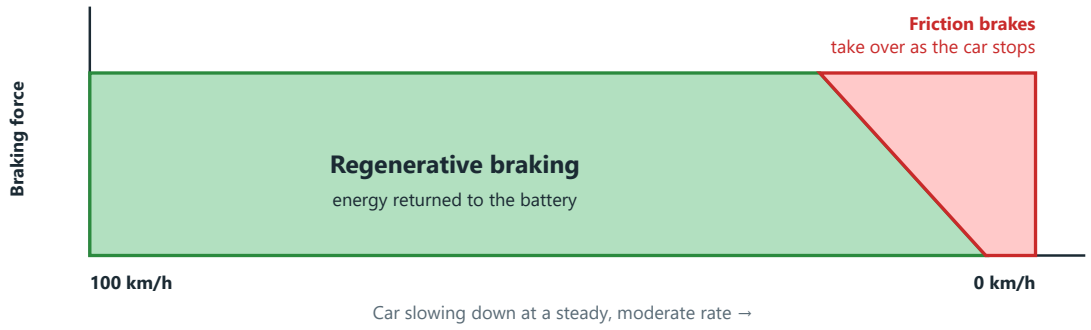


Figure 12.2 Blending during a normal, gentle stop. In a hard stop, friction brakes are added on top from the start.

To make blending possible, and because there is no engine vacuum for a normal brake servo, EVs use an **electric brake booster** or a full **brake-by-wire** unit. The pedal is connected to a sensor and a "pedal feel simulator"; an electric motor or pump creates the hydraulic pressure.

12.4 Brake service on EVs

Because the motor does most of the braking, pads and discs last a very long time. The problem is the opposite of wear: **lack of use**.

- Discs rust, especially at the rear, and after the monsoon.
- Caliper slide pins seize; pads stick in their carriers; brakes drag and waste range.
- **Service:** at least once a year, remove the pads, clean and lubricate the slide pins and pad contact points, and check the discs.
- **Brake fluid** still absorbs moisture. Change it every 2 years or as specified.
- Electric parking brakes need the scan tool's **service mode** to wind back the rear calipers before changing pads.
- Bleeding a brake-by-wire system often needs a scan-tool procedure.

12.5 Tyres, suspension and lifting

- EVs are **300–500 kg heavier** than similar petrol cars and have instant torque. Tyres wear faster. Fit tyres with the correct **load index**; many are marked for EV use (lower rolling resistance, lower noise).
- Rotate tyres regularly and check alignment. Poor alignment costs range as well as rubber.
- Suspension bushes, ball joints and dampers carry more weight; on rough roads they wear sooner.

DANGER — LIFTING AND JACKING

- **Never place a jack or lift pad under the battery pack.** You can crush cells and cause an internal short. Use only the lifting points shown in the manual, usually on the sills.
- Watch for speed breakers and stones: check the battery under-tray for dents and scrapes at every service.
- Removing a battery pack (300–600 kg) needs a proper battery lifting table. It also changes the car's balance on the lift.
- Most EVs must be transported on a flatbed, not towed with driven wheels on the ground.

IN PAKISTAN

High speed breakers, broken roads and flooded underpasses are a real threat to under-floor batteries. An under-body inspection — looking for dents, torn covers, damaged coolant pipes and water marks — is a valuable service you can offer at every visit.

SUMMARY

- One fixed reduction gear of about 10:1; reverse by spinning the motor backwards.
- Regenerative braking turns motion back into battery energy.
- Regeneration is limited when the battery is full, cold or hot, at very low speed, and in emergency stops.
- Friction brakes last long but seize from lack of use: clean and lubricate yearly; change fluid every 2 years.
- EVs are heavy: correct tyres, correct lifting points, never jack under the battery.

Quiz 12

Choose one answer for each question.

1. **Why does an EV need only one gear?**

A Gears are too expensive

B The motor gives useful torque over a very wide speed range, from 0 rpm

C EVs are slow

D The battery cannot power a gearbox

2. **How does an EV reverse?**

A A reverse gear is engaged

B The wheels are disconnected

C The motor spins in the opposite direction

D The battery polarity is swapped

3. **When is regenerative braking reduced?**

A When the battery is at 50%

B When the battery is fully charged

C When the A/C is on

D When the headlights are on

4. **What is the most common brake problem on EVs?**

A Pads wearing out very fast

B Rust and seized sliders from lack of use

C Boiling brake fluid

D Broken brake pedal

5. Where must you never place a jack?

- **A** Under the sill lifting points
- **B** Under the subframe point shown in the manual
- **C** Under the battery pack
- **D** On a level floor

6. Why do EVs use an electric brake booster?

- **A** To charge the battery
- **B** There is no engine vacuum, and it allows blending with regenerative braking
- **C** To make the pedal harder
- **D** To cool the brakes

PART E · HYBRIDS

13 Hybrid systems in detail

Toyota, Honda, Hyundai/Kia and the Chinese plug-in systems: same goal, different engineering.

YOU WILL LEARN

- How a hybrid saves fuel
- How the Toyota power-split system works
- How Honda, Hyundai/Kia and Chinese DHT plug-in hybrids differ
- Typical hybrid faults and service points

13.1 How a hybrid saves fuel

1. **Engine off when not needed** — at traffic lights, in slow traffic, when coasting.
2. **Electric drive at low speed**, where a petrol engine is least efficient.
3. **Regenerative braking** — recovering energy that brakes would waste.
4. **Engine kept at its most efficient load.** If the engine makes more power than needed, the extra charges the battery; if it makes less, the motor adds the rest.
5. **Efficient engine design.** Hybrid engines usually use the *Atkinson cycle* (late-closing intake valves): less power but better economy. The motor covers the missing power.

These gains are biggest in stop-start city traffic — exactly the conditions of Karachi, Lahore and Rawalpindi.

13.2 Toyota Hybrid System (power split)

Used in the Prius, Aqua, Yaris, Corolla Cross, Camry and Lexus hybrids. There is no clutch, no stepped gearbox, no starter motor and no alternator. One **planetary gear set** joins three things:

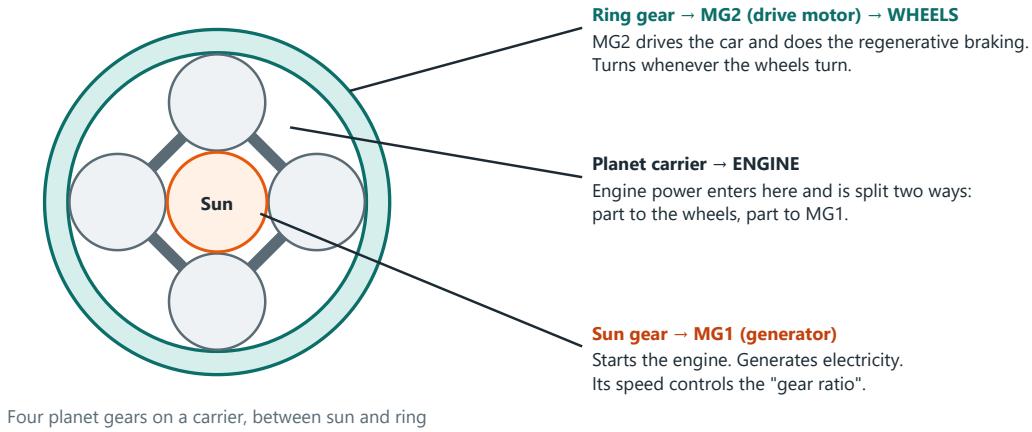


Figure 13.1 The power-split device. Because the electronics control MG1's speed, the engine can run at any speed for any road speed — an "electronic CVT" (e-CVT) with no belts.

Driving situation	What happens
Pulling away gently	Engine off. Battery → MG2 → wheels.
Normal acceleration	MG1 spins the engine to start it. Engine power splits: part to wheels, part through MG1 → MG2.
Hard acceleration	Engine plus battery power through MG2.
Cruising	Mostly engine; surplus charges the battery.
Braking	MG2 generates and charges the battery. Engine off.

Battery: NiMH or lithium-ion, about 1–1.5 kWh. Aqua: 144 V (120 cells × 1.2 V). Prius (third generation): 201.6 V (168 cells). A boost converter raises this to as much as 650 V for the motors. The battery is air-cooled by a fan drawing cabin air.

13.3 Honda e:HEV (two-motor system)

Used in the HR-V e:HEV and other recent Honda hybrids. It has a generator motor, a drive motor, and a single clutch.

- **City driving — series mode.** The engine turns the generator; the drive motor turns the wheels. Engine speed is not tied to road speed.
- **Steady cruising — engine drive.** A clutch connects the engine straight to the wheels through one fixed gear, like top gear.
- **Low load — EV mode.** Battery → drive motor.

> **WORKSHOP TIP — DO NOT CONFUSE THE TWO HONDA SYSTEMS**
>
> Older imported Honda **Vezel and Fit hybrids** (2013–2020) use a different system called **i-DCD**: one motor inside a 7-speed dual-clutch gearbox. These are known for clutch judder and gearbox actuator faults. The newer **e:HEV** has no such gearbox. Always identify which system you have before diagnosing.

13.4 Hyundai / Kia parallel hybrid

Used in the Tucson, Santa Fe and Sportage hybrids. The layout looks familiar to any mechanic:

Turbo petrol engine → engine clutch → electric motor → 6-speed automatic gearbox → wheels

- The motor (about 44 kW) sits where the torque converter would be.
- With the clutch open, the motor alone drives through the gearbox (EV mode). With it closed, engine and motor work together.
- A second small motor, the **HSG** (hybrid starter-generator), is belt-driven on the engine to start it smoothly and to generate.
- Battery: lithium-ion polymer, about 1.5 kWh, 270 V.

13.5 Chinese plug-in hybrids (DHT systems)

BYD (**DM-i**), GWM Haval (**Hi4 / DHT**), and Chery / Jaecoo (**CSH / SHS**) all use a **dedicated hybrid transmission (DHT)**: a compact unit containing a generator, a drive motor, a clutch, and one to three gears for the engine. They behave mainly like an electric car, with the engine helping when needed.

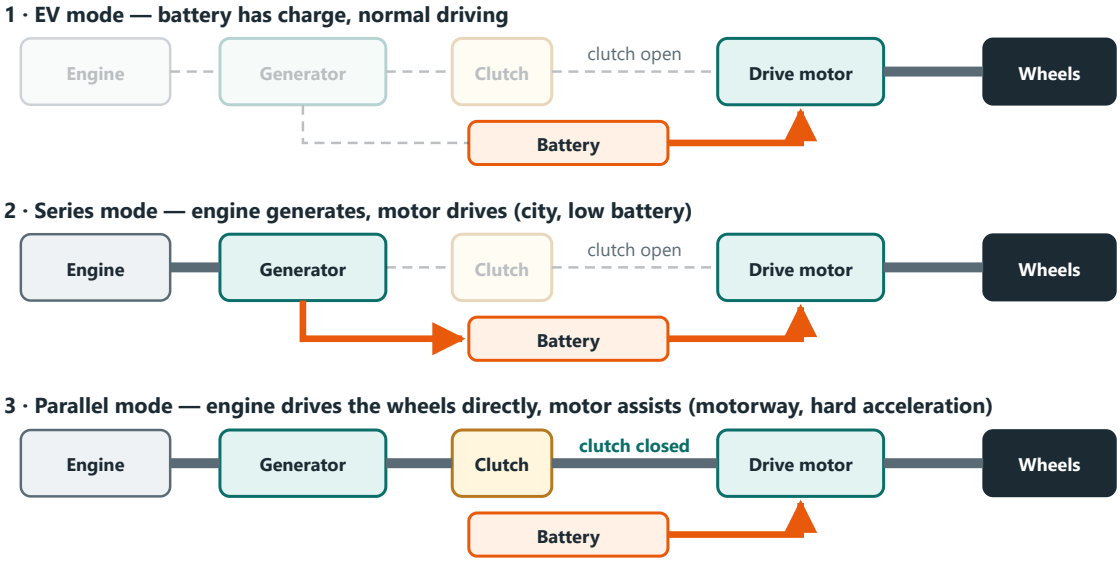


Figure 13.2 The three main modes of a DHT plug-in hybrid. The car's computer switches between them automatically.

Table 13.1 — Plug-in hybrids sold in Pakistan (approximate data)

Model	Engine	Battery	Electric range (test figure)
BYD Sealion 6	1.5 L	18.3 kWh LFP Blade	about 70–80 km
BYD Shark 6 pickup	1.5 L turbo	about 30 kWh LFP Blade	about 100 km
Haval H6 PHEV	1.5 L turbo	about 19 kWh	about 70–100 km
Jaecoo J7 PHEV	1.5 L turbo	about 18 kWh	about 80–90 km
Chery Tiggo 8 PHEV	1.5 L turbo	about 18 kWh	about 75–90 km

A **range-extender EV** (such as the Deepal S05 REEV, about 27 kWh) is a DHT hybrid with mode 3 removed: the engine can only generate.

13.6 What is extra on a plug-in hybrid

- A much bigger battery, on-board charger and charge port — everything from the EV modules applies.
- **Driver modes:** EV (electric first), HEV (automatic), and often "save" or "charge" (use the engine to hold or raise the battery level).
- **Engine that rarely runs.** If the owner drives mostly on electricity, petrol can sit in the tank for months and go stale; oil can collect moisture because the engine never fully warms up. Many PHEVs force the engine to run occasionally for this reason. Oil changes are still needed by *time*, even if the engine hours are low.
- **Sealed, pressurised fuel tank** on many PHEVs: the fuel flap must be released by a button and may take a few seconds.

13.7 Hybrid faults and service

A hybrid needs **all the service of a petrol car** (oil, filters, plugs, coolant) **plus** care of the high-voltage system.

Table 13.2 — Common faults on hybrids

Fault	Symptoms	Notes
HV battery worn (older Toyota, Honda)	Warning triangle, fault code P0A80, engine runs constantly, poor fuel economy, charge gauge swings rapidly up and down	Use scan tool to compare block voltages. One or more weak modules.
Blocked battery cooling fan	Battery overheats; fan noisy; early battery failure	Clean fan and ducts at every service
Inverter coolant pump failure	Overheat warning, power cut	Check flow in reservoir
Weak 12 V battery	Will not go to READY; strange warning lights	The most common "hybrid fault" of all
Brake actuator / accumulator	Brake warning, buzzing pump, hard pedal	Safety-critical; follow manual
Engine issues from stop-start use	Rough running at start, blocked EGR, oil sludge	Engine runs cold more often
Dual-clutch gearbox (Honda i-DCD)	Judder, no drive, gearbox warning	Clutch and actuator learning via scan tool

IN PAKISTAN — HYBRID BATTERY "REPAIR"

Many shops replace one or two weak modules in an old Aqua or Prius battery. It is cheap and gets the car moving. But the remaining modules are just as old, and new modules that do not match the old ones go out of balance quickly, so the fault often returns within months. A good technician tests **all** modules, matches replacements by capacity, cleans the bus bars and fan, and is **honest with the customer** about how long the repair may last compared with a new or fully reconditioned pack.

DANGER

A hybrid in READY mode is silent but **on**. The engine can start at any moment without warning. Before working in the engine bay, switch fully off and keep the key away. And remember: a "small" hybrid battery at 144–270 V is every bit as deadly as a big EV pack.

SUMMARY

- Hybrids save fuel by stopping the engine, driving electrically at low speed, regenerating, and running the engine efficiently.
- Toyota: planetary power split, two motor-generators, e-CVT.
- Honda e:HEV: mostly series, with a direct-drive clutch for cruising. Older Vezel/Fit i-DCD is different.
- Hyundai/Kia: parallel, motor between engine and 6-speed automatic.
- Chinese PHEVs: DHT with EV, series and parallel modes, 18–30 kWh batteries.
- Hybrids need petrol-car servicing plus HV care. Check the 12 V battery first.

Quiz 13

Choose one answer for each question.

1. In the Toyota power-split device, what is connected to the planet carrier?

- **A** MG1
- **B** MG2
- **C** The engine
- **D** The battery

2. Which component starts the engine in a Toyota hybrid?

- **A** A normal 12 V starter motor
- **B** MG1
- **C** The driver's foot
- **D** The DC-DC converter

3. In series mode on a DHT plug-in hybrid, what drives the wheels?

- **A** The engine through the clutch
- **B** The drive motor only
- **C** The generator
- **D** The 12 V battery

4. Which layout do Hyundai and Kia hybrids use?

- **A** Series only
- **B** Power split
- **C** Parallel, with the motor between engine and automatic gearbox
- **D** Hub motors

5. A hybrid will not go to READY and shows many warning lights. What do you check first?

- **A** HV battery modules
- **B** Inverter
- **C** The 12 V battery
- **D** The planetary gear

6. Why do PHEV engines still need oil changes by time, even with few engine hours?

- **A** The oil evaporates
- **B** Short, cold engine runs let moisture and fuel collect in the oil
- **C** The battery heats the oil
- **D** They do not need oil changes

7. Why is a hybrid in READY mode dangerous to work on?

- **A** The radio may come on
- **B** The engine can start by itself without warning
- **C** The tyres may deflate
- **D** It uses more fuel

PART F · SUPPORTING SYSTEMS

14 Charging

AC and DC charging, plugs and sockets, how the car and charger talk, and home charging on Pakistani wiring.

YOU WILL LEARN

- The difference between AC and DC charging
- The connector types used in Pakistan
- How to estimate charging time
- What a safe home charging installation needs
- Common charging faults

14.1 AC charging and DC charging

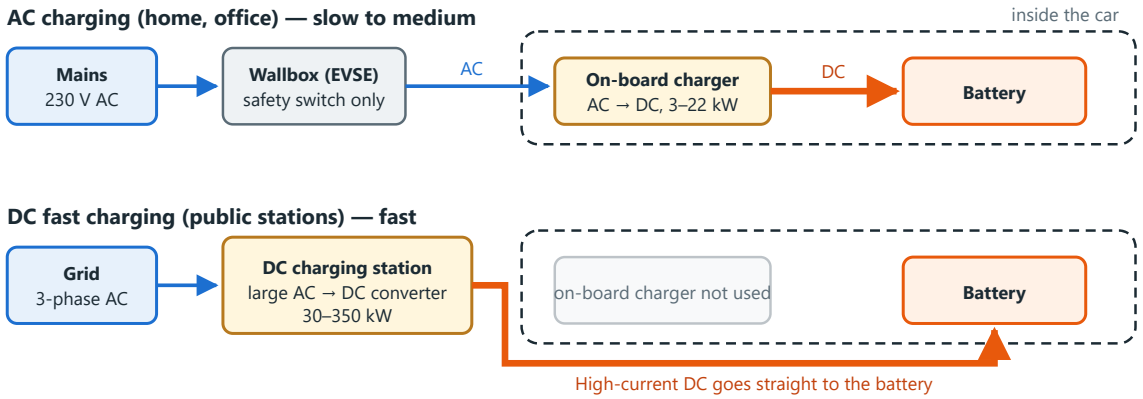


Figure 14.1 In AC charging the conversion happens in the car. In DC charging it happens in the station.

Table 14.1 — Ways to charge

Method	Power	Typical use	Adds per hour (approx.)
Portable charger in a 3-pin socket ("granny cable")	1.8–3 kW	Overnight top-up, emergencies, PHEVs	10–18 km
AC wallbox, single-phase	7 kW	Home, overnight	40–45 km
AC wallbox, three-phase	11–22 kW	Offices, malls, homes with 3-phase	60–130 km
DC fast charger	30–60 kW	City stations	180–350 km
DC rapid charger	120–350 kW	Motorway service areas	10–80% in 20–40 minutes

The actual speed is always the **lowest** of: what the charger can give, what the cable can carry, and what the car can accept. A car with a 7 kW on-board charger takes only 7 kW from a 22 kW wallbox.

14.2 Connectors

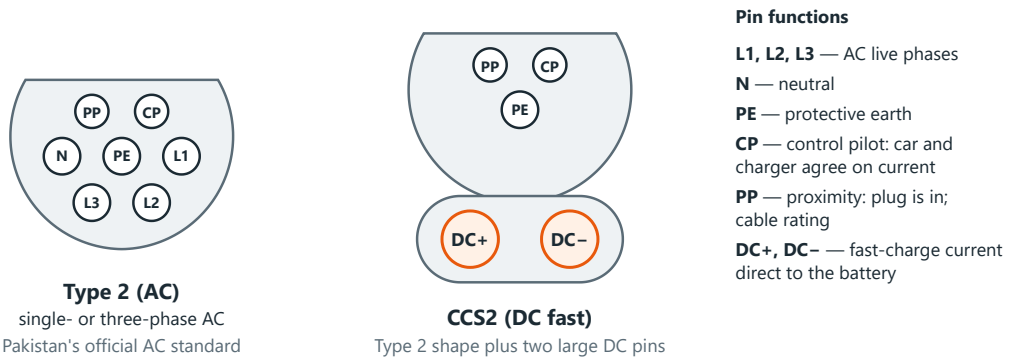


Figure 14.2 The vehicle inlets you will see most on officially imported and locally assembled cars.

Connector	Type	Where you will see it
Type 2	AC	Official Pakistan-market cars (BYD, MG, Deepal and others); all AC wallboxes sold here
CCS2	DC	Official Pakistan-market cars; public fast chargers
GB/T AC and GB/T DC	AC / DC	The Chinese domestic standard. Found on **grey-import** cars brought directly from China. The AC plug looks like Type 2 but does **not** fit. Needs a GB/T charger or a quality adapter.
CHAdeMO	DC	Older Japanese EVs (Nissan Leaf imports)

IN PAKISTAN

NEPRA's rules for public charging specify Type 2 for AC and CCS2 for DC. Many public DC stations also carry a GB/T gun because so many cars come from China. Before advising a customer on a home charger, **look at the car's inlet**. Cheap, unbranded GB/T-to-CCS2 or Type 2 adapters are a common cause of overheated, melted charge ports.

The Chinese GB/T connectors

Training in China is done on GB/T equipment, and cars imported directly from China have GB/T inlets, so you must know these as well.

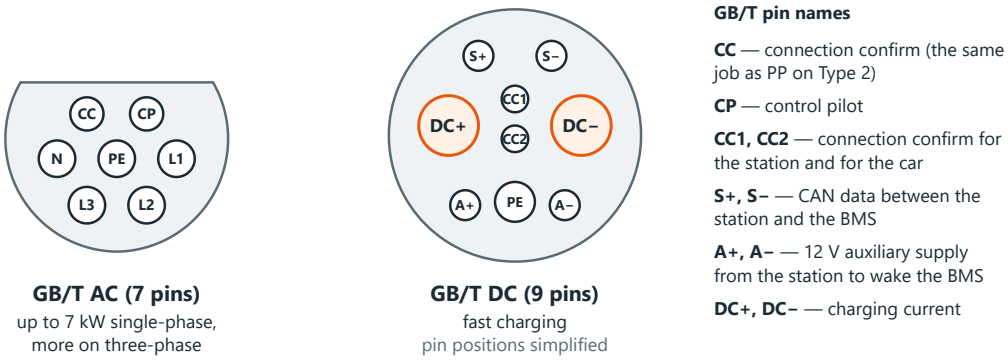


Figure 14.3 The Chinese-standard connectors. The AC plug resembles Type 2 but the two do not fit each other.

Table 14.2 — Which plug, where

Region	AC	DC
China	GB/T AC	GB/T DC
Europe and most export markets, including Pakistan	Type 2	CCS2

Region	AC	DC
North America	Type 1 or NACS	CCS1 or NACS
Japan	Type 1	CHAdeMO

14.3 How the car and charger talk

> **PICTURE IT — KNOCKING ON THE DOOR**
>
> A polite visitor knocks, waits to hear "who is it?", answers, and only walks in when invited. A charger does the same. It "knocks" with a small signal, the car answers, they agree how much current is allowed, the plug locks — and only then does the power come on. Pull the plug and the conversation ends instantly, so the pins are never live in your hand.

No power flows the moment you plug in. First there is a **handshake** on the control pilot (CP) wire:

1. Charger puts +12 V on CP. Not connected.
2. Plug in: a resistor in the car pulls CP down to **9 V**. Charger knows a car is present and starts a 1 kHz signal whose pulse width says "you may take up to X amps".
3. Car is ready: it pulls CP down to **6 V**. The plug is locked.
4. Only now does the charger close its relay and switch on the power.
5. If anything is wrong — earth missing, plug pulled, fault — CP changes and the power is cut in milliseconds.

For **DC charging** the car's BMS and the station exchange digital messages continuously: battery voltage, maximum current, temperature. **The BMS is in charge**; the station supplies exactly the voltage and current the BMS requests.

Measuring the charging signals

When a car will not charge, measuring these low-voltage signals at the inlet tells you whether car and charger are "talking". They are safe, low-voltage tests.

Signal	What to expect
CP to PE (AC charging, Type 2 and GB/T)	+12 V with nothing connected → 9 V when plugged in → 6 V when the car is ready. During charging it is a 1 kHz square wave; the wider the pulse, the more current is allowed.
PP (Type 2) or **CC** (GB/T AC) to PE	A fixed resistor in the plug that tells the car the cable's rating. Roughly: 1.5 kΩ = light-duty cable, 680 Ω = 16–20 A, 220 Ω = 32 A, 100 Ω = 63 A.
CC1 (GB/T DC)	The station watches this point: about 12 V unplugged, 6 V while the latch button is pressed, 4 V when fully connected.
S+ / S− (GB/T DC)	A CAN bus: about 60 Ω between them with power off; messages visible with a CAN tool.

14.4 The charging curve

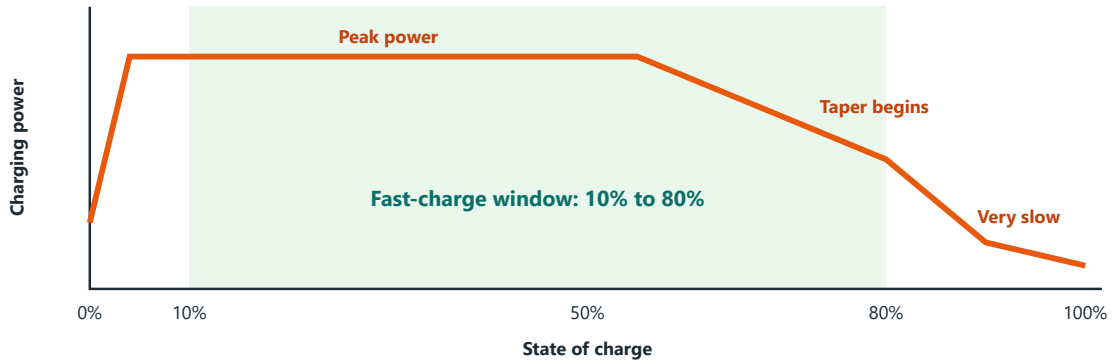


Figure 14.4 A typical DC charging curve. Above about 80% the BMS reduces current to protect the cells.

Like filling a glass of water, you pour fast at first and slow down near the top. Charging from 80% to 100% can take as long as 10% to 80%. Heat and cold also slow charging: a battery at 45 °C or 5 °C will be limited by the BMS.

> **WORKED EXAMPLE — CHARGING TIME**
>
> `Time (h) ≈ Energy needed (kWh) ÷ Charger power (kW)`, plus about 10% for losses.
>
> A 60 kWh battery from 20% to 80% needs 60% × 60 = 36 kWh.
> On a 7 kW wallbox: 36 ÷ 7 × 1.1 ≈ **5.7 hours**.
> On a 60 kW DC charger: 36 ÷ 60 × 1.1 ≈ **40 minutes**.

14.5 Home charging in Pakistan

An EV wallbox is the biggest continuous electrical load most houses will ever have: **32 A for many hours**. Wiring that copes with an air-conditioner cycling on and off may overheat under this.

Table 14.3 — Requirements for a 7 kW (32 A) home charger

Item	Requirement
Sanctioned load	The house connection must support the extra 7 kW. Many single-phase connections do not; a load extension or three-phase meter may be needed.
Dedicated circuit	Its own cable direct from the distribution board. Nothing else on it.
Cable	Copper, at least 6 mm^2; 10 mm^2 for long runs. Good-quality branded cable.
Breaker	40 A MCB for a 32 A charger
Earth-leakage protection	RCD (Type A with 6 mA DC detection in the charger, or Type B)
Earthing	A real, tested earth connection. **Essential.**
Location	Protected from rain and direct sun; cable not lying in water

> **DANGER — EARTHING**
>
> Many Pakistani houses have a poor earth or none at all. A proper EV charger checks the earth and **refuses to charge** without it. Never bypass this check, and never fit a "no-earth" cheat adapter. Without earth, a fault can make the whole car body live. Fix the earth instead.

- **3-pin sockets:** a portable charger draws 8–13 A for hours. Worn sockets and thin extension leads overheat and burn. Use a good 16 A socket on its own circuit; never a multi-plug extension.
- **Voltage swings and load-shedding:** chargers pause when voltage is too low or too high and restart when power returns. Frequent trips point to supply problems, not always the car.
- **Solar:** charging in daylight from rooftop solar is the cheapest energy available. Some wallboxes can follow the solar output.
- **UPS and small generators:** a domestic UPS cannot charge a car. Generators often have poor earthing and unstable voltage; most chargers will reject them.

14.6 Charging faults

Complaint	Check, in this order
Will not start charging	Charging timer or schedule set in the car? Plug fully in and locked? Try another charger and cable. Earth at the socket. Charger error light. Fault codes.
Charges slowly on AC	Current limit set in car or on portable charger. Single-phase supply. Car's on-board charger rating. Hot battery.
Charging stops by itself	Supply voltage dips. Overheating plug or socket (look for brown marks). Battery over-temperature. Charge limit reached.

Complaint	Check, in this order
AC works, DC does not (or the reverse)	AC only → DC contactors or communication. DC only → on-board charger.
Plug stuck in the car	Unlock the car; use the manual release cable (location in the handbook).
Burnt or melted pins	Poor contact, cheap adapter, water or dirt. Replace the inlet **and** the cable; a damaged plug will destroy a new inlet.

SUMMARY

- AC charging: conversion in the car's on-board charger, 2–22 kW. DC charging: conversion in the station, 30–350 kW.
- Pakistan standard: Type 2 (AC) and CCS2 (DC). Grey imports from China use GB/T.
- No power flows until the CP handshake is complete. In DC charging the BMS controls the station.
- Fast-charge from 10% to 80%; it slows sharply after that.
- Home charging needs a dedicated circuit, correct cable and breaker, RCD and a real earth.

Quiz 14

Choose one answer for each question.

1. **During AC charging, where is AC converted to DC?**

   A In the wallbox
   B In the car's on-board charger
   C In the cable
   D In the motor

2. **Which DC fast-charging connector is the official standard in Pakistan?**

   A CHAdeMO
   B Type 1
   C CCS2
   D 3-pin plug

3. **A car with a 7 kW on-board charger is plugged into a 22 kW AC wallbox. How fast does it charge?**

   A 22 kW
   B 15 kW
   C 7 kW
   D 29 kW

4. **How long does a 7 kW charger take to add 35 kWh (ignoring losses)?**

   A 2 hours
   B 5 hours
   C 7 hours
   D 35 hours

5. **Why does DC charging slow down above about 80%?**

   A The station runs out of power
   B The cable gets too long
   C The BMS reduces current to protect the cells as they near full
   D The motor needs to cool

6. **A home charger refuses to charge and shows an earth fault. What is the correct action?**

   A Use an adapter that bypasses the earth check
   B Have the house earthing tested and repaired
   C Use a thinner cable
   D Charge from a UPS instead

7. What does the control pilot (CP) wire do?

A Carries the main charging current

B Earths the car body

C Lets car and charger confirm connection and agree the maximum current

D Locks the steering

PART F · SUPPORTING SYSTEMS

15 Cooling, heating and air-conditioning

Thermal management and A/C in detail: one of the most common sources of EV workshop jobs.

YOU WILL LEARN

- The three things an EV must keep at the right temperature
- How the refrigeration cycle makes cold, stage by stage
- How the electric compressor works and what it needs before it will run
- How to read system pressures and what each pattern means
- The correct way to recover, evacuate and charge refrigerant
- How heat pumps and PTC heaters work
- A step-by-step method for A/C faults

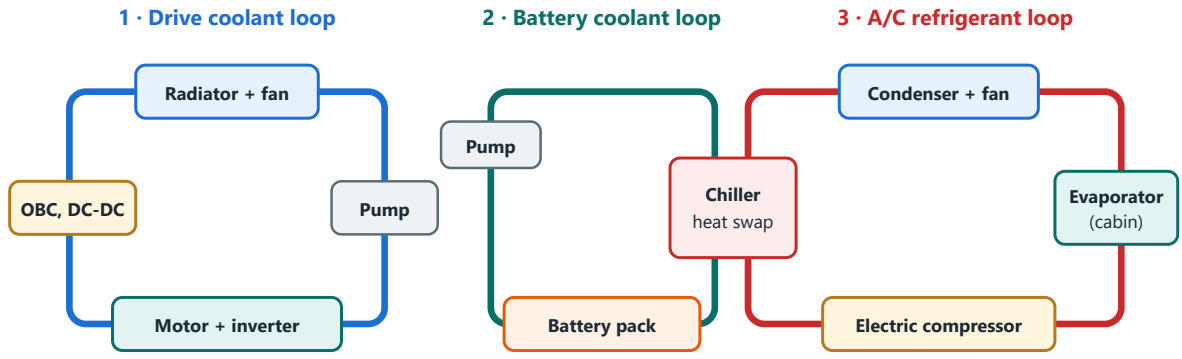
15.1 Why this module matters so much

In a petrol car the A/C is a comfort item. In an EV it is part of the **drive system**, because the same A/C circuit usually cools the battery. That is why A/C work is among the most frequent jobs on electric cars:

- It runs far more hours: while driving, while parked with the cabin pre-cooled, and during fast charging.
- The compressor contains high-voltage electronics that are damaged by moisture and wrong oil.
- There are more valves, sensors and pipes than in a petrol car, so more places to leak or stick.
- When it is weak, the customer notices twice: a warm cabin *and* slow charging or reduced power.

What must be kept at temperature	Ideal range	If too hot
Battery	15–35 °C	Faster ageing, reduced power, slow charging, safety risk
Motor and power electronics	Coolant below about 65 °C	Power reduced; electronics fail
Cabin	Comfortable for people	Unhappy customer, and the A/C uses range

15.2 The loops



The chiller lets cold refrigerant take heat out of the battery coolant. One A/C system cools both people and battery.

Figure 15.1 A typical three-loop layout. Many cars add valves so that loops can be joined or separated as needed.

- **Loop 1** cools the motor, inverter, charger and DC-DC converter through a radiator at the front.
- **Loop 2** circulates coolant through the battery's cold plate. In mild weather it may use a radiator; in hot weather it goes through the **chiller**.
- **Loop 3** is the air-conditioning. A second branch feeds the chiller (or, on direct-cooled packs such as BYD, plates inside the battery).

Each coolant loop has its own **electric pump**. There is no engine-driven water pump and often no thermostat; instead there are electric **multi-way valves**.

15.3 How A/C makes cold: the refrigeration cycle

PICTURE IT — SWEAT

When sweat dries on your skin, you feel cool. The liquid takes heat from your body to turn into vapour. An A/C system does exactly this, on purpose and in a closed loop: a liquid **evaporates** inside the cabin unit and takes heat from the air, then is squeezed and **condensed** back to liquid outside, giving that heat away. A/C does not "make cold". It **carries heat** from inside to outside.

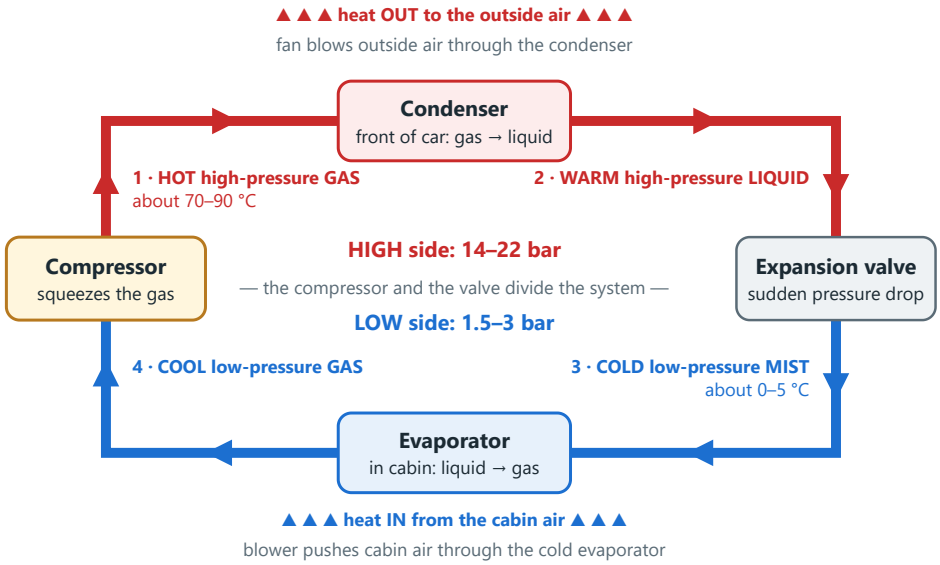


Figure 15.2 The refrigeration cycle. Red is the high-pressure side; blue is the low-pressure side. Pressures shown are typical for R134a on a warm day.

Table 15.1 — The four stages

Stage	What happens	Refrigerant leaves as
1 · Compression	The compressor sucks in cool gas and squeezes it. Squeezing a gas makes it hot (feel a bicycle pump).	Hot, high-pressure gas
2 · Condensation	In the condenser, outside air takes the heat away. The gas turns to liquid.	Warm, high-pressure liquid
3 · Expansion	The liquid is forced through a tiny opening. Pressure falls suddenly and it becomes very cold.	Cold, low-pressure mist
4 · Evaporation	In the evaporator, the cold mist boils to gas by taking heat from the cabin air. The air leaves cold and dry.	Cool, low-pressure gas

KEY IDEA — REMEMBER "C-C-E-E"

Compress → **C**ondense → **E**xpand → **E**vaporate. Hot and high pressure from the compressor to the valve; cold and low pressure from the valve back to the compressor. If you can draw Figure 15.2 from memory, you can diagnose most A/C faults.

15.4 The parts and what goes wrong with each

Table 15.2 — A/C components

Part	Job	Typical faults

Part	Job	Typical faults
Electric compressor	Pumps the refrigerant. Scroll type, with built-in motor and inverter.	Insulation fault, control board failure, seized scroll, noise
Condenser + fan	Gives heat to outside air	Blocked by dust, insects, plastic bags; stone damage and leaks; failed fan
Receiver-drier / desiccant	Stores liquid; a drying bag removes moisture	Saturated after the system has been open; must be renewed after major repairs
Expansion valve (thermal TXV, or electronic EXV)	Meters refrigerant into the evaporator. EVs often have several EXVs: cabin, battery chiller, heat pump.	Blocked by debris or ice; EXV motor sticking
Evaporator + blower	Cools and dries cabin air	Leaks (hard to see), bad smell from mould, blocked water drain, blocked cabin filter
Chiller	Second "evaporator" that cools battery coolant	Internal leak; its EXV sticking
Pressure and temperature sensors	Tell the controller the system state	Wrong reading stops the compressor
Pipes, hoses, O-rings, service ports	Carry refrigerant	The most common leak points

15.5 The electric compressor

Inside one sealed aluminium housing are three things:

1. **The scroll pump** — two spiral plates, one fixed and one moving in a small circle. Gas is trapped between the spirals and squeezed towards the centre. Smooth, quiet, few moving parts.
2. **A three-phase motor** — a small version of the drive motor, cooled by the refrigerant flowing over it.
3. **An inverter and control board** — a small version of the drive inverter, bolted to the housing.

It has two electrical connections: an **orange high-voltage connector** (300–400 V DC, or more), and a **low-voltage connector** carrying 12 V, earth, and a **LIN or CAN** data line. The A/C controller does not switch the compressor on and off like a clutch; it sends a **speed command** (for example 800 to 8,000 rpm) and the compressor reports back its speed, current and any fault.

What the compressor needs before it will run

- A cooling request (cabin A/C on, or the battery asking for cooling)
- High voltage present: car in READY or charging, pre-charge complete, HV fuse good
- HVIL loop closed and no insulation fault
- 12 V supply, earth and LIN/CAN communication good
- Refrigerant pressure inside limits (not empty, not too high)
- Evaporator not near freezing; battery state of charge not too low

If any one is missing, the compressor stays off. So "compressor not running" is usually a missing condition, not a dead compressor.

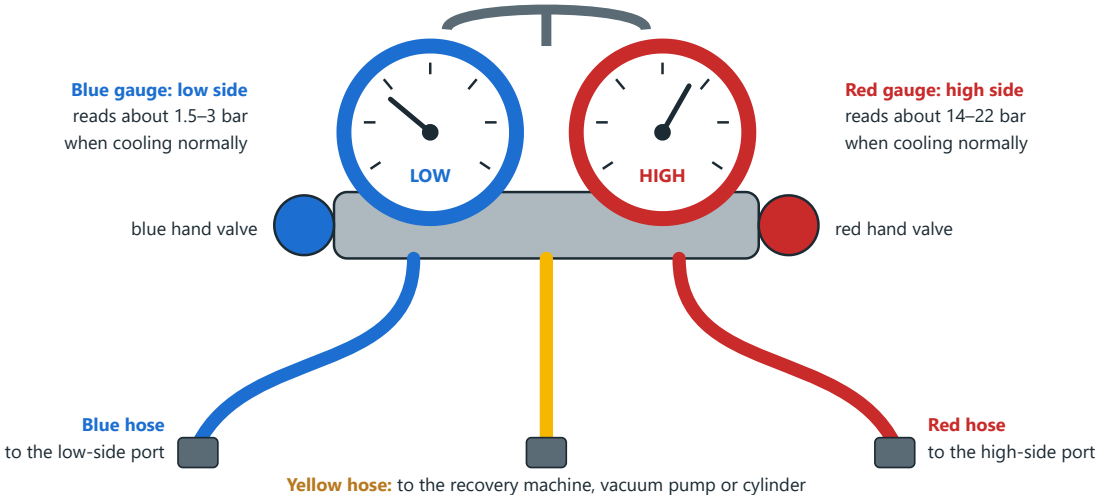
> **DANGER — COMPRESSOR OIL**
>
> In an electric compressor, the oil touches the motor windings. It must be a special **insulating oil** (POE type, such as ND-11 — check the label under the bonnet). Normal **PAG oil** from petrol-car systems conducts electricity. Even a small amount of PAG oil — from a shared A/C machine, hose or oil injector — can cause an **insulation fault** that stops the whole car, and can make the compressor casing live.
>
> Use dedicated or properly flushed A/C equipment for EVs and hybrids, only the specified oil and refrigerant, and only leak-detection dye that is approved for electric compressors.

15.6 Pressures and what they tell you

Pressure units: `1 MPa = 10 bar ≈ 145 psi`. Chinese equipment and manuals use MPa; many gauges show bar and psi.

Table 15.3 — Typical healthy values (R134a, compressor running, warm day)

Reading	Typical value
Low-side pressure	1.5–3 bar (0.15–0.30 MPa)
High-side pressure	14–22 bar (1.4–2.2 MPa); higher in very hot weather
Evaporator outlet temperature	1–5 °C
Air from the centre vent	about 4–10 °C
Both sides, system off and settled	Equal, and matching the outside temperature (about 6–7 bar at 30 °C)



Know your tools A manifold gauge set. Blue is always the low side and red the high side; the two service ports on the car are different sizes so the hoses cannot be swapped.

Most EVs have pressure sensors, so you can read these on the scan tool's live data before connecting any gauges. Always compare with the maker's values for the car and the day's temperature.

Table 15.4 — Reading the pattern

Low side	High side	Likely cause	Think of Figure 15.2
▼ **low**	▼ **low**	Low refrigerant charge (a leak)	Not enough refrigerant anywhere
▲ **high**	▲ **high**	Condenser blocked or fan not working; system overcharged; air in the system	Heat cannot get out at the condenser
▲ **high**	▼ **low**	Compressor weak, or running too slowly	The pump is not creating a pressure difference
▼ **very low** (may go to vacuum)	normal or high	Expansion valve blocked or stuck closed; moisture freezing in the valve; blocked drier	Refrigerant is stuck behind the valve
equal	equal	Compressor not running at all	Go to the electrical checks

15.7 Refrigerant service: recover, evacuate, charge

Refrigerant	Notes
R134a	Common on existing cars. Not flammable. A strong greenhouse gas.
R1234yf	Newer cars. Mildly flammable. Different service ports and its own service machine.

The type and the exact **charge weight in grams** are printed on a label under the bonnet. Never mix types. Never release refrigerant to the air: recover it with a machine.

1. **Recover.** The machine draws out the old refrigerant and measures how much came out. Compare with the label: a low amount confirms a leak.
2. **Repair** the fault. Fit new O-rings with the correct oil. Renew the drier or desiccant if the system has been open for long or the compressor failed.
3. **Evacuate.** Pull a deep vacuum for at least 30 minutes. This boils away moisture and removes air.
4. **Vacuum hold test.** Close the valves and watch for 10–15 minutes. If the vacuum is lost, there is still a leak. Do not charge.
5. **Charge by weight.** Put in exactly the amount on the label. EV systems are very sensitive: too little and too much both cause faults.
6. **Verify.** Run the A/C: check pressures, vent temperature, compressor current and fault codes.

Finding leaks

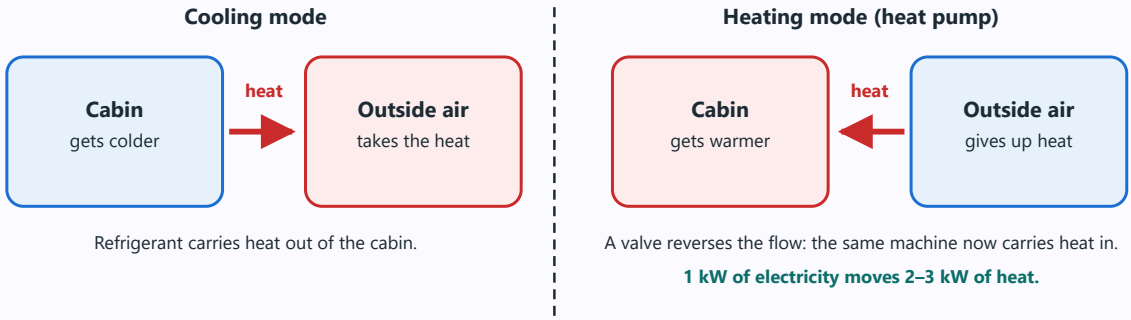
- **Look first:** oily, dusty patches at pipe joints, the condenser and service ports.
- **Electronic leak detector:** a "sniffer" moved slowly under each joint (refrigerant is heavier than air).
- **Nitrogen pressure test:** fill the empty system with dry nitrogen, watch the gauge and use soapy water on joints. Good for larger leaks.
- **UV dye:** only a dye approved for electric compressors.

> **WORKSHOP TIP**
> "Just top it up" is poor practice on any car and worse on an EV. You cannot know how much is in the system without recovering and weighing it, and every top-up through a contaminated machine risks the compressor. Recover, fix the leak, evacuate, charge by weight.

15.8 Heating: heat pump and PTC

There is no hot engine to warm the cabin, so EVs use one or both of these.

- A **PTC heater** — an electric heating element running on high voltage (3–6 kW). Simple, but it takes 1 kW of battery energy to make 1 kW of heat.
- A **heat pump** — the A/C system running in reverse.



A heat pump is an A/C that can run in both directions. A home "inverter A/C" with heating does the same.

Picture it A heat pump does not make heat. It moves heat, which is why it uses so much less energy than a PTC heater.

Extra parts in a heat pump system: a **reversing valve** (or a set of shut-off valves) to change the direction of flow, several **electronic expansion valves**, an inside condenser for heating, and an **accumulator** that protects the

compressor from sucking in liquid. Typical faults: stuck in one mode, an EXV sticking, valve fault codes, refrigerant charge slightly wrong.

PTC heater checks (system isolated): HV fuse, resistance of the element compared with the manual, insulation resistance to the case, and its control signal. A PTC heater that has let coolant inside is a classic cause of insulation faults.

15.9 Coolant

- Use only the **specified coolant**. Battery loops need low-conductivity coolant because it flows close to high-voltage parts.
- Never top up with tap water. Minerals raise conductivity and cause corrosion and scale.
- These systems trap air easily. Many need a **vacuum filling tool** and a **scan-tool bleed routine** that runs the pumps and opens the valves.
- A falling coolant level with no visible leak is serious: coolant may be going **into** the battery pack, motor or chiller.

15.10 Finding A/C and thermal faults

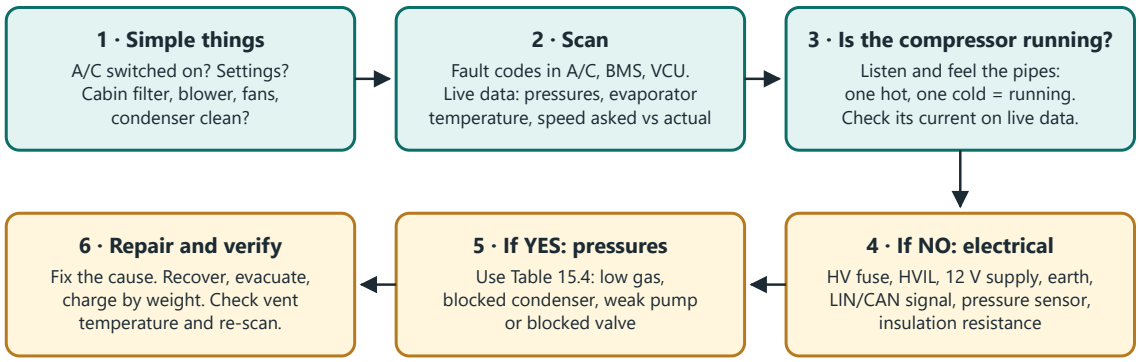


Figure 15.3 A method for "A/C not cooling". The key question is step 3: is the compressor actually running?

Table 15.5 — Thermal symptoms and causes

Symptom	Possible causes
A/C not cold at all	Compressor not running (see step 4); refrigerant lost; blend-door fault sending air through the heater
A/C weak, worse in traffic	Condenser blocked or fan failed; low charge; blocked cabin filter
Cold at first, then warm	Evaporator icing (sensor fault); moisture freezing in the expansion valve
Slow DC charging in summer; reduced power; battery temperature warning	Weak A/C; battery coolant pump; air in the battery loop; chiller valve stuck
"Motor/inverter overheating"	Low coolant; drive-loop pump; radiator blocked; radiator fan
Insulation fault that appears when A/C or heater is switched on	Compressor contaminated with wrong oil or failing internally; PTC heater leaking
Noise from the compressor	Internal wear; loose mounting; liquid refrigerant returning (overcharge)
Bad smell; water on the cabin floor	Mould on the evaporator; blocked evaporator drain; old cabin filter
Coolant loss, no external leak	Internal leak in chiller, battery cold plate or motor — stop and investigate

15.11 When the compressor itself has failed

Failure	How you find it
Motor winding insulation breakdown	Insulation tester from the HV pins and the three motor terminals to the housing; low reading
Control board fault (water, burnt parts)	No communication on LIN/CAN with supplies good; visible corrosion or burning
Scroll seized or worn	High current then shut-down; noise; metal particles in the oil

Replacing a compressor properly:

1. Isolate the HV system and prove dead. Recover the refrigerant.
2. Inspect the old oil and the drier. If there are metal particles, the debris has travelled through the system.
3. **Flush** the pipes and heat exchangers if contaminated; replace the expansion valve and the drier or desiccant.
4. Fit the new compressor with the **correct type and quantity of oil** (new units are often pre-filled; follow the instructions).
5. New O-rings, correct torque. Evacuate, vacuum hold test, charge by weight.
6. Check insulation resistance, clear codes, and verify performance.

Skipping the flush and drier is the usual reason a new compressor fails again within weeks.

IN PAKISTAN

Dust, insects and plastic bags block condensers and radiators quickly here. Washing the front heat exchangers gently with water (from the back towards the front, low pressure) and changing the cabin filter are cheap services that directly protect the battery. Many roadside A/C shops share one machine and one oil between all cars; on an EV that can destroy the compressor. In our climate, **an EV with a weak A/C is an EV with a battery problem coming.**

SUMMARY

- In an EV the A/C cools the battery as well as the cabin, so A/C faults affect charging and power.
- Compress → condense → expand → evaporate. High side hot, low side cold.
- The electric compressor has a scroll pump, motor and inverter; it is speed-controlled over LIN/CAN and needs many conditions before it runs.
- Only insulating POE oil. PAG oil causes insulation faults.
- Read both pressures and match the pattern to the cause.
- Recover, repair, evacuate, hold, charge by weight, verify. Never "top up".
- A heat pump moves heat and is 2–3 times more efficient than a PTC heater.
- Use only specified coolant and bleed with the proper procedure.

Quiz 15

Choose one answer for each question.

1. **What is the ideal working temperature range for a lithium battery?**

A −10 to 5 °C
B 15 to 35 °C
C 50 to 70 °C
D 80 to 100 °C

2. **What is the correct order of the refrigeration cycle?**

A Evaporate, expand, compress, condense
B Compress, condense, expand, evaporate
C Condense, compress, evaporate, expand
D Expand, compress, condense, evaporate

3. **In which part does the refrigerant take heat from the cabin air?**

- **A** Condenser
- **B** Compressor
- **C** Evaporator
- **D** Receiver-drier

4. **Why must normal PAG oil never enter an electric A/C compressor?**

- **A** It is too expensive
- **B** It freezes
- **C** It conducts electricity and causes insulation faults
- **D** It smells bad

5. **Both the low-side and high-side pressures are low with the compressor running. What is the most likely cause?**

- **A** Blocked condenser
- **B** Low refrigerant charge
- **C** Overcharged system
- **D** Failed radiator fan

6. **Both pressures are too high. What should you check first?**

- **A** Cabin filter
- **B** Tyre pressure
- **C** Condenser blockage and condenser fan
- **D** The 12 V battery

7. **How should refrigerant be put into the system?**

- **A** Top up until the pipe feels cold
- **B** After evacuation, by exact weight from the label
- **C** Until the gauge reaches maximum
- **D** Any amount; the system adjusts

8. **The compressor is not running. Which is a possible reason other than a failed compressor?**

- **A** Worn brake pads
- **B** Open HVIL, blown HV fuse, lost LIN/CAN signal or low refrigerant pressure
- **C** Low washer fluid
- **D** Tyre wear

9. **Why is a heat pump more efficient than a PTC heater?**

- **A** It uses petrol
- **B** It runs at 12 V
- **C** It moves heat instead of making it: 1 kW of electricity moves 2–3 kW of heat
- **D** It has no moving parts

10. **An EV fast-charges very slowly on hot days and the A/C is weak. What is the likely link?**

- **A** No link
- **B** The battery cannot be cooled properly, so the BMS limits charging
- **C** The tyres are too soft
- **D** The 12 V battery is overcharged

11. **After a compressor has failed with metal particles in the oil, what must be done before fitting the new one?**

- **A** Nothing extra
- **B** Add more refrigerant
- **C** Flush the system and renew the drier and expansion valve
- **D** Change the cabin filter only

PART F · SUPPORTING SYSTEMS

16 The 12 V system, networks and software

The low-voltage side: small battery, many computers, and the wires that let them talk.

YOU WILL LEARN

- Why the 12 V battery is critical in an EV
- What the main control units do
- How CAN bus works and how to check it
- Why software and calibration are part of modern repair

16.1 The 12 V battery

Every EV and hybrid still has a 12 V battery. It powers the computers, lights, locks, wipers and — most importantly — the **contactors**. If the 12 V battery is flat, the computers cannot wake up and the contactors cannot close. The car is dead, even with a full high-voltage battery.

KEY IDEA

The most common reason an EV or hybrid "will not start" is a weak 12 V battery. Check it first, every time. It is the cheapest and easiest fix in the whole car.

- The 12 V battery in an EV is small because it never has to crank an engine. It is usually AGM lead-acid; some cars (BYD) use a small 12 V lithium battery.
- It is charged by the DC-DC converter only when the car is in READY or charging.
- Typical life is 3–5 years; less in heat.
- **Jump-starting:** you can normally boost an EV's 12 V battery from another car or a jump pack to wake it up. Do not use an EV to jump-start a petrol car unless the handbook allows it; the starter current can damage the DC-DC converter.

IN PAKISTAN

Aftermarket **trackers, alarms, dash cameras and LED accessories** are fitted to almost every car here. Wired badly, they drain the small 12 V battery while the car is parked, or disturb the data network. When you see a repeated flat 12 V battery or strange electrical faults, look for added wiring first.

Finding a parked drain

Connect an ammeter in series with the 12 V negative lead (or use a DC clamp meter). Lock the car and wait 20–30 minutes for the computers to go to sleep. A healthy car draws less than about **30–50 mA**. If it is higher, pull fuses one at a time until the current drops to find the circuit responsible.

16.2 Control units

Unit	Role
VCU — vehicle control unit	The manager. Reads accelerator, brake and gear selector; decides torque; coordinates all the others.
BMS	Battery state, limits and protection.
MCU — motor control unit (inverter)	Delivers the torque the VCU asks for.
OBC / DC-DC controller	Charging and 12 V supply.
TMS / HVAC controller	Pumps, valves, compressor, fans.
ABS / ESC / brake controller	Braking, stability, brake blending.
EPS	Electric power steering.

Unit	Role
BCM — body control module	Lights, locks, windows, wipers.
Gateway	Connects the different networks and the diagnostic socket.

16.3 CAN bus

Instead of a separate wire for every signal, control units share a pair of wires called a **CAN bus** and send each other short digital messages, thousands per second.

> **PICTURE IT — THE GROUP CHAT**
>
> A CAN bus works like a group chat. Every member sees every message. Each message starts with a label saying what it is about ("battery temperature", "wheel speed"), and each control unit simply ignores the ones it does not need. If one member goes silent, the others notice and complain: that is a "lost communication" fault code. If someone shouts nonsense continuously (a short circuit on the wires), nobody can talk at all.

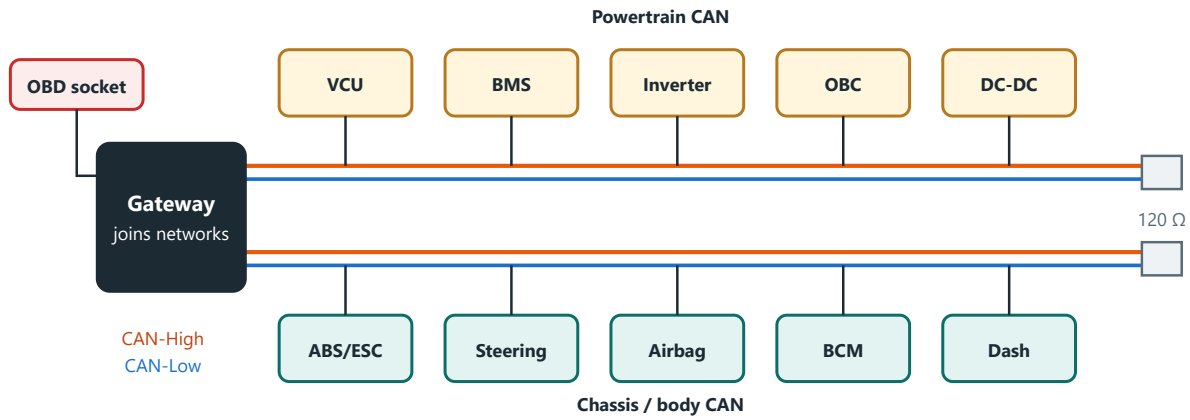


Figure 16.1 A simplified network. Real cars have several CAN buses plus LIN (single-wire, for simple devices) and sometimes Ethernet (for cameras and screens).

Basic CAN checks with a multimeter

Test	Good result	Meaning if wrong
Resistance between CAN-H and CAN-L, 12 V battery disconnected	About **60 Ω**	120 Ω → one terminating resistor or part of the bus is open. Near 0 Ω → wires shorted together.
Voltage CAN-H to earth, ignition on	About 2.5–3.5 V	0 V or 12 V → short to earth or to supply
Voltage CAN-L to earth, ignition on	About 1.5–2.5 V	As above

At the **OBD socket** (16 pins, under the dashboard): pin 16 is battery positive, pins 4 and 5 are earth, pins 6 and 14 are CAN-High and CAN-Low. This is where the scan tool connects.

A fault code beginning with **U** (for example U0100, "lost communication with…") means a control unit is not being heard on the network. Check its power supply, earth and connectors before condemning the unit itself.

16.4 Software, coding and calibration

- **Software updates.** Makers fix faults and add features by updating software, at the dealer or "over the air" (OTA). Some customer complaints are cured only by an update.
- **Coding and pairing.** Many replacement parts (BMS, inverter, charger, even 12 V batteries on some cars) must be registered to the car with a scan tool before they will work.
- **Calibration.** After replacing or disturbing certain parts you must run a learning routine: resolver angle offset after motor work, steering angle sensor after alignment, brake pedal sensor, and so on.

- **Driver assistance (ADAS).** Front cameras and radars for automatic braking and lane keeping need re-calibration after windscreen replacement, bumper repair, suspension or alignment work.

WORKSHOP TIP

Learn to read **wiring diagrams**. They are the map for every electrical fault. Start by tracing simple circuits (a horn, a brake light) until the symbols for fuses, relays, earths, connectors and splices are familiar. This one skill separates a diagnostician from a parts-changer.

16.5 Reading the instrument panel

The dashboard is the car's first report. Always ask the customer which lights and messages appeared, or look for yourself, before connecting the scan tool. Symbols differ a little between makers, but the **colours** follow one rule everywhere.

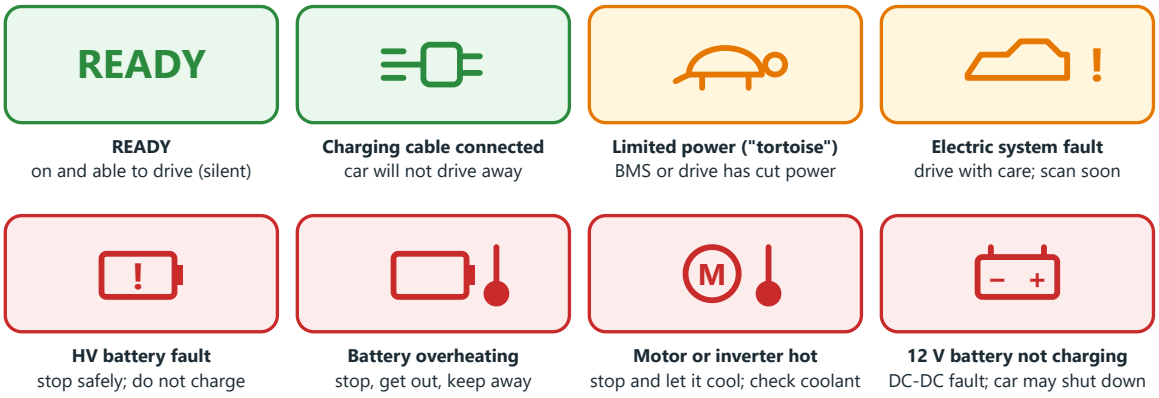


Figure 16.2 Common EV indicators (simplified drawings). **Green** = information. **Amber** = caution, have it checked. **Red** = stop safely now.

- **READY** replaces engine noise. A car showing READY will move if the accelerator is touched.
- The **power gauge** replaces the rev counter: it swings one way for power used and the other way for regeneration.
- Many cars show **SOC in %**, estimated range, and average consumption in kWh per 100 km. Note these on the job card.
- A red light together with a **text message** ("Stop safely", "Check EV system") is the car's own summary of the fault. Write it down word for word.

SUMMARY

- The 12 V battery wakes the car and closes the contactors. Check it first.
- Added accessories often cause drains and network faults.
- Control units share data on CAN bus: two wires, about 60 Ω between them.
- "U" codes are communication faults: check power, earth and wiring first.
- Modern repair includes software updates, coding and calibration.

Quiz 16

Choose one answer for each question.

1. **An EV with a fully charged HV battery is completely dead. What is the most likely cause?**

A Failed motor
B Worn tyres
C Flat 12 V battery
D Low brake fluid

2. What charges the 12 V battery in an EV?

- **A** An alternator
- **B** The DC-DC converter
- **C** The inverter
- **D** Regenerative braking directly

3. What resistance should you measure between CAN-H and CAN-L with the battery disconnected?

- **A** 0 Ω
- **B** About 60 Ω
- **C** 120 Ω
- **D** 1 MΩ

4. Which control unit decides how much torque to request from the motor?

- **A** BCM
- **B** BMS
- **C** VCU
- **D** EPS

5. A fault code U0100 "lost communication" is stored. What do you check first?

- **A** Replace the control unit immediately
- **B** The unit's power supply, earth and connectors
- **C** Tyre pressures
- **D** The motor windings

6. What is an acceptable parked current drain on the 12 V battery after the car has gone to sleep?

- **A** 5 A
- **B** 1 A
- **C** Less than about 50 mA
- **D** 500 mA

PART G · IN THE WORKSHOP

17 Tools and test equipment

What you need, what each tool tells you, and how to use it without hurting yourself or the car.

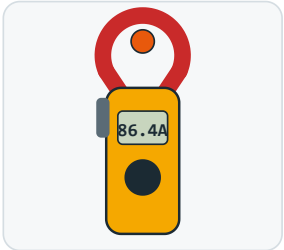
YOU WILL LEARN

- The essential test instruments for EV and hybrid work
- How to measure voltage, current and resistance correctly
- What an insulation tester does
- What a scan tool can show you

17.1 The equipment list

Table 17.1 — Equipment, in the order you should acquire it

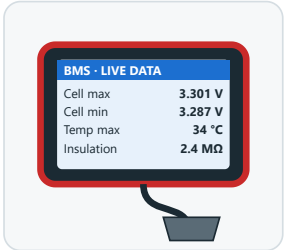
Stage	Equipment	Used for
1 · Basic low-voltage and service work	Digital multimeter, **CAT III 1,000 V**, with matching leads	All voltage, resistance and continuity tests
	Scan tool with EV / hybrid coverage	Fault codes, live data, service functions
	12 V battery tester and charger	The most common fault of all
	DC clamp meter	Current without breaking the circuit; parked drain
	Infrared thermometer	Hot connectors, brakes, radiators
2 · High-voltage after practical training	Class 0 gloves + leather protectors, face shield, safety shoes	Personal protection
	Insulated tool set (1,000 V)	Work near HV parts
	Insulation tester (megohmmeter), 500 V / 1,000 V	Finding insulation faults
	Warning signs, barriers, lock-out box, rescue hook, insulating mat	Safe working area
	Coolant vacuum filler; A/C machine dedicated to POE oil	Thermal system service
3 · Advanced specialist battery and drive work	Battery lifting table	Removing packs (300–600 kg)
	Milliohm meter	Motor windings, bus-bar joints
	Oscilloscope	CAN signals, resolver, sensors
	Module charger / balancer, thermal camera	Battery repair



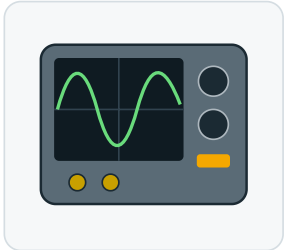
DC clamp meter
jaw closes round ONE cable; reads current without contact



Insulation tester
press TEST: applies 500–1000 V and reads megohms



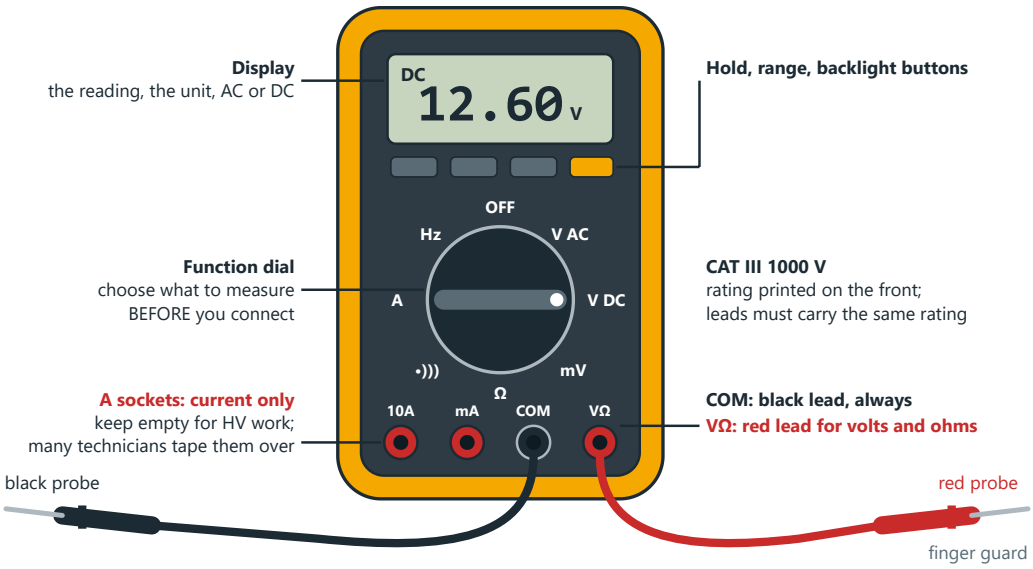
Scan tool
plugs into the OBD socket; codes and live data



Oscilloscope
draws voltage against time; for fast signals

Know your tools The other instruments you will handle in training. Look these up online and in tool shops so that the real ones are familiar when you first hold them.

17.2 The multimeter



Know your tools A typical digital multimeter. Makes differ in colour and layout, but every professional meter has these same parts.

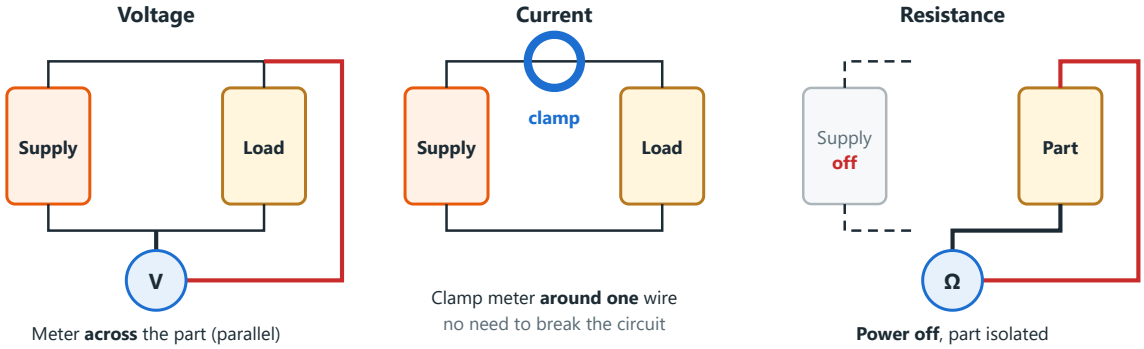


Figure 17.1 The three basic measurements. Red lead to the positive side, black lead to the negative side.

Rules that prevent accidents

- **Check the CAT rating.** For vehicle high voltage use CAT III 1,000 V (or CAT IV 600 V). Cheap unrated meters can explode in your hand on a high-energy circuit.
- **Leads in the right sockets.** If the red lead is left in the amps (A) socket and you touch it across a voltage, the meter is a dead short. On an HV battery this causes an arc flash. Many technicians tape over the amps sockets on their HV meter and use a clamp for current.
- **Select the function before you connect.** DC volts for batteries; AC volts for mains.
- **Never measure resistance on a live circuit.**
- **Inspect the leads** every time. Use leads with shrouded plugs and finger guards.
- **One hand where possible.** Clip one lead on, then probe with one hand, so current cannot pass hand to hand across your chest.

17.3 The insulation tester

A multimeter measures resistance using a few volts. Damaged insulation can look perfect at 3 V and break down at 400 V. An **insulation tester** (megohmmeter, often called a "megger") applies **500 V or 1,000 V DC** at a tiny, safe current and measures resistance in megohms.

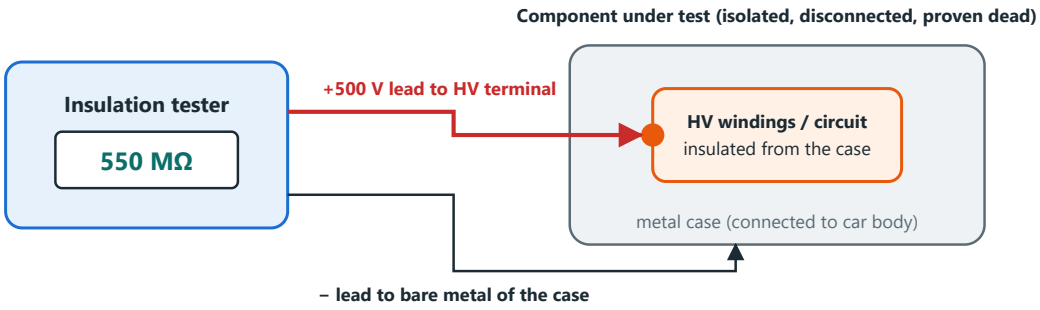


Figure 17.2 Insulation testing a component such as an A/C compressor, motor or cable.

- Used only on components that are **isolated, disconnected and proven dead**, following the manual. Sensitive electronics may need to be unplugged first.
- Healthy parts normally read tens or hundreds of megohms. The service manual gives the minimum acceptable value.
- The tester's output is itself high voltage. Keep hands clear and let it discharge the part before removing the leads.
- **Never** use an insulation tester across battery cells or on live circuits.

17.4 The scan tool

The scan tool plugs into the OBD socket and talks to every control unit. It is your window into the car.

Function	What it gives you
Read / clear DTCs	Fault codes from every module, with status (current, stored, pending)
Freeze-frame data	A snapshot of conditions at the moment a fault was recorded
Live data	Real-time values: cell voltages, temperatures, currents, insulation resistance, contactor states
Actuator tests	Command pumps, fans, valves and contactors on and off to test them
Service functions	Brake service mode, coolant bleed, resolver learning, battery registration, coding

Table 17.2 — Live data worth learning

Value	Healthy sign
Cell voltage maximum / minimum	Small difference at rest (a few hundredths of a volt or less)
Battery temperature maximum / minimum	Within a few degrees of each other
SOC and SOH	SOH appropriate for age
Insulation resistance	High (hundreds of kΩ to MΩ); stable
Pack voltage vs inverter input voltage	Equal when contactors are closed
12 V system voltage in READY	13.5–14.8 V
Motor, inverter and coolant temperatures	Stable under load

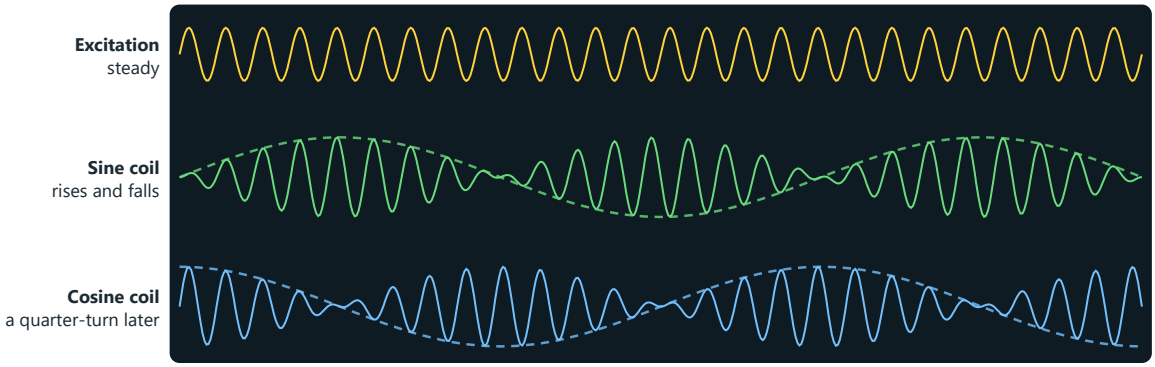
> **WORKSHOP TIP — CHOOSING A SCAN TOOL**
>
> A cheap Bluetooth OBD reader shows only basic engine codes. For EV work you need a tool that reads **all modules** on the brands you service. Check coverage for Chinese brands (BYD, Changan/Deepal, MG, GWM, Chery) before buying; it varies a lot between tool makers and software versions. For older Toyota hybrids, Toyota's own software (Techstream) with a suitable interface is very capable.

17.5 Component tests you will meet in training

These are standard exercises in hands-on courses. All high-voltage parts are tested only when isolated and proven dead.

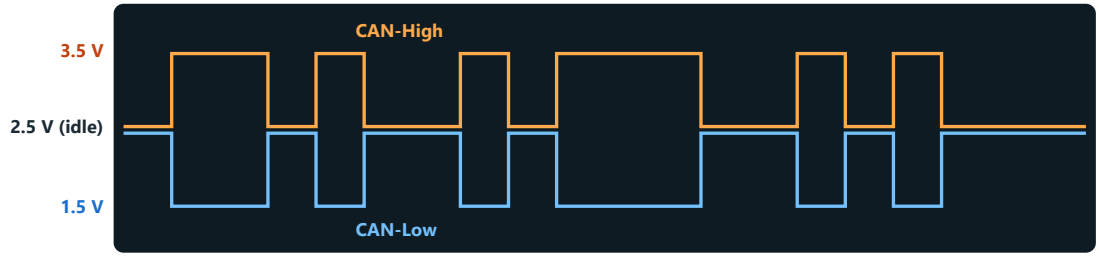
Table 17.3 — Common component tests

Component	Test	Good result
Contactor	Coil resistance across the two small control terminals	Matches the manual (often tens of ohms). Open circuit = failed coil.
	Contact check: energise the coil from 12 V and measure across the two large terminals	Clicks; near 0 Ω closed, open circuit when released. Continuity with the coil off = welded contacts.
Pre-charge resistor	Resistance	Matches the manual (commonly tens of ohms)
HV fuse	Continuity, removed from circuit	Near 0 Ω
HVIL loop	Continuity through the whole loop, then connector by connector	Continuous. The break is at the connector where continuity stops.
Motor windings	Resistance U–V, V–W, W–U with a milliohm meter; insulation to the case	Three equal readings; high insulation resistance
Resolver	Resistance of its three coils (excitation, sine, cosine); signals on an oscilloscope	Coil resistances match the manual; clean signals as in Figure 17.3
Temperature sensor (NTC)	Resistance at a known temperature	Matches the table in the manual; resistance falls as temperature rises
CAN bus	60 Ω test, voltages, oscilloscope picture	As in Module 16 and Figure 17.4



Rotor turning steadily → the dashed outline of each signal follows the rotor angle

Figure 17.3 Resolver signals on an oscilloscope. The inverter compares the sine and cosine outputs to know the exact rotor angle.



A healthy bus: clean, square, and the two lines are mirror images of each other

Figure 17.4 CAN bus on an oscilloscope. If one line is flat, or the two are not mirror images, there is a wiring fault.

Oscilloscope and CAN tools

- A multimeter shows one number. An **oscilloscope** draws voltage against time, so you can see signals that change thousands of times a second: resolver outputs, CAN messages, the control pilot's square wave, sensor signals.
- A **CAN interface** (training centres often use one called PCAN) connects a laptop to the bus and lists every message with its label and data. Decoding messages is advanced work, but simply seeing whether a unit is transmitting at all is a powerful test.

SUMMARY

- Start with a CAT III 1,000 V meter, a capable scan tool and a 12 V battery tester.
- Voltage: across. Current: clamp around one wire. Resistance: power off.
- Wrong lead socket + high voltage = arc flash.
- An insulation tester finds faults a multimeter cannot see.
- Live data — especially cell voltage spread and insulation resistance — is the heart of EV diagnosis.

Quiz 17

Choose one answer for each question.

1. **What rating should a multimeter have for vehicle high-voltage work?**

A CAT I 300 V
B Any rating
C CAT III 1,000 V
D 12 V automotive only

2. **How is a voltmeter connected?**

A In series with the load
B Across (in parallel with) the part being measured
C Around the wire
D To earth only

3. **What happens if you measure voltage with the red lead in the amps socket?**

A The reading is slightly low
B Nothing
C The meter becomes a short circuit — possible arc flash
D The battery charges

4. **Why is an insulation tester needed as well as a multimeter?**

A It is cheaper
B It tests insulation at high voltage, revealing faults that do not show at a few volts
C It measures current
D It reads fault codes

5. **Which scan-tool function shows cell voltages in real time?**

A Clear DTCs
B Freeze frame
C Live data
D Coding

6. **Before using an insulation tester on a component, the component must be…**

A hot
B connected to the battery
C isolated, disconnected and proven dead
D charging

PART G · IN THE WORKSHOP

18 Fault finding

A method for diagnosis, how to read fault codes, and six real-world cases.

YOU WILL LEARN

- A step-by-step diagnostic method
- How fault codes are structured
- How to approach the most common EV and hybrid complaints
- When to stop and refer the job

18.1 The method

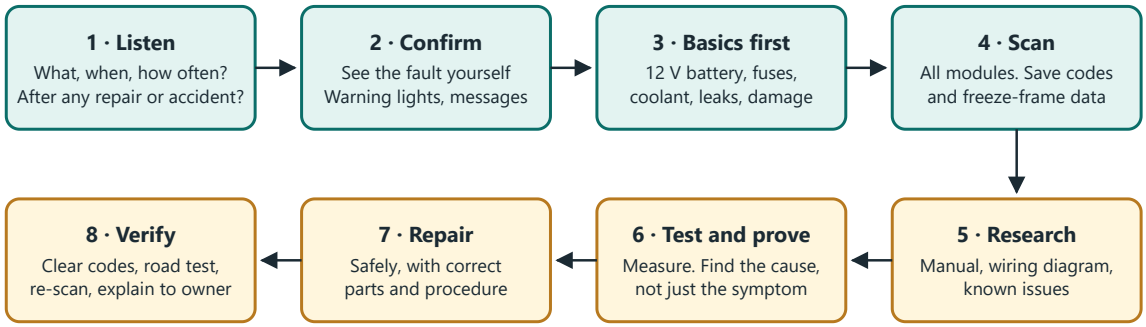


Figure 18.1 The diagnostic method. Skipping steps is how parts get replaced for nothing.

KEY IDEA

A fault code is a clue, not a verdict. The code tells you which circuit or value the computer is unhappy with. It does not tell you which part to replace. "Battery over-temperature" might be the battery — or a blocked fan, a failed pump, low refrigerant, or a faulty sensor. Your job is to prove which.

18.2 Reading fault codes

Standard diagnostic trouble codes (DTCs) have a letter and four characters.

First letter	System	Examples
P	Powertrain (includes HV battery, motor, inverter, charging)	P0A80 — replace hybrid battery pack P0AA6 — HV system insulation fault P0A0A — HV interlock (HVIL) circuit P0A94 — DC-DC converter performance P0AA1 — battery positive contactor stuck closed
B	Body	Airbags, locks, climate
C	Chassis	ABS, steering, brake system
U	Network	U0100 — lost communication with a control unit

Codes beginning **P0** are standard across makers; many hybrid and EV faults sit in the **P0A–P0E** range. Codes beginning **P1**, and many others, are specific to the maker; you need that maker's information to interpret them.

18.3 Six common cases

Case 1 — "The car is completely dead / will not go to READY"

1. Measure the 12 V battery. Below about 12.0 V → charge or replace, then retest.

2. Check main 12 V fuses and battery terminals.
3. Brake pedal switch (the car needs to see the pedal pressed), key detection, gear in P.
4. Scan for codes: HVIL, pre-charge failure, contactor, insulation fault, BMS.
5. Has anyone worked on the car recently? A connector not fully seated opens the HVIL.

Case 2 — "Insulation fault" / "leakage" warning

1. Ask **when** it appears: in rain? after washing? when the A/C or heater is switched on? only when charging?
2. Read the live insulation resistance value. Watch it while switching HV consumers on and off (A/C, heater).
3. With the system safely isolated, disconnect HV components **one at a time** and insulation-test each: A/C compressor, PTC heater, on-board charger, motor, cables.
4. The reading that is low identifies the faulty part or cable.
5. If everything outside the pack tests good, the fault is inside the battery — a specialist job.

Frequent causes: wrong oil in the A/C compressor, water in a connector or the charge port, coolant leaking into a component, rodent-damaged cables.

Case 3 — "Range has dropped"

1. Ask about recent changes: season, route, speed, new tyres, roof rack.
2. Tyre pressures, brake drag (spin each wheel; feel for hot wheels after a drive), alignment.
3. Scan: SOH, cell voltage spread, battery temperatures.
4. If SOH is good and the cells are balanced, the battery is healthy. Explain consumption and conditions.

Case 4 — "It will not charge"

Follow the table in Module 14.6. The quickest test is **substitution**: try a different charger and cable. If the car charges elsewhere, the fault is in the customer's equipment or house supply, not the car.

Case 5 — "Low power / tortoise light, mostly in summer"

1. Scan for temperature-related codes and look at live temperatures for battery, motor and inverter.
2. Coolant levels in each reservoir. Pumps running? Radiator and condenser clean? Fans working?
3. A/C performance (because the battery depends on it).
4. Hybrids: battery fan and ducts.

Case 6 — Fault appears straight after another repair

Go back to what was touched. An unplugged or half-latched connector, a pinched wire, an earth strap left off, air left in a coolant loop, or a part that needed coding. The most recent work is the most likely cause.

18.4 Know your limits

STOP AND REFER THE JOB WHEN…

- the fault is **inside the battery pack** and you are not trained and equipped for pack work;
- the car has **crash or flood damage** to HV components;
- you see signs of **overheating, swelling, leaking or smell** from the battery;
- you do not have the **service information** for that model;
- the repair needs a tool or software you do not have.

Saying "this is beyond what I can do safely; here is who can help" is professional. Guessing with 400 volts is not.

SUMMARY

- Listen, confirm, basics, scan, research, test, repair, verify.
- Codes are clues. Prove the cause by measurement.
- P = powertrain, B = body, C = chassis, U = network.
- Always start with the 12 V battery and a visual check.
- For insulation faults, isolate and test one component at a time.
- Know when to stop.

Quiz 18

Choose one answer for each question.

1. **What is the first step of the diagnostic method?**

   A Replace the most likely part
   B Clear all codes
   C Listen to the customer
   D Remove the battery

2. **What does a DTC starting with "U" indicate?**

   A Engine fault
   B Brake fault
   C Network communication fault
   D Airbag fault

3. **A code says "battery over-temperature". What should you do?**

   A Replace the battery
   B Check cooling fan, pump, coolant, A/C and sensors to find the real cause
   C Ignore it
   D Disconnect the temperature sensor

4. **An insulation fault appears only when the A/C is switched on. Which component is the first suspect?**

   A Motor
   B On-board charger
   C A/C compressor
   D 12 V battery

5. **A car will not charge at home but charges normally at a public charger. Where is the fault most likely?**

   A In the car's battery
   B In the home charger, cable or house supply
   C In the motor
   D In the tyres

6. **A warning light comes on the day after another workshop replaced a coolant pump. What do you check first?**

   A The oldest parts of the car
   B Everything that was touched during that repair
   C The paintwork
   D The spare wheel

7. **When should you refer a job to a specialist?**

   A Never
   B Only if the customer insists
   C When the fault is inside the pack, or you lack the training, information or tools to do it safely
   D When it is nearly closing time

PART G · IN THE WORKSHOP

19 Maintenance and inspection

EVs need less servicing than petrol cars — but not none. What to check, and when.

YOU WILL LEARN

- A typical EV and hybrid service schedule
- How to carry out a battery health check
- How to inspect a used EV before purchase

19.1 A typical service schedule

Always follow the maker's schedule for the model. This table shows what is typical.

Table 19.1 — Typical maintenance for a battery electric car

Interval	Work
Every service (12 months or 15,000–20,000 km)	Scan all modules for fault codes · Battery health report (SOH, cell spread) · 12 V battery test · Tyres: pressure, wear, rotate · Brakes: inspect, clean and lubricate sliders · Suspension and steering joints · Under-body and battery case inspection · Coolant levels (all reservoirs) · Check HV cables and connectors visually for damage (do not disconnect) · Charge port: clean, check pins and seal · Cabin filter · Clean radiator and condenser · Wipers, lights, washer fluid · Software update check
Every 2 years	Brake fluid change · A/C performance check · Door and charge-flap seals and drains
Every 4–5 years (or as specified)	Coolant change with correct fluid and bleed procedure · Reduction gear oil (if specified) · 12 V battery likely due for replacement · A/C desiccant where specified

Hybrids and plug-in hybrids: add everything a petrol engine needs — engine oil and filter (by time as well as distance), air filter, spark plugs, engine coolant, fuel system, exhaust — plus the hybrid battery fan and filter, and the transmission (DHT / e-CVT) fluid.

19.2 The battery health check

This is a service customers value and will pay for. With a capable scan tool:

1. Record **SOH** and odometer.
2. Record **highest and lowest cell voltage** at rest, and the difference.
3. Record **highest and lowest battery temperature**.
4. Record the **insulation resistance**.
5. Note any stored battery or charging fault codes.
6. If possible, repeat the cell voltage reading during a short acceleration: a weak cell drops much more than the others.
7. Give the customer a simple written report, and keep a copy. Next year you can show the trend.

19.3 Inspecting a used EV or hybrid

The used market for EVs is beginning. Buyers are nervous about batteries and will want an independent check.

Table 19.2 — Pre-purchase inspection checklist

Area	Check
Documents	Official import or local assembly? Warranty remaining and transferable? Service history.
Charging	Inlet type (Type 2 / CCS2 or GB/T). Test AC charging; DC if possible. Condition of pins. Cables supplied.
Battery	Full health check as above. Compare displayed range at 100% with the original figure.
Under-body	Battery case dents, scrapes, corrosion, water marks, non-original sealant (sign of opening).

Area	Check
Accident and flood signs	Paint differences, panel gaps, silt or rust under carpets and in connectors, musty smell.
Cooling and A/C	Coolant levels and colour; A/C cold at idle; fans working.
Fault codes	Full scan. Be suspicious if all codes were cleared just before sale (readiness data will show).
Road test	Smooth drive, no whine or clunk, regeneration working, straight braking, no warning lights.
12 V and accessories	12 V battery health; quality of any added wiring.

19.4 Storage and customer advice

- Leaving a car for weeks: store at 40–60% charge, in shade. Do not leave it plugged into a poor-quality supply.
- The 12 V battery can go flat during long storage. Advise a periodic wake-up drive or a 12 V maintainer if the maker allows.
- Keep records of every job: date, kilometres, readings, parts. Good records build customer trust and protect you if there is a dispute.

SUMMARY

- EV service is inspection-heavy: scan, battery report, 12 V, tyres, brakes, coolant, under-body.
- Brake fluid every 2 years; coolant and gear oil as specified.
- Hybrids need all petrol-engine servicing as well.
- A written battery health report is valuable to owners and buyers.

Quiz 19

Choose one answer for each question.

1. How often should brake fluid typically be changed on an EV?

A Never
B About every 2 years
C Every 3 months
D Only when the pads are changed

2. Which of these belongs in a battery health report?

A Tyre brand
B SOH and the difference between highest and lowest cell voltage
C Radio stations
D Paint colour

3. What extra servicing does a plug-in hybrid need compared with a BEV?

A None
B Only tyres
C All the petrol-engine items such as oil, filters and spark plugs
D A second charge port

4. On a used EV, non-original sealant around the battery case suggests…

A a factory upgrade
B the pack may have been opened or repaired
C the car is new
D nothing important

5. What is the best state of charge for storing an EV for several weeks?

A 0%
B 40–60%
C 100%
D It makes no difference

PART G · IN THE WORKSHOP

20 Electric bikes and rickshaws

The biggest EV market in Pakistan by numbers, and the easiest place to start earning.

YOU WILL LEARN

- The parts of an electric two- or three-wheeler
- The battery options and how to compare them
- How to diagnose the most common faults
- The safety points specific to small EVs

20.1 Why this market matters

More than 200,000 electric bikes and rickshaws were produced in Pakistan in 2025–26, over three times the year before. Government subsidies under the PAVE scheme support models with lithium batteries. These vehicles are simple, the tools are cheap, and the customers are everywhere. For a new technician, this is the fastest route to real experience and income, and every principle is the same as in a car — just smaller.

20.2 The system

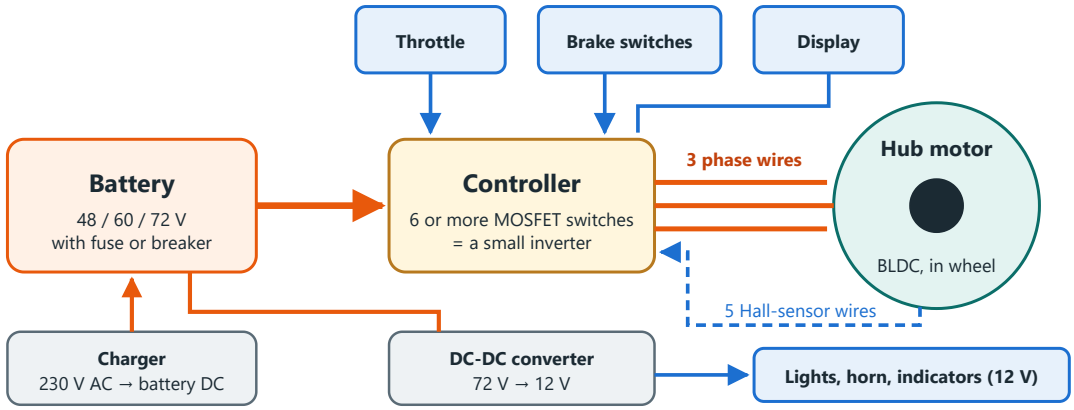


Figure 20.1 An electric motorcycle or scooter. A rickshaw is the same with a larger battery and a motor driving a rear axle with differential.

Part	Notes
Battery	48, 60 or 72 V. Capacity 20–50 Ah on bikes; more on rickshaws (5–9 kWh).
Controller	A small inverter. Rated by voltage and maximum current (for example 72 V, 50 A). Has a low-voltage cut-off to protect the battery.
Hub motor	BLDC, 1–3 kW, built into the wheel. Three thick phase wires and five thin Hall-sensor wires (5 V, earth, three signals).
Throttle	A Hall sensor giving roughly 0.8 V (closed) to 4.2 V (full) on a 5 V supply.
Brake switches	Tell the controller to cut power when a brake lever is pulled.
DC-DC converter	Steps battery voltage down to 12 V for lights and horn.
Charger	External. **Must match the battery's voltage and chemistry.**
BMS	Inside every lithium pack. Lead-acid packs have none.

DANGER — "SMALL" DOES NOT MEAN SAFE

- A "72 V" pack reaches **84–88 V** when fully charged or charging. That is above the 60 V high-voltage limit. A "60 V" pack reaches about 70 V.
- These batteries can deliver **hundreds of amps** into a short circuit. A spanner across the terminals becomes red-hot instantly. Burns and fires are far more common on bikes than shocks.
- Remove rings and watches. Disconnect the battery (or switch off its breaker) before working. Insulate the terminals.

20.3 Battery choices

Table 20.1 — E-bike battery types

	Lead-acid ("graphene")	Lithium (LFP or NMC)
Build	Five or six 12 V blocks in series (60 V or 72 V)	One sealed pack with built-in BMS
Weight for the same energy	About 3 times heavier	Light
Life	300–500 cycles (1–2 years of daily use)	1,000–3,000 cycles
Purchase cost	Low	Higher (lower cost per km over its life)
Charging time	6–8 hours	2–5 hours
Main risks	Acid, heavy, short life if deeply discharged	Fire if poor quality, damaged or wrongly charged

"**Graphene battery**" in this market means a lead-acid battery with a carbon additive in the plates. It is a little better than plain lead-acid, but it is still lead-acid.

WORKED EXAMPLE — RANGE

Battery 72 V, 32 Ah → 72 × 32 = 2,304 Wh. Consumption about 35 Wh/km.

Range = 2,304 ÷ 35 ≈ **66 km** when new (lead-acid gives less in practice because it should not be fully discharged).

20.4 Common faults

Complaint	Checks
No drive, display on	Brake switch stuck on (unplug to test) · Side-stand switch · Throttle signal voltage (0.8 → 4.2 V) · Hall-sensor signals (each should switch between about 0 V and 5 V as the wheel is turned slowly by hand) · Controller
Completely dead	Battery voltage · Main fuse / breaker · Key switch · Battery connector burnt · Lithium BMS has cut off (very low cell)
Wheel hard to turn by hand	Shorted controller MOSFET or phase wires touching. Unplug the three phase wires: if the wheel now turns freely, the controller is at fault; if not, the motor.
Short range	Lead-acid: measure each 12 V block after charging *and under load*. One weak block (drops far below the others) ruins the whole string. Lithium: capacity fade or cell imbalance. Also tyre pressure and dragging brakes.
Jerky or noisy motor	One Hall sensor failed · Phase connector loose or burnt · Water in the motor
No lights or horn but it drives	DC-DC converter or its fuse
Will not charge	Charger output voltage · Charge port and fuse · Lithium BMS charge protection

IN PAKISTAN

Water is the great enemy of cheap e-bikes. Riding through monsoon water floods hub motors, controllers and connectors. Opening, drying, cleaning and re-sealing a hub motor, and replacing rusted bearings and Hall sensors, is steady work every rainy season.

20.5 Lead-acid to lithium conversions

Many owners will ask you to replace worn lead-acid blocks with a lithium pack. It can be a good upgrade if done properly:

- Choose a pack with a real BMS, from a supplier who states the chemistry, cell count and maximum current.
- The BMS continuous current rating must exceed the controller's maximum current.
- **Replace the charger.** A lead-acid charger must never be used on a lithium pack. Use the charger designed for that pack's chemistry and cell count.
- Check the controller's low-voltage cut-off suits the new pack.
- Mount the pack firmly, protected from water and impact, with a fuse close to the battery.

DANGER — CHARGING FIRES

Most e-bike battery fires happen **while charging**, usually indoors at night, with a cheap pack, a wrong charger, or a damaged battery. Advise every customer: use only the correct charger; charge in an open, ventilated place away from the exit and from anything that burns; do not charge a swollen, dropped or water-damaged battery; do not leave it charging all night unattended.

SUMMARY

- Battery → controller → BLDC hub motor, with throttle and brake switches as inputs.
- "72 V" really means up to 88 V and very high short-circuit current.
- Lead-acid ("graphene") is cheap and short-lived; lithium is lighter and lasts longer but must be good quality.
- Test lead-acid blocks individually under load.
- Match the charger to the battery. Most fires happen during charging.

Quiz 20

Choose one answer for each question.

1. What type of motor is used in most electric scooter wheels?

A Induction | **B** BLDC hub motor | **C** Petrol | **D** Stepper

2. A "72 V" battery can reach about what voltage when charging?

A 48 V | **B** 60 V | **C** 72 V exactly | **D** 84–88 V

3. In Pakistan's e-bike market, a "graphene battery" is…

A a lithium battery | **B** a lead-acid battery with a carbon additive | **C** a sodium battery | **D** a fuel cell

4. An e-bike has a 60 V, 30 Ah battery and uses 30 Wh/km. What is its approximate range?

A 30 km | **B** 60 km | **C** 90 km | **D** 180 km

5. **A lead-acid e-bike has short range. What is the correct test?**

A Measure only the total pack voltage

B Measure each 12 V block under load to find a weak one

C Replace the motor

D Change the tyres

6. **The wheel is hard to turn by hand, but turns freely when the three phase wires are unplugged. What is faulty?**

A The motor bearings

B The tyre

C The controller (shorted MOSFET)

D The throttle

7. **When converting from lead-acid to lithium, what must always be changed as well?**

A The seat

B The tyres

C The charger

D The horn

PART G · IN THE WORKSHOP

21 Your path forward

From this book to hands-on training, and from training to a career.

YOU WILL LEARN

- The levels of EV technician skill
- How to prepare for practical training
- Where the work will come from
- The habits of a professional

21.1 The skill ladder

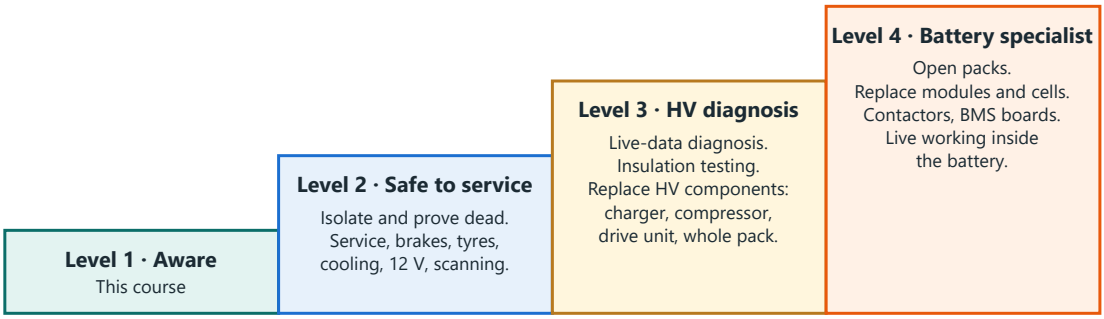


Figure 21.1 The skill ladder. Most everyday work — and most income — is at Levels 2 and 3.

21.2 Preparing for hands-on training

Before you travel for practical training, you should be able to tick every box below **without looking at the book**.

Table 21.1 — Self-assessment: "I can…"

☐	Skill	Module
☐	Use Ohm's law and the power formula to solve simple problems	2
☐	Explain the difference between kW and kWh	2
☐	State the high-voltage limit and list the three ways HV can injure	3
☐	Write out the safe isolation procedure from memory	3
☐	Explain "prove, test, prove"	3
☐	Describe MHEV, HEV, PHEV, REEV and BEV	4
☐	Draw the main components of a BEV and the energy flow between them	5
☐	Compare LFP and NMC cells	6
☐	Calculate pack voltage and energy from cell data	7
☐	Explain what the BMS, contactors and pre-charge circuit do	8
☐	Sketch a motor's torque–speed curve and explain it	9
☐	Explain what the inverter, DC-DC converter and on-board charger each do	10
☐	Calculate range from battery energy and consumption	11
☐	Explain regenerative braking and when it is limited	12
☐	Describe the three modes of a DHT plug-in hybrid	13

☐	Skill	Module
☐	Identify Type 2, CCS2 and GB/T, and estimate charging time	14
☐	Draw the refrigeration cycle, and explain why A/C compressor oil matters	15
☐	Explain why the 12 V battery is checked first	16
☐	Describe the eight-step diagnostic method	18

Practise now, safely, on low voltage

- Buy a decent multimeter and practise on 12 V: battery voltage, charging voltage, fuse checks, voltage drop across a switch, continuity of a wire.
- Build simple circuits with a 12 V battery, bulbs, switches and a relay.
- Read any wiring diagram you can find.
- Watch a scan tool being used on any car, and learn to navigate its menus.
- Take a basic first-aid and CPR class.

21.3 Words you will see in Chinese

Labels, scan-tool screens and service documents on Chinese cars are sometimes not translated. These are worth recognising.

Chinese	Pronunciation	Meaning	Chinese	Pronunciation	Meaning
高压	gāo yā	High voltage	危险	wēi xiǎn	Danger
电池	diàn chí	Battery	电机	diàn jī	Motor
充电	chōng diàn	Charging	故障	gù zhàng	Fault
电压	diàn yā	Voltage	电流	diàn liú	Current
温度	wēn dù	Temperature	绝缘	jué yuán	Insulation
禁止	jìn zhǐ	Prohibited	断开	duàn kāi	Disconnect
三电	sān diàn	Three-electric system	电控	diàn kòng	Electronic control
压缩机	yā suō jī	Compressor	空调	kōng tiáo	Air-conditioning
冷却液	lěng què yè	Coolant	逆变器	nì biàn qì	Inverter

21.4 Where the work will come from

Service	Skill level	Why there is demand
E-bike and rickshaw repair, battery testing, lithium conversions	1–2	Huge numbers; low-cost tools
Hybrid battery testing, fan cleaning, 12 V batteries, inverter coolant pumps	2–3	Large fleet of ageing Japanese hybrids
EV servicing: brakes, tyres, alignment, coolant, A/C (with correct equipment)	2	Owners leaving dealer servicing after warranty
Battery health reports and pre-purchase inspections	2–3	Growing used-EV market; nervous buyers
Charging fault diagnosis; home charger checks with a qualified electrician	2–3	Poor house wiring and earthing
HV component replacement and insulation fault-finding	3	High dealer prices
Battery pack repair	4	Highest value; needs most training and equipment

21.5 Habits of a professional

1. **Safety first, every time.** No shortcut is worth your life. Stop anyone who is working unsafely, including your seniors.
2. **Measure, don't guess.** Diagnose before you replace.
3. **Be honest.** Tell the customer what you found, what you are sure of, and what you are not. Never invent a fault. Reputation is your real capital.
4. **Keep records.** Readings, photos, parts fitted.
5. **Keep it clean.** A tidy bay, clean tools, covered seats. HV connectors must be spotless and dry.
6. **Keep learning.** This technology changes every year. Read service information, follow training, ask questions.
7. **Know your limits** — and say so.

A FINAL WORD

You are entering this trade at the right moment. In ten years, electric and hybrid vehicles will be ordinary in Pakistan, and the people who learned early — properly and safely — will be the ones training others and running the workshops. Study this book until you own it. Then go and get your hands dirty, with your gloves on.

Final assessment — 20 mixed questions

Do this only after completing all modules. Pass mark: 16 out of 20, and every safety question correct.

1. Above what DC voltage must high-voltage precautions be used?

- **A** 12 V
- **B** 48 V
- **C** 60 V
- **D** 400 V

2. A 400 V system delivers 200 A. What is the power?

- **A** 2 kW
- **B** 8 kW
- **C** 80 kW
- **D** 800 kW

3. Which unit describes how far a car can go?

- **A** kW
- **B** Nm
- **C** kWh
- **D** rpm

4. What is the last step before starting work on an isolated HV system?

- **A** Remove gloves
- **B** Prove dead with a meter that has been checked before and after
- **C** Reconnect the 12 V battery
- **D** Start the engine

5. Which vehicle type has a charge port and an engine that only generates electricity?

- **A** HEV
- **B** MHEV
- **C** REEV
- **D** BEV

6. 108 LFP cells (3.2 V each) in series give a nominal pack voltage of about…

- **A** 108 V
- **B** 346 V
- **C** 400 V
- **D** 800 V

7. Which chemistry is generally the most resistant to thermal runaway?

- **A** NMC
- **B** LFP
- **C** Both the same
- **D** Neither can burn

8. The cell voltage spread on a pack is 0.25 V at rest. This suggests…

- **A** a healthy pack
- **B** a weak or failing cell or module
- **C** a flat 12 V battery
- **D** a worn tyre

9. What happens during pre-charge?

- **A** The 12 V battery is charged
- **B** The inverter capacitor is charged gently through a resistor
- **C** The cabin is cooled
- **D** The cells are balanced

10. In the constant-power region of a motor, as speed rises the torque…

- **A** rises
- **B** stays the same
- **C** falls
- **D** becomes zero

11. Which part converts DC to three-phase AC?

- **A** DC-DC converter
- **B** On-board charger
- **C** Inverter
- **D** Contactor

12. A car has 45 kWh usable and uses 150 Wh/km. Range is…

- **A** 150 km
- **B** 300 km
- **C** 450 km
- **D** 675 km

13. Regenerative braking is weakest when…

- **A** the battery is at 100%
- **B** the battery is at 50%
- **C** the road is dry
- **D** the lights are on

14. In a Toyota hybrid, which unit drives the wheels electrically?

- **A** MG1
- **B** MG2
- **C** The starter
- **D** The alternator

15. Which connector pair is Pakistan's official standard?

- **A** Type 1 and CHAdeMO
- **B** Type 2 and CCS2
- **C** GB/T only
- **D** 3-pin only

16. What must never be used in an electric A/C compressor?

- **A** POE oil
- **B** The specified refrigerant
- **C** PAG oil
- **D** A clean hose

17. An EV is dead with a full HV battery. First check?

- **A** Motor
- **B** Inverter
- **C** 12 V battery
- **D** Tyres

18. You are asked to jack up an EV. Where must the jack not go?

- **A** Sill jacking point
- **B** Under the battery pack
- **C** Subframe point in manual
- **D** On a hard floor

19. A colleague is being shocked. What is your first action?

- **A** Pull them off by hand
- **B** Do not touch; isolate the power or use an insulated hook
- **C** Throw water
- **D** Take a photo

20. A fault code tells you…

- **A** exactly which part to replace
- **B** which circuit or value the computer is unhappy with — a clue to test
- **C** the price of the repair
- **D** nothing useful

Appendix A — Formula sheet

To find	Formula	Units
Voltage	`V = I × R`	V, A, Ω
Current	`I = V ÷ R` or `I = P ÷ V`	A
Power	`P = V × I`	W (÷1,000 for kW)
Heat loss in a cable	`P = I² × R`	W
Energy	`E = P × t`	kWh = kW × hours
Battery energy	`Wh = V × Ah`	Wh (÷1,000 for kWh)
Pack voltage	`cells in series × cell V`	V
Pack capacity	`cells in parallel × cell Ah`	Ah
C-rate	`power (kW) ÷ energy (kWh)`	C
State of health	`capacity now ÷ capacity new × 100`	%
Motor power	`kW = Nm × rpm ÷ 9,550`	kW
Wheel torque	`motor Nm × gear ratio`	Nm
Range	`usable Wh ÷ Wh per km`	km
Charging time	`kWh needed ÷ charger kW × 1.1`	hours
Horsepower	`1 hp ≈ 0.746 kW` · `1 kW ≈ 1.34 hp`	

Numbers to remember

High-voltage limit	60 V DC / 30 V AC	CAN bus resistance	about 60 Ω
LFP cell	3.2 V nominal (2.5–3.65 V)	12 V system in READY	13.5–14.8 V
NMC cell	3.6–3.7 V nominal (3.0–4.2 V)	Parked 12 V drain	under 50 mA
NiMH cell	1.2 V	Battery ideal temperature	15–35 °C
Pakistan mains	230 V, 50 Hz	Fast-charge window	10–80%
Gloves	Class 0 (1,000 V AC)	Meter	CAT III 1,000 V

Appendix B — Abbreviations

Short	Full	Short	Full
AC	Alternating current	LFP	Lithium iron phosphate
ADAS	Advanced driver-assistance systems	MCU	Motor control unit
AGM	Absorbent glass mat (12 V battery)	MG1 / MG2	Motor-generator 1 / 2
BEV	Battery electric vehicle	MHEV	Mild hybrid electric vehicle
BLDC	Brushless DC (motor)	MSD	Manual service disconnect
BMS	Battery management system	NEV	New energy vehicle
CAN	Controller area network	NiMH	Nickel-metal hydride
CCS	Combined charging system	NMC	Nickel manganese cobalt
CP / PP	Control pilot / proximity pilot	OBC	On-board charger
CTP / CTB	Cell-to-pack / cell-to-body	OBD	On-board diagnostics
DC	Direct current	PDU	Power distribution unit
DHT	Dedicated hybrid transmission	PHEV	Plug-in hybrid electric vehicle
DoD	Depth of discharge	PMSM	Permanent-magnet synchronous motor
DTC	Diagnostic trouble code	PPE	Personal protective equipment
EVSE	Electric vehicle supply equipment (charger)	PTC	Positive temperature coefficient (heater)
HEV	Hybrid electric vehicle	PWM	Pulse-width modulation
HV / LV	High voltage / low voltage	RCD	Residual current device
HVIL	High-voltage interlock loop	REEV	Range-extender electric vehicle
IGBT	Insulated-gate bipolar transistor	SiC	Silicon carbide
IP67	Dust-tight, protected against immersion	SOC / SOH	State of charge / state of health
kW / kWh	Kilowatt / kilowatt-hour	V2L	Vehicle-to-load
LOTO	Lock out, tag out	VCU	Vehicle control unit

Appendix C — English–Urdu glossary

Technical words are kept in English throughout the course because scan tools, manuals and trainers use them. This list gives the Urdu meaning to help you understand; learn the English word.

English	اردو	English	اردو
Voltage	وولٹیج — بجلی کا دباؤ	Current	کرنٹ — بجلی کا بہاؤ
Resistance	مزاحمت — بہاؤ میں رکاوٹ	Power	طاقت (پاور)
Energy	توانائی	Capacity	گنجائش
Battery	بیٹری	Cell	سیل — بیٹری کی سب سے چھوٹی اکائی
Charging	چارجنگ	Discharge	ڈسچارج — بیٹری سے توانائی کا نکلنا
Series	سیریز — ایک کے بعد ایک	Parallel	متوازی — ساتھ ساتھ
High voltage	ہائی وولٹیج — خطرناک وولٹیج	Electric shock	بجلی کا جھٹکا
Danger	خطرہ	Safety	حفاظت
Insulation	انسولیشن — بجلی کو روکنے والی تہہ	Insulating gloves	حفاظتی ربڑ کے دستانے
Short circuit	شارٹ سرکٹ	Earth (ground)	ارتھ
Fuse	فیوز	Contactor / relay	کنٹیکٹر / ریلے — بجلی سے چلنے والا سوئچ
Motor	موٹر	Engine	انجن
Torque	ٹارک — گھمانے کی قوت	Speed	رفتار
Magnet	مقناطیس	Gear	گیئر
Inverter	انورٹر — ڈی سی کو اے سی میں بدلتا ہے	Charger	چارجر
Hybrid	ہائبرڈ — انجن اور موٹر دونوں والی گاڑی	Range	رینج — ایک چارج پر طے ہونے والا فاصلہ
Regenerative braking	بریک لگاتے وقت توانائی واپس بیٹری میں	Efficiency	کارکردگی
Temperature	درجہ حرارت	Heat	حرارت / گرمی
Coolant	کولنٹ — ٹھنڈا کرنے والا مائع	Cooling	ٹھنڈا کرنا
Sensor	سینسر	Connector	کنیکٹر
Cable / wire	کیبل / تار	Wiring diagram	وائرنگ کا نقشہ
Fault	خرابی	Diagnosis	تشخیص
Repair	مرمت	Maintenance	دیکھ بھال

English	اردو	English	اردو
Measure	پیمائش کرنا	Tool	اوزار
Balance	توازن	Leak	رساؤ
Fire	آگ	Warranty	وارنٹی
Technician	ٹیکنیشن	Workshop	ورکشاپ

Appendix D — Quiz answers

Mark your own work honestly. For any wrong answer, go back to that section, read it again, and write the correct answer in your own words.

Quiz	Answers
Quiz 1	1 B · 2 C · 3 D · 4 A · 5 B
Quiz 2	1 B · 2 C · 3 B · 4 B · 5 C · 6 A
Quiz 3	1 C · 2 D · 3 C · 4 C · 5 B · 6 C · 7 C · 8 B
Quiz 4	1 D · 2 C · 3 B · 4 A · 5 B
Quiz 5	1 C · 2 B · 3 A · 4 C · 5 C · 6 B
Quiz 6	1 C · 2 B · 3 D · 4 A · 5 C · 6 B
Quiz 7	1 B · 2 C · 3 C · 4 B · 5 C · 6 C
Quiz 8	1 C · 2 B · 3 D · 4 B · 5 C · 6 B · 7 C
Quiz 9	1 B · 2 C · 3 C · 4 B · 5 C · 6 C
Quiz 10	1 C · 2 B · 3 C · 4 B · 5 B · 6 C
Quiz 11	1 C · 2 B · 3 C · 4 B · 5 C · 6 C · 7 B
Quiz 12	1 B · 2 C · 3 B · 4 B · 5 C · 6 B
Quiz 13	1 C · 2 B · 3 B · 4 C · 5 C · 6 B · 7 B
Quiz 14	1 B · 2 C · 3 C · 4 B · 5 C · 6 B · 7 C
Quiz 15	1 B · 2 B · 3 C · 4 C · 5 B · 6 C · 7 B · 8 B · 9 C · 10 B · 11 C
Quiz 16	1 C · 2 B · 3 B · 4 C · 5 B · 6 C
Quiz 17	1 C · 2 B · 3 C · 4 B · 5 C · 6 C
Quiz 18	1 C · 2 C · 3 B · 4 C · 5 B · 6 B · 7 C
Quiz 19	1 B · 2 B · 3 C · 4 B · 5 B
Quiz 20	1 B · 2 D · 3 B · 4 B · 5 B · 6 C · 7 C
Final assessment	1 C · 2 C · 3 C · 4 B · 5 C · 6 B · 7 B · 8 B · 9 B · 10 C · 11 C · 12 B · 13 A · 14 B · 15 B · 16 C · 17 C · 18 B · 19 B · 20 B

Appendix E — Sources and further reading

Market and policy information in this course was compiled in November 2026 from the following. Prices and specifications change; check current sources.

- PakWheels — Electric car prices in Pakistan 2026; plug-in hybrid guide 2026; model specification pages (pakwheels.com)
- ProPakistani — BYD, Haval, Jaecoo and hybrid market coverage, 2024–2026 (propakistani.pk)
- Profit / Pakistan Today — National EV Policy 2025–30 launch; PAVE scheme; NEPRA charging tariff; Chery PHEV launch (profit.pakistantoday.com.pk)
- Arab News Pakistan — NEV policy; charging station targets; electric two- and three-wheeler production figures (arabnews.pk)
- The Express Tribune, Dawn, Business Recorder — charging tariff, PAVE scheme, EV transition reporting
- NEPRA — Tariff determination for EV charging stations (April 2025); public EV charging station rules (nepra.org.pk)
- CnEVPost — BYD Shark launch in Pakistan (cnevpost.com)
- Lok Sujag — reporting on e-bike batteries, e-rickshaws and lithium battery policy (loksujag.com)
- TechJuice — Chinese auto brands in Pakistan; PHEV comparisons (techjuice.pk)
- Training syllabuses from a Guangzhou training provider (7-day NEV, 3-day battery repair, 3-day A/C), used to align topics and typical test limits

Standards referred to

- UN ECE Regulation 100 — electrical safety of electric vehicles (voltage classes, insulation resistance)
- IEC 62196 — charging connectors (Type 2, CCS2) · IEC 61851 — conductive charging systems · GB/T 20234 — Chinese charging connectors
- IEC 60903 / ASTM D120 — insulating gloves · IEC 61010 — test instrument measurement categories (CAT ratings)
- ISO 15031 / SAE J2012 — diagnostic trouble codes

REMINDER

This course is general education. It does not replace the manufacturer's service information for a specific vehicle, or supervised practical training. Never work on high-voltage systems until you are trained, assessed, equipped and authorised to do so.

Appendix F — Pictures to remember

Every hard idea in this course has an everyday picture. Cover the right-hand column and test yourself: can you explain the idea using only the picture?

Idea	Everyday picture	Module
Voltage, current, resistance	Water tank height, flow in the pipe, a narrow section of pipe	2
kW and kWh	Width of the pipe and size of the tank	2
Series and parallel	Stacking torch cells end to end (brighter) or side by side (lasts longer)	2
Why hand-to-hand shock kills	The path between the hands runs through the heart	3
"Prove, test, prove"	A torch with a dead cell shows darkness in a lit room too: check the torch first	3
Battery under the floor	A skateboard	5
Lithium-ion cell	Ions as passengers travelling between two stations, back and forth	6
Usable capacity	A water tank you never drain to the mud or fill to the brim	7
Weakest cell	The weakest link in a chain	8
Pre-charge	Opening a small side gate before the main canal gate	8
Rotating field	The donkey following the carrot	9
PWM	A tap that is only fully on or off, opened in longer or shorter bursts	10
Performance limit	A chain of pipes: the narrowest one sets the flow	11
Regenerative braking	A bicycle dynamo making pedalling harder	12
Charging handshake	Knocking and waiting to be invited in	14
Charging curve	Filling a glass: fast at first, slow near the top	14
Refrigeration	Sweat cooling the skin as it evaporates	15
Heat pump	An A/C that carries heat in either direction	15
CAN bus	A group chat where everyone hears every message	16
Fault code	A patient saying "my chest hurts": a clue for the doctor, not the diagnosis	18

HOW TO MAKE IT STICK

- **Draw it.** After each module, close the book and redraw its main diagram from memory. Check, correct, repeat the next day.
- **Teach it.** Explain one "picture" to a friend or family member who knows nothing about cars. If they understand, you do.
- **Find it.** Stand next to any EV or hybrid and name what you can see without touching: orange cables, charge port, coolant reservoirs, the compressor, the 12 V battery.
- **Come back to it.** Repeat each quiz after one day, one week and one month. Memory needs spaced repeats, not one long sitting.